

On some equations concerning the Ramanujan notebooks, the Instantons and Magnetic Monopoles. New possible mathematical connections with various parameters of Theoretical Physics, String Theory and Supersymmetry Breaking. IV

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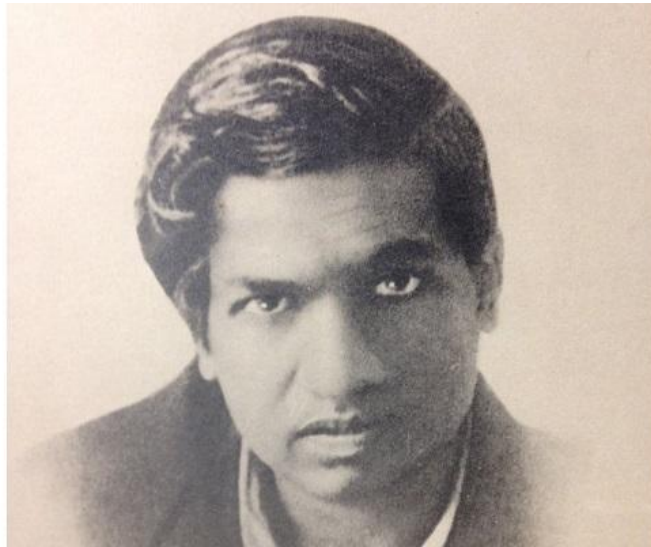
Abstract

In this research thesis (part IV), we analyze some equations concerning the Ramanujan notebooks, the Instantons and Magnetic Monopoles. We describe the new possible mathematical connections with various parameters of Theoretical Physics, String Theory and Supersymmetry Breaking.

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Srinivasa Ramanujan (1887-1920)



<https://www.moduscc.it/ramanujan-il-grande-matematico-indiano-13453-131115/>

Vesuvius landscape with gorse – Naples



<https://www.pinterest.it/pin/95068242114589901/>

From:

A. A. Karatsuba, “On the zeros of arithmetic Dirichlet series without Euler product,” *Izv. Ross. Akad. Nauk, Ser. Mat.* 57 (5), 3–14 (1993)

On the Zeros of the Davenport Heilbronn Function

S. A. Gritsenko - Received May 15, 2016 - ISSN 0081-5438, Proceedings of the Steklov Institute of Mathematics, 2017, Vol. 296, pp. 65–87.

We have:

Let

$$\varkappa = \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}$$

and χ_1 be a character modulo 5 such that $\chi_1(2) = i$.

The Davenport–Heilbronn function $f(s)$ is defined by the equality

$$f(s) = \frac{1 - i\varkappa}{2} L(s, \chi_1) + \frac{1 + i\varkappa}{2} L(s, \bar{\chi}_1), \quad \text{where} \quad L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

The function $f(s)$ satisfies the Riemann-type functional equation

$$g(s) = g(1 - s), \quad \text{where} \quad g(s) = \left(\frac{\pi}{5}\right)^{-s/2} \Gamma\left(\frac{s+1}{2}\right) f(s),$$

but there is no Euler product for it.

$$(\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1) = \kappa$$

Input:

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}$$

Decimal approximation:

0.2840790438404122960282918323931261690910880884457375827591626661

...

$$0.28407904384\dots = \kappa$$

Alternate forms:

$$\frac{1}{4} \left(\sqrt{10 - 2\sqrt{5}} - 2\sqrt{5} + \sqrt{5(10 - 2\sqrt{5})} - 2 \right)$$

$$\frac{1}{4} (1 + \sqrt{5}) \left(\sqrt{10 - 2\sqrt{5}} - 2 \right)$$

$$\frac{1}{2} \left(-1 - \sqrt{5} + \sqrt{2(5 + \sqrt{5})} \right)$$

Minimal polynomial:

$$x^4 + 2x^3 - 6x^2 - 2x + 1$$

Expanded forms:

$$\frac{\sqrt{10 - 2\sqrt{5}}}{\sqrt{5} - 1} - \frac{2}{\sqrt{5} - 1}$$

$$\frac{1}{4} \sqrt{10 - 2\sqrt{5}} + \frac{1}{4} \sqrt{5(10 - 2\sqrt{5})} + \frac{1}{2} (-1 - \sqrt{5})$$

For $((((\sqrt{10-2\sqrt{5}}-2)/(\sqrt{5}-1)))) = 8\pi G$; $G = 0.011303146014$

Indeed:

$$(((\sqrt{10-2\sqrt{5}}-2)/(\sqrt{5}-1)))/(8\pi)$$

Input:

$$\frac{\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}}{8\pi}$$

Result:

$$\frac{\sqrt{10-2\sqrt{5}}-2}{8(\sqrt{5}-1)\pi}$$

Decimal approximation:

0.0113031460140052147973750129442035744685760313920017808594909667
...

0.01130314.... = g (gravitational coupling constant)

Property:

$$\frac{-2 + \sqrt{10-2\sqrt{5}}}{8(-1+\sqrt{5})\pi} \text{ is a transcendental number}$$

Alternate forms:

$$\frac{\sqrt{10-2\sqrt{5}}-2\sqrt{5}+\sqrt{5(10-2\sqrt{5})}-2}{32\pi}$$

$$-\frac{1 + \sqrt{5} - \sqrt{2(5 + \sqrt{5})}}{16\pi}$$

$$\frac{-1 - \sqrt{5} + \sqrt{2(5 + \sqrt{5})}}{16\pi}$$

Expanded forms:

$$-\frac{1}{16\pi} - \frac{\sqrt{5}}{16\pi} + \frac{\sqrt{10 - 2\sqrt{5}}}{32\pi} + \frac{\sqrt{5(10 - 2\sqrt{5})}}{32\pi}$$

$$\frac{\sqrt{10 - 2\sqrt{5}}}{8(\sqrt{5} - 1)\pi} - \frac{1}{4(\sqrt{5} - 1)\pi}$$

Series representations:

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{(8\pi)(\sqrt{5} - 1)} = \frac{-2 + \sqrt{9 - 2\sqrt{5}} \sum_{k=0}^{\infty} \binom{\frac{1}{2}}{k} (9 - 2\sqrt{5})^{-k}}{8\pi \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right)}$$

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{(8\pi)(\sqrt{5} - 1)} = \frac{-2 + \sqrt{9 - 2\sqrt{5}} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (9 - 2\sqrt{5})^{-k}}{k!}}{8\pi \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{4}\right)^k \left(-\frac{1}{2}\right)_k}{k!} \right)}$$

$$\frac{\sqrt{10-2\sqrt{5}}-2}{(8\pi)(\sqrt{5}-1)} = \frac{-2+\sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (10-2\sqrt{5}-z_0)^k z_0^{-k}}{k!}}{8\pi \left(-1+\sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (5-z_0)^k z_0^{-k}}{k!}\right)}$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

We note that:

$$(((\sqrt{10-2\sqrt{5}}-2))/(\sqrt{5}-1))*((2i(\sqrt{5}-1)t + \sqrt{5}-1)/(2(\sqrt{2(5-\sqrt{5})}-2))) - 2)))$$

Input:

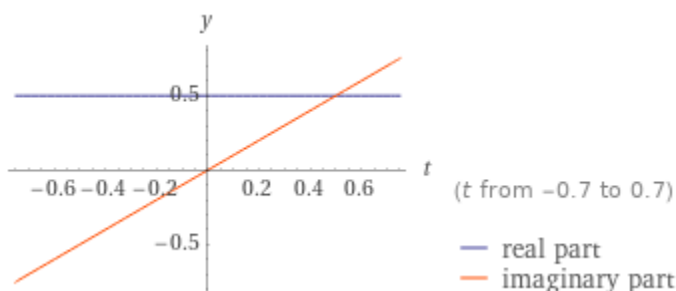
$$\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1} \times \frac{2i(\sqrt{5}-1)t + \sqrt{5}-1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)}$$

i is the imaginary unit

Exact result:

$$\frac{\left(\sqrt{10-2\sqrt{5}}-2\right)\left(2i\left(\sqrt{5}-1\right)t + \sqrt{5}-1\right)}{2\left(\sqrt{5}-1\right)\left(\sqrt{2\left(5-\sqrt{5}\right)}-2\right)}$$

Plot:



Alternate form assuming $t > 0$:

$$\begin{aligned} & \frac{i \sqrt{10 - 2\sqrt{5}} t}{\sqrt{2(5 - \sqrt{5})} - 2} - \frac{2 i t}{\sqrt{2(5 - \sqrt{5})} - 2} + \\ & \frac{\sqrt{5(10 - 2\sqrt{5})}}{2(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} - \frac{\sqrt{10 - 2\sqrt{5}}}{2(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} - \\ & \frac{\sqrt{5}}{(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} + \frac{1}{(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} \end{aligned}$$

Alternate forms:

$$\frac{1}{8} (1 + \sqrt{5}) \left(2 i \sqrt{2(3 - \sqrt{5})} t + \sqrt{5} - 1 \right)$$

$$\frac{1}{2} (1 + 2 i t)$$

$$\frac{1}{2} + i t$$

$1/2 + it$ = real part of every nontrivial zero of the Riemann zeta function

Derivative:

$$\frac{d}{dt} \left(\frac{\left(\sqrt{10 - 2\sqrt{5}} - 2 \right) (2 i (\sqrt{5} - 1) t + \sqrt{5} - 1)}{(\sqrt{5} - 1) \left(2 \left(\sqrt{2(5 - \sqrt{5})} - 2 \right) \right)} \right) = i$$

Indefinite integral:

$$\int \frac{(\sqrt{10-2\sqrt{5}}-2)(2i(\sqrt{5}-1)t+\sqrt{5}-1)}{(\sqrt{5}-1)\left(2\left(\sqrt{2(5-\sqrt{5})}-2\right)\right)} dt = \frac{t}{2} + \frac{it^2}{2} + \text{constant}$$

And again:

$$(((\sqrt{10-2\sqrt{5}}-2))/(2x))) * ((2i(\sqrt{5}-1)t + \sqrt{5}-1)/(2(\sqrt{2(5-\sqrt{5})}-2))) = (1/2 + it)$$

Input:

$$\frac{\sqrt{10-2\sqrt{5}}-2}{2x} \times \frac{2i(\sqrt{5}-1)t + \sqrt{5}-1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)} = \frac{1}{2} + it$$

i is the imaginary unit

Exact result:

$$\frac{(\sqrt{10-2\sqrt{5}}-2)(2i(\sqrt{5}-1)t + \sqrt{5}-1)}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} = \frac{1}{2} + it$$

Alternate form assuming t and x are real:

$$\frac{\sqrt{5}-1}{x} = 2$$

Alternate form:

$$\frac{(\sqrt{5} - 1)(1 + 2it)}{4x} = \frac{1}{2} + it$$

Alternate form assuming t and x are positive:

$$2x + 1 = \sqrt{5}$$

Expanded forms:

$$\begin{aligned} & \frac{i\sqrt{5(10-2\sqrt{5})}t}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{i\sqrt{10-2\sqrt{5}}t}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{i\sqrt{5}t}{\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \\ & \frac{it}{\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \frac{\sqrt{5(10-2\sqrt{5})}}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{\sqrt{10-2\sqrt{5}}}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \\ & \frac{\sqrt{5}}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \frac{1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} = \frac{1}{2} + it \end{aligned}$$

$$\frac{i\sqrt{5}t}{2x} - \frac{it}{2x} + \frac{\sqrt{5}}{4x} - \frac{1}{4x} = \frac{1}{2} + it$$

Solutions:

$$t = \frac{i}{2}, \quad x \neq 0$$

$$x = \frac{\sqrt{5}}{2} - \frac{1}{2}$$

Input:

$$\frac{\sqrt{5}}{2} - \frac{1}{2}$$

Decimal approximation:

0.6180339887498948482045868343656381177203091798057628621354486227

...

$$0.6180339887\dots = \frac{1}{\phi}$$

Solution for the variable x:

$$x = \frac{-2i\sqrt{5}t + 2it - \sqrt{5} + 1}{-2 - 4it}$$

Implicit derivatives:

$$\frac{\partial x(t)}{\partial t} = \frac{2(-1 + \sqrt{5} - 2x)x}{(-1 + \sqrt{5})(-i + 2t)}$$

$$\frac{\partial t(x)}{\partial x} = \frac{(-1 + \sqrt{5})(-i + 2t)}{2(-1 + \sqrt{5} - 2x)x}$$

From:

MAGNETIC MONOPOLES IN UNIFIED GAUGE THEORIES

G. 't HOOFT - CERN, Geneva - Received 31 May 1974

We have that:

From (2.9) we find the Lagrange equation

$$\frac{d}{dr} \left(2r^4 \frac{dW}{dr} + 4r^3 W \right) = r^2 \left[4r \frac{dW}{dr} + 12W + 6er^2 W^2 + 2e^2 r^4 W^3 + 2er^2 Q^2 + 2e^2 r^4 WQ^2 \right]. \quad (2.13)$$

So, substituting (2.11) and (2.12),

$$(3-n)(4-2n)ar^{2-n} \xrightarrow{r \rightarrow \infty} -4nar^{2-n} + 12ar^{2-n} + 6ea^2r^{4-2n} + 2e^2a^3r^{6-3n} + 2eF^2r^2 + 2e^2aF^2r^{4-n}. \quad (2.14)$$

The only solution is

$$n = 2, \quad a = -1/e. \quad (2.15)$$

$$-8/e - 12/e + 6e \times \frac{1}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + 2e^2 \left(-\frac{1}{e} \right) x^2$$

Input

$$-\frac{8}{e} - \frac{12}{e} + 6e \times \frac{1}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + 2e^2 \left(-\frac{x^2}{e} \right)$$

Result

$$-\frac{16}{e}$$

Decimal approximation

$$-5.886071058743077145528380322583373879132978096508285352125388827$$

...

$$-5.886071058....$$

Property

$-\frac{16}{e}$ is a transcendental number

Alternative representation

$$-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{(2e^2)(-1)x^2}{e} = -\frac{8}{\exp(z)} - \frac{12}{\exp(z)} + \frac{6\exp(z)}{\exp^2(z)} + 2\exp^2(z) \left(-\frac{1}{\exp(z)}\right)^3 + 2\exp(z)x^2 + \frac{(2\exp^2(z))(-1)x^2}{\exp(z)} \text{ for } z = 1$$

Series representations

$$-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{(2e^2)(-1)x^2}{e} = -16 \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$$

$$-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{(2e^2)(-1)x^2}{e} = -\frac{16}{\sum_{k=0}^{\infty} \frac{1}{k!}}$$

$$-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{(2e^2)(-1)x^2}{e} = -\frac{32}{\sum_{k=0}^{\infty} \frac{1+k}{k!}}$$

From which:

$$e*[-(-8/e-12/e+6*e*1/e^2+2*e^2*(-1/e)^3+2*e*x^2+2*e^2(-1/e)*x^2)]$$

Input

$$e \left(- \left(-\frac{8}{e} - \frac{12}{e} + 6e \times \frac{1}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + 2e^2 \left(-\frac{x^2}{e} \right) \right) \right)$$

Result

16

16

Alternative representation

$$e(-1) \left(-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + \frac{2e^2(-x^2)}{e} \right) = \\ \exp(z)(-1) \left(-\frac{8}{\exp(z)} - \frac{12}{\exp(z)} + \frac{6\exp(z)}{\exp^2(z)} + \right. \\ \left. 2\exp^2(z) \left(-\frac{1}{\exp(z)} \right)^3 + 2\exp(z)x^2 + \frac{2\exp^2(z)(-x^2)}{\exp(z)} \right) \text{ for } z = 1$$

$$27 \times 4e * [(-8/e - 12/e + 6 * e * 1/e^2 + 2 * e^2 * (-1/e)^3 + 2 * e * x^2 + 2 * e^2 * (-1/e) * x^2)]$$

Input

$$27 \times 4e \left(- \left(-\frac{8}{e} - \frac{12}{e} + 6e \times \frac{1}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + 2e^2 \left(-\frac{x^2}{e} \right) \right) \right)$$

Result

1728

1728

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

Alternative representation

$$27(-1) \left(-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + \frac{(2e^2)(-1)x^2}{e} \right) 4e =$$

$$27(-1) \left(-\frac{8}{\exp(z)} - \frac{12}{\exp(z)} + \frac{6\exp(z)}{\exp^2(z)} + 2\exp^2(z) \left(-\frac{1}{\exp(z)} \right)^3 + \right.$$

$$\left. 2\exp(z)x^2 + \frac{(2\exp^2(z))(-1)x^2}{\exp(z)} \right) 4\exp(z) \text{ for } z = 1$$

$$((27*4e*[-(-8/e-12/e+6*e*1/e^2+2*e^2*(-1/e)^3+2*e*x^2+2*e^2(-1/e)*x^2)]))^{1/15}$$

Input

$$\sqrt[15]{27 \times 4e \left(-\left(-\frac{8}{e} - \frac{12}{e} + 6e \times \frac{1}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + 2e^2 \left(-\frac{x^2}{e} \right) \right) \right)}$$

Exact result

$$2^{2/5} \sqrt[5]{3}$$

Decimal approximation

1.6437518295172257623084979362309795173834925899454752004110297675

...

$$1.643751829\dots \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934\dots \text{ (trace of the instanton shape)}$$

$$((4e*[-(-8/e-12/e+6*e*1/e^2+2*e^2*(-1/e)^3+2*e*x^2+2*e^2(-1/e)*x^2)]))^{2+276}$$

Input

$$\left(4e \left(-\left(-\frac{8}{e} - \frac{12}{e} + 6e \times \frac{1}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + 2e^2 \left(-\frac{x^2}{e} \right) \right) \right) \right)^2 + 276$$

Result

4372

4372

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

with a total flux

$$4\pi/e.$$

$$(4\pi) / ((-16*1/(-8/e-12/e+6*e*1/e^2+2*e^2*(-1/e)^3+2*e*x^2+2*e^2(-1/e)*x^2)))$$

Input

$$-\frac{4\pi}{16 \times \frac{1}{-\frac{8}{e}-\frac{12}{e}+6e \times \frac{1}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+2e^2\left(-\frac{x^2}{e}\right)}}$$

Result

$$\frac{4\pi}{e}$$

Decimal approximation

4.6229093991636868716403727332507851964834040926576632819988261413

...

4.622909399....

Alternative representations

$$-\frac{\frac{4\pi}{16}}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}} = -\frac{720^\circ}{\frac{16}{-\frac{20}{e}+2ex^2-\frac{2e^2x^2}{e}+2e^2\left(-\frac{1}{e}\right)^3+\frac{6e}{e^2}}}$$

$$-\frac{\frac{4\pi}{16}}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}} = \frac{-4i\log(-1)}{\frac{16}{-\frac{20}{e}+2ex^2-\frac{2e^2x^2}{e}+2e^2\left(-\frac{1}{e}\right)^3+\frac{6e}{e^2}}}$$

$$-\frac{\frac{4\pi}{16}}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}} = -\frac{8i\log\left(\frac{1-i}{1+i}\right)}{\frac{16}{-\frac{20}{e}+2ex^2-\frac{2e^2x^2}{e}+2e^2\left(-\frac{1}{e}\right)^3+\frac{6e}{e^2}}}$$

Series representations

$$-\frac{\frac{4\pi}{16}}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}} = \frac{16\sum_{k=0}^{\infty}\frac{(-1)^k}{1+2k}}{e}$$

$$-\frac{4\pi}{\frac{16}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}}} = 4\pi \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$$

$$-\frac{4\pi}{\frac{16}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}}} = \sum_{k=0}^{\infty} \frac{1}{k!}$$

Integral representations

$$-\frac{4\pi}{\frac{16}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}}} = \frac{16}{e} \int_0^1 \sqrt{1-t^2} \, dt$$

$$-\frac{4\pi}{\frac{16}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}}} = \frac{8}{e} \int_0^1 \frac{1}{\sqrt{1-t^2}} \, dt$$

$$-\frac{4\pi}{\frac{16}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2\left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}}} = \frac{8}{e} \int_0^{\infty} \frac{1}{1+t^2} \, dt$$

From which:

$$\left(\left(\left(\left(4\pi\right) / \left(\left(-16 * 1 / \left(-8 / e-12 / e+6 * e * 1 / e^2+2 * e^2 * \left(-1 / e\right)^3+2 * e * x^2+2 * e^2 * \left(-1 / e\right) * x^2\right)\right)\right)\right)\right)^{1 / \left(\pi\right)}$$

Input

$$\sqrt[\pi]{-\frac{4\pi}{16 \times \frac{1}{-\frac{8}{e}-\frac{12}{e}+6e \times \frac{1}{e^2}+2e^2 \left(-\frac{1}{e}\right)^3+2ex^2+2e^2 \left(-\frac{x^2}{e}\right)}}$$

Exact result

$$\sqrt[\pi]{\frac{4\pi}{e}}$$

Decimal approximation

1.6279802788808392329148815396473608021583860286191167202640384745
...

1.62798027888.... result near to the mean between $\zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$ and the value of golden ratio 1.61803398..., i.e. 1.63148399

Alternative representations

$$\sqrt[\pi]{-\frac{4\pi}{16 \times \frac{1}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2 \left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}}}} = \sqrt[180^\circ]{-\frac{720^\circ}{16 \times \frac{1}{-\frac{20}{e}+2ex^2-\frac{2e^2x^2}{e}+2e^2 \left(-\frac{1}{e}\right)^3+\frac{6e}{e^2}}}}$$

$$\sqrt[\pi]{-\frac{4\pi}{16 \times \frac{1}{-\frac{8}{e}-\frac{12}{e}+\frac{6e}{e^2}+2e^2 \left(-\frac{1}{e}\right)^3+2ex^2+\frac{2e^2(-x^2)}{e}}}} = \left(\frac{-4i \log(-1)}{-\frac{20}{e}+2ex^2-\frac{2e^2x^2}{e}+2e^2 \left(-\frac{1}{e}\right)^3+\frac{6e}{e^2}} \right)^{-1/(i \log(-1))}$$

$$\sqrt[\pi]{-\frac{4\pi}{16} - \frac{-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{2e^2(-x^2)}{e}}}{\sqrt[\pi]{-\frac{4\pi}{16} - \frac{-\frac{8}{\exp(z)} - \frac{12}{\exp(z)} + \frac{6\exp(z)}{\exp^2(z)} + 2\exp^2(z) \left(-\frac{1}{\exp(z)}\right)^3 + 2\exp(z)x^2 + \frac{2\exp^2(z)(-x^2)}{\exp(z)}}} = \text{for } z = 1$$

Series representations

$$\sqrt[\pi]{-\frac{4\pi}{16} - \frac{-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{2e^2(-x^2)}{e}}}{= \sqrt[\pi]{4\pi} \sqrt[\pi]{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}}$$

$$\sqrt[\pi]{-\frac{4\pi}{16} - \frac{-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{2e^2(-x^2)}{e}}}{= \sqrt[\pi]{4\pi} \sqrt[\pi]{\frac{1}{\sum_{k=0}^{\infty} \frac{1}{k!}}}$$

$$\sqrt[\pi]{-\frac{4\pi}{16} - \frac{-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{2e^2(-x^2)}{e}}}{= \sum_{k=0}^{\infty} \frac{(-1)^k}{1+2k} \sqrt[2]{2} e^{-1/\left(4 \sum_{k=0}^{\infty} \frac{(-1)^k}{1+2k}\right)} \sqrt[4]{\sum_{k=0}^{\infty} \frac{(-1)^k}{1+2k}}$$

Integral representations

$$\sqrt[\pi]{-\frac{4\pi}{16} - \frac{-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e}\right)^3 + 2ex^2 + \frac{2e^2(-x^2)}{e}}}{= 2^{3/\left(2 \int_0^{\infty} \frac{1}{1+t^2} dt\right)} e^{-1/\left(2 \int_0^{\infty} \frac{1}{1+t^2} dt\right)} \sqrt[2]{\int_0^{\infty} \frac{1}{1+t^2} dt}$$

$$\sqrt[4]{\frac{4\pi}{16 \left(-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + \frac{2e^2(-x^2)}{e} \right)}} = 2^{3/2} \left(2 \int_0^\infty \frac{\sin(t)}{t} dt \right) e^{-1/2} \left(2 \int_0^\infty \frac{\sin(t)}{t} dt \right)^2 \int_0^\infty \frac{\sin(t)}{t} dt \sqrt{\int_0^\infty \frac{\sin(t)}{t} dt}$$

$$\sqrt[4]{\frac{4\pi}{16 \left(-\frac{8}{e} - \frac{12}{e} + \frac{6e}{e^2} + 2e^2 \left(-\frac{1}{e} \right)^3 + 2ex^2 + \frac{2e^2(-x^2)}{e} \right)}} = \int_0^1 \sqrt{1-t^2} dt \sqrt[4]{2} e^{-1/4} \left(4 \int_0^1 \sqrt{1-t^2} dt \right)^4 \int_0^1 \sqrt{1-t^2} dt \sqrt{\int_0^1 \sqrt{1-t^2} dt}$$

From:

$$M_m = \frac{4\pi}{e^2} M_W C(\beta) ,$$

$$1/2 [((4\pi)/e^2 * 80.385 * 1.1) + ((4\pi)/e^2 * 80.385 * 1.44)]$$

Input interpretation

$$\frac{1}{2} \left(\frac{4\pi}{e^2} \times 80.385 \times 1.1 + \frac{4\pi}{e^2} \times 80.385 \times 1.44 \right)$$

Result

173.620...

173.620....

Alternative representations

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = \frac{73\,504.^\circ}{e^2}$$

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = - \frac{408.356 i \log(-1)}{e^2}$$

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = \frac{816.712 i \log\left(\frac{1-i}{1+i}\right)}{e^2}$$

Series representations

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = \frac{1633.42 \sum_{k=0}^{\infty} \frac{(-1)^k}{1+2k}}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^2}$$

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = 1633.42 \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{1+2k} \right) \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \right)^2$$

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = \frac{1633.42 \sum_{k=0}^{\infty} \frac{(-1)^k}{1+2k}}{\left(\sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} \right)^2}$$

Integral representations

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = \frac{816.712}{e^2} \int_0^{\infty} \frac{1}{1+t^2} dt$$

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = \frac{1633.42}{e^2} \int_0^1 \sqrt{1-t^2} dt$$

$$\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) = \frac{816.712}{e^2} \int_0^\infty \frac{\sin(t)}{t} dt$$

From which:

$$233 \times \frac{1}{\left(\left(\left(\frac{1}{2} \left[\frac{(4\pi)}{e^2} \times 80.385 \times 1.1 + \frac{(4\pi)}{e^2} \times 80.385 \times 1.44 \right] \right) - 29 \right) \right)}$$

Input interpretation

$$233 \times \frac{1}{\frac{1}{2} \left(\frac{4\pi}{e^2} \times 80.385 \times 1.1 + \frac{4\pi}{e^2} \times 80.385 \times 1.44 \right) - 29}$$

Result

1.6111193045309765814916239420598468263422206406010937159123479041

...

1.61111930453.... result that is a very good approximation to the value of the golden ratio 1.618033988749...

Alternative representations

$$\frac{233}{\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) - 29} = \frac{233}{-29 + \frac{73.504^\circ}{e^2}}$$

$$\frac{233}{\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) - 29} = \frac{233}{-29 - \frac{408.356 i \log(-1)}{e^2}}$$

$$\frac{233}{\frac{1}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) - 29} = \frac{233}{-29 + \frac{816.712 i \log\left(\frac{1-i}{1+i}\right)}{e^2}}$$

Series representations

$$\frac{\frac{233}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) - 29}{0.142645} =$$

$$-0.0177541 + \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{1+2k} \right) \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \right)^2$$

$$\frac{\frac{233}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) - 29}{0.570581} =$$

$$-0.0710165 + \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \right)^2 \sum_{k=1}^{\infty} 4^{-k} (-1 + 3^k) \zeta(1+k)$$

$$\frac{\frac{233}{2} \left(\frac{(4\pi) 80.385 \times 1.1}{e^2} + \frac{(4\pi) 80.385 \times 1.44}{e^2} \right) - 29}{0.142645} =$$

$$-0.0177541 + \left(\sum_{k=1}^{\infty} \tan^{-1} \left(\frac{1}{F_{1+2k}} \right) \right) \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \right)^2$$

Now, we have:

$$L = \int \mathcal{L} d^3\mathbf{x} = 4\pi \int_0^{\infty} r^2 dr \left[-r^2 \left(\frac{dW}{dr} \right)^2 - 4rW \frac{dW}{dr} - 6W^2 - 2er^2W^3 \right.$$

$$\left. - \frac{1}{2}e^2r^4W^4 - \frac{1}{2}r^2 \left(\frac{dQ}{dr} \right)^2 - rQ \frac{dQ}{dr} - \frac{3}{2}Q^2 - 2er^2WQ^2 - e^2r^4W^2Q^2 \right.$$

$$\left. + \frac{1}{4}\lambda F^2r^2Q^2 - \frac{1}{8}\lambda r^4Q^4 - \frac{1}{8}\lambda F^4 \right],$$

$$Q(r) \rightarrow F/r .$$

For F = 64 ; Q = 4

$$W(r) \rightarrow ar^{-n} .$$

$$-1/e * 16^{-2}$$

Input

$$-\frac{1}{e \times 16^2}$$

Result

$$-\frac{1}{e \ 256}$$

Decimal approximation

-0.001437029067075946568732514727193206513460199730592843103546237

...

-0.001437029....

Property

$-\frac{1}{256 e}$ is a transcendental number

For:

$$n = 2 , \quad a = -1/e .$$

$$r \rightarrow \infty$$

$$Q = 4 \quad W = -0.001437 \quad r = 16$$

From:

$$L = \int \mathcal{L} d^3x = 4\pi \int_0^\infty r^2 dr \left[-r^2 \left(\frac{dW}{dr} \right)^2 - 4rW \frac{dW}{dr} - 6W^2 - 2er^2W^3 \right. \\ \left. - \frac{1}{2}e^2r^4W^4 - \frac{1}{2}r^2 \left(\frac{dQ}{dr} \right)^2 - rQ \frac{dQ}{dr} - \frac{3}{2}Q^2 - 2er^2WQ^2 - e^2r^4W^2Q^2 \right. \\ \left. + \frac{1}{4}\lambda F^2r^2Q^2 - \frac{1}{8}\lambda r^4Q^4 - \frac{1}{8}\lambda F^4 \right],$$

[256(256(-0.001437/16)^2-4(16*(-0.001437))*(-0.001437/16)-6(-0.001437)^2-2*e*256*(-0.001437)^3-1/2*e^2*(16^4*(-0.001437)^4)-1/2*256(4/16)^2-64*(4/16)-3/2(4^2)-2*e*256*16*(-0.001437)-e^2*16^4*(-0.001437)^2*16+1/4*64^2*16^2*4^2-1/8*16^4*4^4-1/8*64^4]

Input

$$256 \left(256 \left(-\frac{0.001437}{16} \right)^2 - 4 (16 \times (-0.001437)) \left(-\frac{0.001437}{16} \right) - 6 (-0.001437)^2 - 2e \times 256 (-0.001437)^3 - \frac{1}{2} e^2 (16^4 (-0.001437)^4) \right)$$

Result

-0.003964751172855216068454314594122419530100151901301187497971977

...

-0.0039647511....

-0.003964751172855216-(((1/2*256(4/16)^2-64*(4/16)-3/2(4^2)-2*e*256*16*(-0.001437)-e^2*16^4*(-0.001437)^2*16+1/4*64^2*16^2*4^2-1/8*16^4*4^4-1/8*64^4)))

Input interpretation

$$-0.003964751172855216 - \left(\frac{1}{2} \times 256 \left(\frac{4}{16} \right)^2 - 64 \times \frac{4}{16} - \frac{3}{2} \times 4^2 - 2e \times 256 \times 16 \times (-0.001437) - e^2 \times 16^4 (-0.001437)^2 \times 16 + \frac{1}{4} \times 64^2 \times 16^2 \times 4^2 - \frac{1}{8} \times 16^4 \times 4^4 - \frac{1}{8} \times 64^4 \right)$$

Result

15.996035255373379422571207390898147281073391521540001183703000410

...

15.99603525....

Alternative representation

$$\begin{aligned} & -0.0039647511728552160000 - \\ & \left(\frac{256}{2} \left(\frac{4}{16} \right)^2 - \frac{64 \times 4}{16} - \frac{4^2 \times 3}{2} - 2e^{256 \times 16(-0.001437)} - \right. \\ & \left. e^2 16^4 (-0.001437)^2 16 + \frac{1}{4} \times 64^2 \times 16^2 \times 4^2 - \frac{16^4 \times 4^4}{8} - \frac{64^4}{8} \right) = \\ & -0.0039647511728552160000 - \left(\frac{256}{2} \left(\frac{4}{16} \right)^2 - \frac{64 \times 4}{16} - \frac{4^2 \times 3}{2} - \right. \\ & \left. 2 \exp(z) 256 \times 16(-0.001437) - \exp^2(z) 16^4 (-0.001437)^2 16 + \right. \\ & \left. \frac{1}{4} \times 64^2 \times 16^2 \times 4^2 - \frac{16^4 \times 4^4}{8} - \frac{64^4}{8} \right) \text{ for } z = 1 \end{aligned}$$

Series representations

$$\begin{aligned} & -0.0039647511728552160000 - \\ & \left(\frac{256}{2} \left(\frac{4}{16} \right)^2 - \frac{64 \times 4}{16} - \frac{4^2 \times 3}{2} - 2e^{256 \times 16(-0.001437)} - \right. \\ & \left. e^2 16^4 (-0.001437)^2 16 + \frac{1}{4} \times 64^2 \times 16^2 \times 4^2 - \frac{16^4 \times 4^4}{8} - \frac{64^4}{8} \right) = \\ & 31.9960352488271447840000 - 11.7719e + 2.16528 \sum_{k=0}^{\infty} \frac{2^k}{k!} \end{aligned}$$

$$\begin{aligned} & -0.0039647511728552160000 - \\ & \left(\frac{256}{2} \left(\frac{4}{16} \right)^2 - \frac{64 \times 4}{16} - \frac{4^2 \times 3}{2} - 2e^{256 \times 16(-0.001437)} - \right. \\ & \left. e^2 16^4 (-0.001437)^2 16 + \frac{1}{4} \times 64^2 \times 16^2 \times 4^2 - \frac{16^4 \times 4^4}{8} - \frac{64^4}{8} \right) = \\ & 2.16528 \left(14.7769 - 5.43667 \sum_{k=0}^{\infty} \frac{1}{k!} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^2 \right) \end{aligned}$$

$$\begin{aligned}
 & -0.0039647511728552160000 - \\
 & \left(\frac{256}{2} \left(\frac{4}{16} \right)^2 - \frac{64 \times 4}{16} - \frac{4^2 \times 3}{2} - 2 e^{256 \times 16} (-0.001437) - \right. \\
 & \quad \left. e^2 16^4 (-0.001437)^2 16 + \frac{1}{4} \times 64^2 \times 16^2 \times 4^2 - \frac{16^4 \times 4^4}{8} - \frac{64^4}{8} \right) = \\
 & 0.541319 \left(59.1075 - 10.8733 \sum_{k=0}^{\infty} \frac{1+k}{k!} + \left(\sum_{k=0}^{\infty} \frac{1+k}{k!} \right)^2 \right)
 \end{aligned}$$

4Pi Integrate[15.99603525]x

Input interpretation

$$4 \pi \int 15.99603525 x \, dx$$

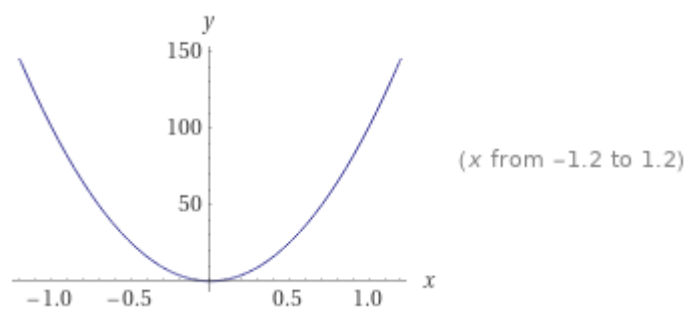
Result

$$100.506 x^2$$

For x = 1:

100.506

Plot



The energy of the system is given by:

$$E = -L$$

$$E = -100.506$$

Now, we have that:

Let us introduce dimensionless parameters:

$$w = W/F^2 e, \quad q = Q/F^2 e,$$

$$x = eFr, \quad \beta = \lambda/e^2 = M_H^2/M_W^2.$$

We obtain:

$$Q = 4; \quad W = -0.001437; \quad r = 16$$

$$e \cdot 64 \cdot 16$$

$$e \times 64 \times 16$$

$$1024 \, e$$

$$2783.5205923420623210089343706651263977034210239487586047661748507$$

...

$$2783.5205923 \dots$$

$1024 \, e$ is a transcendental number

$$x = 2783.5205923$$

$$4/(64^2 \cdot e)$$

$$\frac{4}{64^2 \, e}$$

$$\frac{1}{1024 e}$$

0.0003592572667689866421831286817983016283650499326482107758865593

...

$$q = 0.00035925726$$

$$-0.001437/(64^2 * e)$$

$$-\frac{0.001437}{64^2 e}$$

$$-1.29063... \times 10^{-7}$$

$$w = -1.29063 * 10^{-7}$$

$$(125.10^2 / 80.379^2)$$

$$\frac{125.10^2}{80.379^2}$$

2.4223083250591087236812176047917899933278387893317785225614243719

...

$$\beta = 2.422308325 ;$$

For:

$$\beta = 2.422308325 ; \quad q = 0.00035925726 ; \quad x = 2783.5205923 ; \quad w = -1.29063 * 10^{-7}$$

From:

$$\begin{aligned} & \frac{4\pi M_W}{e^2} \int_0^\infty x^2 dx \left[x^2 \left(\frac{dw}{dx} \right)^2 + 4xw \frac{dw}{dx} + 6w^2 + 2x^2 w^3 + \frac{1}{2} x^4 w^4 + \frac{1}{2} x^2 \left(\frac{dq}{dx} \right)^2 \right. \\ & \quad \left. + xq \frac{dq}{dx} + \frac{3}{2} q^2 + 2x^2 w q^2 + x^4 w^2 q^2 - \frac{1}{4} \beta x^2 q^2 + \frac{1}{8} \beta x^4 q^4 + \frac{1}{8} \beta \right]. \quad (3.2) \end{aligned}$$

we obtain:

$$2783.5205923^2*(-1.29063e-7/2783.5205923)^2+4*2783.5205923*(-1.29063e-7)*(-1.29063e-7/2783.5205923)+6*(-1.29063e-7)^2+2*(2783.5205923)^2(-1.29063e-7)^3+1/2*(2783.5205923)^4(-1.29063e-7)^4+1/2*(2783.5205923)^2*(0.00035925726/2783.5205923)^2$$

Input interpretation

$$2783.52^2\left(-\frac{1.29063\times 10^{-7}}{2783.52}\right)^2+4\times 2783.52(-1.29063\times 10^{-7})\left(-\frac{1.29063\times 10^{-7}}{2783.52}\right)+6(-1.29063\times 10^{-7})^2+2\times 2783.52^2(-1.29063\times 10^{-7})^3+\frac{1}{2}\times 2783.52^4(-1.29063\times 10^{-7})^4+\frac{1}{2}\times 2783.52^2\left(\frac{0.00035925726}{2783.52}\right)^2$$

Result

$$6.453304767567086713186372379761307131285688844288\times 10^{-8}$$

$$6.45330476\dots\times 10^{-8}$$

$$2783.52*0.000359257(0.000359257/2783.52)+3/2(0.000359257)^2+2*2783.52^2(-1.29063e-7)(0.000359257)^2$$

Input interpretation

$$2783.52\times 0.000359257\times \frac{0.000359257}{2783.52}+\frac{3}{2}\times 0.000359257^2+2\times 2783.52^2(-1.29063\times 10^{-7})\times 0.000359257^2$$

Result

$$6.45384733287751251139592704\times 10^{-8}$$

$$6.4538473328\dots\times 10^{-8}$$

$$6.4538473328775125113e-8+(((0.000359257)^4(-1.29063e-7)^2(0.000359257)^2-1/4(2.4223083)2783.52^2(0.000359257)^2+1/8(2.4223083)2783.52^4(0.000359257)^4+1/8(2.4223083))))$$

Input interpretation

$$6.4538473328775125113 \times 10^{-8} + \left(0.000359257^4 (-1.29063 \times 10^{-7})^2 \times 0.000359257^2 + \frac{1}{4} \times 2783.52^2 \times 0.000359257^2 \times (-2.4223083) + \frac{1}{8} \times 2.4223083 \times 2783.52^4 \times 0.000359257^4 + \frac{1}{8} \times 2.4223083 \right)$$

Result

$$6.4539578763529249077431368550433666068909786255302662647416... \times 10^{-8}$$

$$6.4539578... * 10^{-8}$$

$$6.453304767567086 \times 10^{-8}+(((6.4538473328775125113e-8+(((0.000359257)^4(-1.29063e-7)^2(0.000359257)^2-1/4(2.4223083)2783.52^2(0.000359257)^2+1/8(2.4223083)2783.52^4(0.000359257)^4+1/8(2.4223083))))))$$

Input interpretation

$$6.453304767567086 \times 10^{-8} + \left(6.4538473328775125113 \times 10^{-8} + \left(0.000359257^4 (-1.29063 \times 10^{-7})^2 \times 0.000359257^2 + \frac{1}{4} \times 2783.52^2 \times 0.000359257^2 \times (-2.4223083) + \frac{1}{8} \times 2.4223083 \times 2783.52^4 \times 0.000359257^4 + \frac{1}{8} \times 2.4223083 \right) \right)$$

Result

$$1.2907262643920010907743136855043366606890978625530266264741... \times 10^{-7}$$

$$1.2907262643... * 10^{-7}$$

For the above calculations, from:

$$\frac{4\pi M_W}{e^2} \int_0^\infty x^2 dx \left[x^2 \left(\frac{dw}{dx} \right)^2 + 4xw \frac{dw}{dx} + 6w^2 + 2x^2 w^3 + \frac{1}{2}x^4 w^4 + \frac{1}{2}x^2 \left(\frac{dq}{dx} \right)^2 + xq \frac{dq}{dx} + \frac{3}{2}q^2 + 2x^2 w q^2 + x^4 w^2 q^2 - \frac{1}{4}\beta x^2 q^2 + \frac{1}{8}\beta x^4 q^4 + \frac{1}{8}\beta \right]. \quad (3.2)$$

we obtain:

$$(1/\text{nepier number}^2) * 4\pi * 80.379 * \text{integrate} [2783.52(1.290726264392 \times 10^{-7})]x$$

Input interpretation

$$\frac{1}{e^2} \times 4 (\pi \times 80.379) \int (2783.52 \times 1.290726264392 \times 10^{-7}) x dx$$

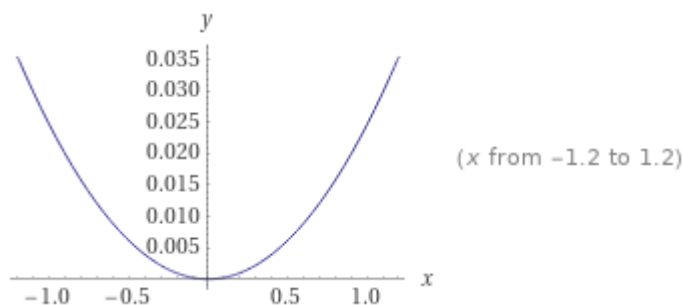
Result

$$0.0245562 x^2$$

For x = 1:

0.0245562

Plot



Alternate form assuming x is real

$$0.0245562 x^2 + 0$$

Indefinite integral assuming all variables are real

$$0.00818542 x^3 + \text{constant}$$

We note that this result 0.0245562 can be expressed also:

$$\frac{1}{18 \times 2^{5/7} e^{8/7} \log^{18/7}(2) \log^{9/7}(3)} \approx 0.024556201015414$$

Indeed:

$$1/(18 \cdot 2^{5/7} e^{8/7} \log^{18/7}(2) \log^{9/7}(3))$$

Input

$$\frac{1}{(18 \times 2^{5/7}) e^{8/7} \log^{18/7}(2) \log^{9/7}(3)}$$

$\log(x)$ is the natural logarithm

Exact result

$$\frac{1}{18 \times 2^{5/7} e^{8/7} \log^{18/7}(2) \log^{9/7}(3)}$$

Decimal approximation

0.0245562010154143622206940380785032244807065915796546210093963934

...

0.024556201....

Alternative representations

$$\frac{1}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = \frac{1}{18 \times 2^{5/7} e^{8/7} \log_e^{18/7}(2) \log_e^{9/7}(3)}$$

$$\frac{1}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = \frac{1}{18 \times 2^{5/7} e^{8/7} (\log(a) \log_a(2))^{18/7} (\log(a) \log_a(3))^{9/7}}$$

$$\frac{1}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = \frac{1}{18 \times 2^{5/7} e^{8/7} (2 \coth^{-1}(2))^{9/7} (2 \coth^{-1}(3))^{18/7}}$$

Series representations

$$\frac{1}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = \frac{1}{18 \times 2^{5/7} e^{8/7} \left(2i\pi \left\lfloor \frac{\arg(2-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k} \right)^{18/7} \left(2i\pi \left\lfloor \frac{\arg(3-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-x)^k x^{-k}}{k} \right)^{9/7}} \text{ for } x < 0$$

$$\frac{1}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = \frac{1}{18 \times 2^{5/7} e^{8/7} \left(2i\pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{18/7} \left(2i\pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{9/7}}$$

$$\frac{1}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} =$$

$$1 / \left(18 \times 2^{5/7} e^{8/7} \left(\left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{18/7} \right.$$

$$\left. \left(\left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{9/7} \right)$$

From which, we obtain:

$$67 \cdot 1 / (18 \cdot 2^{5/7} e^{8/7} \log^{18/7}(2) \log^{9/7}(3))$$

Input

$$67 \times \frac{1}{(18 \times 2^{5/7}) e^{8/7} \log^{18/7}(2) \log^{9/7}(3)}$$

$\log(x)$ is the natural logarithm

Exact result

$$\frac{67}{18 \times 2^{5/7} e^{8/7} \log^{18/7}(2) \log^{9/7}(3)}$$

Decimal approximation

1.6452654680327622687865005512597160402073416358368596076295583578

...

$$1.645265468.... \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934 ... \text{ (trace of the instanton shape)}$$

Alternative representations

$$\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = \frac{67}{18 \times 2^{5/7} e^{8/7} \log_e^{18/7}(2) \log_e^{9/7}(3)}$$

$$\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = \frac{67}{18 \times 2^{5/7} e^{8/7} (\log(a) \log_a(2))^{18/7} (\log(a) \log_a(3))^{9/7}}$$

$$\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = \frac{67}{18 \times 2^{5/7} e^{8/7} (2 \coth^{-1}(2))^{9/7} (2 \coth^{-1}(3))^{18/7}}$$

Series representations

$$\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = 67 / \left(18 \times 2^{5/7} e^{8/7} \left(2i\pi \left\lfloor \frac{\arg(2-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k} \right)^{18/7} \left(2i\pi \left\lfloor \frac{\arg(3-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-x)^k x^{-k}}{k} \right)^{9/7} \right) \text{ for } x < 0$$

$$\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} = 67 / \left(18 \times 2^{5/7} e^{8/7} \left(2i\pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{18/7} \left(2i\pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{9/7} \right)$$

$$\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} =$$

$$67 / \left(18 \times 2^{5/7} e^{8/7} \left(\left\lfloor \frac{\arg(2 - z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(2 - z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2 - z_0)^k z_0^{-k}}{k} \right)^{18/7} \right.$$

$$\left. \left(\left\lfloor \frac{\arg(3 - z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(3 - z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3 - z_0)^k z_0^{-k}}{k} \right)^{9/7} \right)$$

$$(((67*1/(18*2^{5/7}) e^{8/7} \log^{18/7}(2) \log^{9/7}(3))))^{15-24+1}$$

Input

$$\left(67 \times \frac{1}{(18 \times 2^{5/7}) e^{8/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 + 1$$

$\log(x)$ is the natural logarithm

Exact result

$$\frac{2461\,059\,085\,914\,092\,013\,369\,600\,043}{6908\,559\,991\,272\,917\,434\,368 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)} - 23$$

Decimal approximation

1729.0227369224800028605462454985916634950002248425737446488157722
...

1729.0227369....

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

Alternate forms

$$\begin{aligned} & (2461059085914092013369600043 \times 2^{2/7} - \\ & \quad 317793759598554201980928 e^{120/7} \log^{270/7}(2) \log^{135/7}(3)) / \\ & (13817119982545834868736 e^{120/7} \log^{270/7}(2) \log^{135/7}(3)) \end{aligned}$$

$$\begin{aligned} & -((158896879799277100990464 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3) - \\ & \quad 2461059085914092013369600043) / \\ & (6908559991272917434368 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3))) \end{aligned}$$

Alternative representations

$$\begin{aligned} & \left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 + 1 = \\ & -23 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} \log_e^{18/7}(2) \log_e^{9/7}(3)} \right)^{15} \end{aligned}$$

$$\begin{aligned} & \left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 + 1 = \\ & -23 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} (\log(a) \log_a(2))^{18/7} (\log(a) \log_a(3))^{9/7}} \right)^{15} \end{aligned}$$

$$\begin{aligned} & \left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 + 1 = \\ & -23 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} (2 \coth^{-1}(2))^{9/7} (2 \coth^{-1}(3))^{18/7}} \right)^{15} \end{aligned}$$

Series representations

$$\begin{aligned}
 & \left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 + 1 = \\
 & \left(2461\,059\,085\,914\,092\,013\,369\,600\,043 \times 2^{2/7} - 317\,793\,759\,598\,554\,201\,980\,928 \right. \\
 & \quad \left. e^{120/7} \left(2i\pi \left[\frac{\arg(2-x)}{2\pi} \right] + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k} \right)^{270/7} \right. \\
 & \quad \left. \left(2i\pi \left[\frac{\arg(3-x)}{2\pi} \right] + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-x)^k x^{-k}}{k} \right)^{135/7} \right) / \\
 & \left(13817\,119\,982\,545\,834\,868\,736 e^{120/7} \right. \\
 & \quad \left(2i\pi \left[\frac{\arg(2-x)}{2\pi} \right] + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k} \right)^{270/7} \\
 & \quad \left. \left(2i\pi \left[\frac{\arg(3-x)}{2\pi} \right] + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-x)^k x^{-k}}{k} \right)^{135/7} \right) \text{ for } x < 0
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 + 1 = \\
 & \left(2461\,059\,085\,914\,092\,013\,369\,600\,043 \times 2^{2/7} - \right. \\
 & \quad \left. 317\,793\,759\,598\,554\,201\,980\,928 e^{120/7} \right. \\
 & \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{270/7} \\
 & \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{135/7} \right) / \\
 & \left(13817\,119\,982\,545\,834\,868\,736 e^{120/7} \right. \\
 & \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{270/7} \\
 & \quad \left. \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{135/7} \right)
 \end{aligned}$$

$$\begin{aligned}
& \left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 + 1 = \\
& \left(2461059085914092013369600043 \times 2^{2/7} - \right. \\
& 317793759598554201980928 e^{120/7} \left(\left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \right. \\
& \left. \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{270/7} \\
& \left(\left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log(z_0) - \right. \\
& \left. \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{135/7} \Bigg) / \\
& \left(13817119982545834868736 e^{120/7} \left(\left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \right. \right. \\
& \left. \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{270/7} \\
& \left(\left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log(z_0) - \right. \\
& \left. \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{135/7} \Bigg)
\end{aligned}$$

$\arg(z)$ is the complex argument

$\lfloor x \rfloor$ is the floor function

$$((((67*1/(18*2^{5/7}) e^{8/7} \log^{18/7}(2) \log^{9/7}(3))))^{15-24})^{1/15}$$

Input

$$\sqrt[15]{\left(67 \times \frac{1}{(18 \times 2^{5/7}) e^{8/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24}$$

$\log(x)$ is the natural logarithm

Exact result

$$\sqrt[15]{\frac{2\,461\,059\,085\,914\,092\,013\,369\,600\,043}{6\,908\,559\,991\,272\,917\,434\,368 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)} - 24}$$

Decimal approximation

1.6437532714010389038771871316333922278225036575421375785054843355

...

$1.6437532714\dots \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934\dots$ (trace of the instanton shape)

Alternate forms

$$\left(\frac{(2\,461\,059\,085\,914\,092\,013\,369\,600\,043 - 165\,805\,439\,790\,550\,018\,424\,832 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3))^\wedge (1/15)}{(18 \times 2^{5/7} e^{8/7} \log^{18/7}(2) \log^{9/7}(3))} \right)$$

$$\left(\frac{(2\,461\,059\,085\,914\,092\,013\,369\,600\,043 \times 2^{2/7} - 331\,610\,879\,581\,100\,036\,849\,664 e^{120/7} \log^{270/7}(2) \log^{135/7}(3))^\wedge (1/15)}{(18 \times 2^{11/15} e^{8/7} \log^{18/7}(2) \log^{9/7}(3))} \right)$$

All 15th roots of $2461059085914092013369600043/(6908559991272917434368 \cdot 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)) - 24$

$$e^0 \sqrt[15]{\frac{2\,461\,059\,085\,914\,092\,013\,369\,600\,043}{6\,908\,559\,991\,272\,917\,434\,368 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)} - 24} \approx 1.644$$

(real, principal root)

$$e^{(2i\pi)/15} \sqrt[15]{\frac{2461059085914092013369600043}{6908559991272917434368 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)} - 24} \\ \approx 1.5016 + 0.6686i$$

$$e^{(4i\pi)/15} \sqrt[15]{\frac{2461059085914092013369600043}{6908559991272917434368 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)} - 24} \\ \approx 1.0999 + 1.2215i$$

$$e^{(6i\pi)/15} \sqrt[15]{\frac{2461059085914092013369600043}{6908559991272917434368 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)} - 24} \\ \approx 0.5079 + 1.563i$$

$$e^{(8i\pi)/15} \sqrt[15]{\frac{2461059085914092013369600043}{6908559991272917434368 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)} - 24} \\ \approx -0.1718 + 1.635i$$

Alternative representations

$$\sqrt[15]{\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)}\right)^{15} - 24} = \\ \sqrt[15]{-24 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} \log_e^{18/7}(2) \log_e^{9/7}(3)}\right)^{15}}$$

$$\sqrt[15]{\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)}\right)^{15}} - 24 =$$

$$\sqrt[15]{-24 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} (2 \coth^{-1}(2))^{9/7} (2 \coth^{-1}(3))^{18/7}}\right)^{15}}$$

$$\sqrt[15]{\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)}\right)^{15}} - 24 =$$

$$\sqrt[15]{-24 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} (\log(a) \log_a(2))^{18/7} (\log(a) \log_a(3))^{9/7}}\right)^{15}}$$

Series representations

$$\sqrt[15]{\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)}\right)^{15}} - 24 =$$

$$\left(-24 + 2461\,059\,085\,914\,092\,013\,369\,600\,043 / \left(6\,908\,559\,991\,272\,917\,434\,368 \times\right.\right.$$

$$\left.2^{5/7} e^{120/7} \left(2 i \pi \left\lfloor \frac{\arg(2-x)}{2 \pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k} \right)^{270/7}\right.$$

$$\left.\left(2 i \pi \left\lfloor \frac{\arg(3-x)}{2 \pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-x)^k x^{-k}}{k} \right)^{135/7} \right)^{1/15}$$

15) for $x < 0$

$$\begin{aligned}
& \sqrt[15]{\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)}\right)^{15} - 24} = \\
& \left(-24 + 2461059085914092013369600043 / \right. \\
& \quad \left(6908559991272917434368 \times 2^{5/7} e^{120/7} \left(\log(z_0) + \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \right. \right. \\
& \quad \left. \left. \left(\log\left(\frac{1}{z_0}\right) + \log(z_0) \right) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{270/7} \right. \right. \\
& \quad \left. \left(\log(z_0) + \left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \left(\log\left(\frac{1}{z_0}\right) + \log(z_0) \right) - \right. \right. \\
& \quad \left. \left. \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{135/7} \right) \right) \wedge (1/15)
\end{aligned}$$

$$\begin{aligned}
& \sqrt[15]{\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)}\right)^{15} - 24} = \\
& \left(-24 + 2461059085914092013369600043 / \right. \\
& \quad \left(6908559991272917434368 \times 2^{5/7} e^{120/7} \right. \\
& \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{270/7} \\
& \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \right. \\
& \quad \left. \left. \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{135/7} \right) \right) \wedge (1/15)
\end{aligned}$$

$$((1/27((((67*1/(18*2^{5/7}) e^{(8/7)} \log^{(18/7)}(2) \log^{(9/7)}(3))))^{15-24})))^2+276$$

Input

$$\left(\frac{1}{27}\left(\left(67\times\frac{1}{(18\times 2^{5/7})e^{8/7}\log^{18/7}(2)\log^{9/7}(3)}\right)^{15}-24\right)\right)^2+276$$

$\log(x)$ is the natural logarithm

Exact result

$$276+\frac{1}{729}\left(\frac{2461\,059\,085\,914\,092\,013\,369\,600\,043}{6\,908\,559\,991\,272\,917\,434\,368\times 2^{5/7}e^{120/7}\log^{270/7}(2)\log^{135/7}(3)}-24\right)^2$$

Decimal approximation

4372.1077905638663014371841630721557068266799997585508431983887814

...

$$4372.1077905\dots\approx 4372$$

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned}64g_{22}^{24} &= e^{\pi\sqrt{22}}-24+276e^{-\pi\sqrt{22}}-\dots, \\64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}}+\dots,\end{aligned}$$

so that

$$64(g_{22}^{24}+g_{22}^{-24})=e^{\pi\sqrt{22}}-24+4372e^{-\pi\sqrt{22}}+\dots=64\{(1+\sqrt{2})^{12}+(1-\sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}}=2508951.9982\dots$$

Alternate forms

$$276+\frac{1}{729}\left(24-\frac{2461\,059\,085\,914\,092\,013\,369\,600\,043}{6\,908\,559\,991\,272\,917\,434\,368\times 2^{5/7}e^{120/7}\log^{270/7}(2)\log^{135/7}(3)}\right)^2$$

$$276 + \left(165\,805\,439\,790\,550\,018\,424\,832 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3) - \right. \\ \left. 2\,461\,059\,085\,914\,092\,013\,369\,600\,043 \right)^2 / \\ (69\,587\,717\,281\,098\,571\,698\,664\,621\,889\,807\,980\,894\,661\,640\,192 \times \\ 2^{3/7} e^{240/7} \log^{540/7}(2) \log^{270/7}(3))$$

$$(6056811824360306133893955506189170373758117309785601849 \times 2^{4/7} + \\ 38522385714622962407013847475789998312550962298880 \\ e^{240/7} \log^{540/7}(2) \log^{270/7}(3) - \\ 1632227936362060193108504182376331879481138797871104 \times \\ 2^{2/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)) / \\ (139175434562197143397329243779615961789323280384 \\ e^{240/7} \log^{540/7}(2) \log^{270/7}(3))$$

Expanded form

$$\frac{22420}{81} + \\ 6056811824360306133893955506189170373758117309785601849 / \\ (69587717281098571698664621889807980894661640192 \times \\ 2^{3/7} e^{240/7} \log^{540/7}(2) \log^{270/7}(3)) - \\ \frac{2461059085914092013369600043}{104923754867457433534464 \times 2^{5/7} e^{120/7} \log^{270/7}(2) \log^{135/7}(3)}$$

Alternative representations

$$\left(\frac{1}{27} \left(\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 \right) \right)^2 + 276 = \\ 276 + \left(\frac{1}{27} \left(-24 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} \log_e^{18/7}(2) \log_e^{9/7}(3)} \right)^{15} \right) \right)^2$$

$$\left(\frac{1}{27} \left(\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 \right) \right)^2 + 276 = \\ 276 + \left(\frac{1}{27} \left(-24 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} (2 \coth^{-1}(2))^{9/7} (2 \coth^{-1}(3))^{18/7}} \right)^{15} \right) \right)^2$$

$$\left(\frac{1}{27} \left(\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 \right) \right)^2 + 276 =$$

$$276 + \left(\frac{1}{27} \left(-24 + \left(\frac{67}{18 \times 2^{5/7} e^{8/7} (\log(a) \log_a(2))^{18/7} (\log(a) \log_a(3))^{9/7}} \right)^{15} \right) \right)^2$$

Series representations

$$\left(\frac{1}{27} \left(\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 \right) \right)^2 + 276 =$$

$$\left(6056811824360306133893955506189170373758117309785601849 \times \right.$$

$$2^{4/7} -$$

$$1632227936362060193108504182376331879481138797871104 \times$$

$$2^{2/7} e^{120/7} \left(2i\pi \left\lfloor \frac{\arg(2-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k} \right)^{270/7}$$

$$\left(2i\pi \left\lfloor \frac{\arg(3-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-x)^k x^{-k}}{k} \right)^{135/7} +$$

$$38522385714622962407013847475789998312550962298880$$

$$e^{240/7} \left(2i\pi \left\lfloor \frac{\arg(2-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k} \right)^{540/7}$$

$$\left(2i\pi \left\lfloor \frac{\arg(3-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-x)^k x^{-k}}{k} \right)^{270/7} \Bigg) /$$

$$\left(139175434562197143397329243779615961789323280384 \right.$$

$$e^{240/7} \left(2i\pi \left\lfloor \frac{\arg(2-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k} \right)^{540/7}$$

$$\left. \left(2i\pi \left\lfloor \frac{\arg(3-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-x)^k x^{-k}}{k} \right)^{270/7} \right) \text{ for } x < 0$$

$$\begin{aligned}
& \left(\frac{1}{27} \left(\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 \right) \right)^2 + 276 = \\
& \left(6056811824360306133893955506189170373758117309785601849 \times \right. \\
& \quad 2^{4/7} - \\
& \quad 1632227936362060193108504182376331879481138797871104 \times \\
& \quad 2^{2/7} e^{120/7} \\
& \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2 - z_0)^k z_0^{-k}}{k} \right)^{270/7} \\
& \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3 - z_0)^k z_0^{-k}}{k} \right)^{135/7} + \\
& \quad 38522385714622962407013847475789998312550962298880 e^{240/7} \\
& \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2 - z_0)^k z_0^{-k}}{k} \right)^{540/7} \\
& \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3 - z_0)^k z_0^{-k}}{k} \right)^{270/7} \Bigg) / \\
& \left(139175434562197143397329243779615961789323280384 e^{240/7} \right. \\
& \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2 - z_0)^k z_0^{-k}}{k} \right)^{540/7} \\
& \quad \left(2i\pi \left[\frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right] + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3 - z_0)^k z_0^{-k}}{k} \right)^{270/7} \Bigg)
\end{aligned}$$

$$\begin{aligned}
& \left(\frac{1}{27} \left(\left(\frac{67}{e^{8/7} 18 \times 2^{5/7} \log^{18/7}(2) \log^{9/7}(3)} \right)^{15} - 24 \right) \right)^2 + 276 = \\
& \left(6056811\,824\,360\,306\,133\,893\,955\,506\,189\,170\,373\,758\,117\,309\,785\,601\,849 \times \right. \\
& \quad 2^{4/7} - \\
& \quad 1\,632\,227\,936\,362\,060\,193\,108\,504\,182\,376\,331\,879\,481\,138\,797\,871\,104 \times \\
& \quad 2^{2/7} e^{120/7} \left(\left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log(z_0) - \right. \\
& \quad \left. \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{270/7} \left(\left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \right. \\
& \quad \left. \left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{135/7} + \\
& \quad 38\,522\,385\,714\,622\,962\,407\,013\,847\,475\,789\,998\,312\,550\,962\,298\,880 \\
& \quad e^{240/7} \left(\left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \right. \\
& \quad \left. \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{540/7} \\
& \quad \left(\left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log(z_0) - \right. \\
& \quad \left. \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{270/7} \Bigg) / \\
& \left(139\,175\,434\,562\,197\,143\,397\,329\,243\,779\,615\,961\,789\,323\,280\,384 \right. \\
& \quad e^{240/7} \left(\left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \right. \\
& \quad \left. \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{540/7} \\
& \quad \left(\left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(3-z_0)}{2\pi} \right\rfloor \log(z_0) - \right. \\
& \quad \left. \sum_{k=1}^{\infty} \frac{(-1)^k (3-z_0)^k z_0^{-k}}{k} \right)^{270/7} \Bigg)
\end{aligned}$$

Now, we have that:

For

$$Q = 4 ; \quad W = -0.001437 ; \quad r = 16$$

From:

$$\begin{aligned} & \frac{d}{dr} \left(2r^4 \frac{dW}{dr} + 4r^3 W \right) \\ & = r^2 \left[4r \frac{dW}{dr} + 12W + 6er^2 W^2 + 2e^2 r^4 W^3 + 2er^2 Q^2 + 2e^2 r^4 W Q^2 \right] . \end{aligned} \quad (2.13)$$

$$16^2 [4 \times 16 \times (-0.001437/16) + 12 \times (-0.001437) + 6 \times e \times 16^2 \times (-0.001437)^2 + 2 \times e^2 \times 16^4 \times (-0.001437)^3 + 2 \times e \times 16^2 \times 16 + 2 \times e^2 \times 16^4 \times (-0.001437) \times 16]$$

Input

$$16^2 \left(4 \times 16 \left(-\frac{0.001437}{16} \right) + 12 \times (-0.001437) + 6 \times e \times 16^2 \times (-0.001437)^2 + 2 \times e^2 \times 16^4 \times (-0.001437)^3 + 2 \times e \times 16^2 \times 16 + 2 \times e^2 \times 16^4 \times (-0.001437) \times 16 \right)$$

Result

110.894...

110.894....

Alternative representation

$$\begin{aligned} & 16^2 \left(\frac{1}{16} (4 \times 16) (-0.001437) + 12 (-0.001437) + 6 \times e \times 16^2 \times (-0.001437)^2 + \right. \\ & \quad \left. 2 \times e^2 \times 16^4 \times (-0.001437)^3 + 2 \times e \times 16^2 \times 16 + 2 \times e^2 \times 16^4 \times (-0.001437) \times 16 \right) = \\ & 16^2 \left(\frac{1}{16} (4 \times 16) (-0.001437) + 12 (-0.001437) + \right. \\ & \quad \left. 6 \exp(z) \times 16^2 \times (-0.001437)^2 + 2 \exp^2(z) \times 16^4 \times (-0.001437)^3 + \right. \\ & \quad \left. 2 \exp(z) \times 16^2 \times 16 + 2 \exp^2(z) \times 16^4 \times (-0.001437) \times 16 \right) \text{ for } z = 1 \end{aligned}$$

Series representations

$$16^2 \left(\frac{1}{16} (4 \times 16) (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \\ \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + 2 e^2 16^4 (-0.001437) 16 \right) = \\ -5.88595 + 2.09715 \times 10^6 e - 771484. \sum_{k=0}^{\infty} \frac{2^k}{k!}$$

$$16^2 \left(\frac{1}{16} (4 \times 16) (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \\ \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + 2 e^2 16^4 (-0.001437) 16 \right) = \\ -771484. \left(-2.71833 + \sum_{k=0}^{\infty} \frac{1}{k!} \right) \left(-2.80664 \times 10^{-6} + \sum_{k=0}^{\infty} \frac{1}{k!} \right)$$

$$16^2 \left(\frac{1}{16} (4 \times 16) (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \\ \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + 2 e^2 16^4 (-0.001437) 16 \right) = \\ -192871. \left(-5.43667 + \sum_{k=0}^{\infty} \frac{1+k}{k!} \right) \left(-5.61328 \times 10^{-6} + \sum_{k=0}^{\infty} \frac{1+k}{k!} \right)$$

From which:

$$16((16^2[4*16*(-0.001437/16)+12*(-0.001437)+6*e*16^2(-0.001437)^2+2*e^2*16^4*(-0.001437)^3+2*e*16^2*16+2*e^2*16^4*(-0.001437)*16]))-47+\varphi$$

Input

$$16 \left(16^2 \left(4 \times 16 \left(-\frac{0.001437}{16} \right) + 12 \times (-0.001437) + \right. \right. \\ \left. \left. 6e \times 16^2 (-0.001437)^2 + 2e^2 \times 16^4 (-0.001437)^3 + \right. \right. \\ \left. \left. 2e \times 16^2 \times 16 + 2e^2 \times 16^4 \times (-0.001437) \times 16 \right) \right) - 47 + \phi$$

ϕ is the golden ratio

Result

1728.92...

$$1728.92..... \approx 1729$$

This result is very near to the mass of candidate glueball **f₀(1710) scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

Alternative representations

$$16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6e 16^2 (-0.001437)^2 + 2e^2 16^4 \right. \\ \left. (-0.001437)^3 + 2e 16^2 \times 16 + 2e^2 16^4 (-0.001437) 16 \right) - 47 + \phi = \\ 16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 \exp(z) 16^2 (-0.001437)^2 + \right. \\ \left. 2 \exp^2(z) 16^4 (-0.001437)^3 + 2 \exp(z) 16^2 \times 16 + \right. \\ \left. 2 \exp^2(z) 16^4 (-0.001437) 16 \right) - 47 + 2 \cos\left(\frac{\pi}{5}\right) \text{ for } z = 1$$

$$16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6e 16^2 (-0.001437)^2 + 2e^2 16^4 \right. \\ \left. (-0.001437)^3 + 2e 16^2 \times 16 + 2e^2 16^4 (-0.001437) 16 \right) - 47 + \phi = \\ 16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 \exp(z) 16^2 (-0.001437)^2 + \right. \\ \left. 2 \exp^2(z) 16^4 (-0.001437)^3 + 2 \exp(z) 16^2 \times 16 + \right. \\ \left. 2 \exp^2(z) 16^4 (-0.001437) 16 \right) - 47 - 2 \cos(6 \times 6 \times 6^\circ) \text{ for } z = 1$$

$$\begin{aligned}
& 16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + 2 e^2 16^4 \right. \\
& \quad \left. (-0.001437)^3 + 2 e 16^2 \times 16 + 2 e^2 16^4 (-0.001437) 16 \right) - 47 + \phi = \\
& 16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 \exp(z) 16^2 (-0.001437)^2 + \right. \\
& \quad \left. 2 \exp^2(z) 16^4 (-0.001437)^3 + 2 \exp(z) 16^2 \times 16 + \right. \\
& \quad \left. 2 \exp^2(z) 16^4 (-0.001437) 16 \right) - 47 + \frac{1}{2} \csc\left(\frac{\pi}{10}\right) \text{ for } z = 1
\end{aligned}$$

Series representations

$$\begin{aligned}
& 16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \\
& \quad \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + 2 e^2 16^4 (-0.001437) 16 \right) - \\
& 47 + \phi = -141.175 + \phi + 3.35544 \times 10^7 \sum_{k=0}^{\infty} \frac{1}{k!} - 1.23437 \times 10^7 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^2
\end{aligned}$$

$$\begin{aligned}
& 16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \\
& \quad \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + 2 e^2 16^4 (-0.001437) 16 \right) - 47 + \\
& \phi = -141.175 + \phi + 1.67772 \times 10^7 \sum_{k=0}^{\infty} \frac{1+k}{k!} - 3.08593 \times 10^6 \left(\sum_{k=0}^{\infty} \frac{1+k}{k!} \right)^2
\end{aligned}$$

$$\begin{aligned}
& 16 \times 16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \\
& \quad \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + 2 e^2 16^4 (-0.001437) 16 \right) - \\
& 47 + \phi = -141.175 + \phi + 3.35544 \times 10^7 \sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} - \\
& 1.23437 \times 10^7 \left(\sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} \right)^2
\end{aligned}$$

$$((16((16^2[4*16*(-0.001437/16)+12*(-0.001437)+6*e*16^2(-0.001437)^2+2*e^2*16^4*(-0.001437)^3+2*e*16^2*16+2*e^2*16^4*(-0.001437)*16]))-47+\phi))^{1/15}$$

Input

$$\left(16\left(16^2\left(4\times 16\left(-\frac{0.001437}{16}\right)+12\times(-0.001437)+6e\times 16^2(-0.001437)^2+2e^2\times 16^4(-0.001437)^3+2e\times 16^2\times 16+2e^2\times 16^4\times(-0.001437)\times 16\right)\right)-47+\phi\right)^{(1/15)}$$

ϕ is the golden ratio

Result

1.64381...

$$1.64381\dots \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934\dots \text{ (trace of the instanton shape)}$$

$$((1/27((16((16^2[4*16*(-0.001437/16)+12*(-0.001437)+6*e*16^2(-0.001437)^2+2*e^2*16^4*(-0.001437)^3+2*e*16^2*16+2*e^2*16^4*(-0.001437)*16]))-47+\phi-1))))^{2+276+2/5}$$

Input

$$\left(\frac{1}{27}\left(16\left(16^2\left(4\times 16\left(-\frac{0.001437}{16}\right)+12\times(-0.001437)+6e\times 16^2(-0.001437)^2+2e^2\times 16^4(-0.001437)^3+2e\times 16^2\times 16+2e^2\times 16^4\times(-0.001437)\times 16\right)\right)-47+\phi-1\right)\right)^{2+276+\frac{2}{5}}$$

ϕ is the golden ratio

Result

4372.01...

$$4372.01\dots \approx 4372$$

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

Alternative representations

$$\begin{aligned} &\left(\frac{1}{27}\left(16\left(16^2\left(\frac{4}{16}\times 16(-0.001437) + 12(-0.001437) + 6e16^2(-0.001437)^2 + \right.\right.\right.\right. \\ &\quad \left.\left.\left.2e^216^4(-0.001437)^3 + 2e16^2\times 16 + \right.\right.\right. \\ &\quad \left.\left.\left.2e^216^4(-0.001437)16\right)\right) - 47 + \phi - 1\right)\right)^2 + 276 + \frac{2}{5} = \\ &\left(\frac{1}{27}\left(16\left(16^2\left(\frac{4}{16}\times 16(-0.001437) + 12(-0.001437) + 6\exp(z)16^2\right.\right.\right.\right. \\ &\quad \left.\left.\left.(-0.001437)^2 + 2\exp^2(z)16^4(-0.001437)^3 + \right.\right.\right. \\ &\quad \left.\left.\left.2\exp(z)16^2\times 16 + 2\exp^2(z)16^4(-0.001437)16\right)\right) - \right. \\ &\quad \left.47 + 2\cos\left(\frac{\pi}{5}\right) - 1\right)\right)^2 + 276 + \frac{2}{5} \text{ for } z = 1 \end{aligned}$$

$$\begin{aligned} &\left(\frac{1}{27}\left(16\left(16^2\left(\frac{4}{16}\times 16(-0.001437) + 12(-0.001437) + 6e16^2(-0.001437)^2 + \right.\right.\right.\right. \\ &\quad \left.\left.\left.2e^216^4(-0.001437)^3 + 2e16^2\times 16 + \right.\right.\right. \\ &\quad \left.\left.\left.2e^216^4(-0.001437)16\right)\right) - 47 + \phi - 1\right)\right)^2 + 276 + \frac{2}{5} = \\ &\left(\frac{1}{27}\left(16\left(16^2\left(\frac{4}{16}\times 16(-0.001437) + 12(-0.001437) + 6\exp(z)16^2\right.\right.\right.\right. \\ &\quad \left.\left.\left.(-0.001437)^2 + 2\exp^2(z)16^4(-0.001437)^3 + \right.\right.\right. \\ &\quad \left.\left.\left.2\exp(z)16^2\times 16 + 2\exp^2(z)16^4(-0.001437)16\right)\right) - \right. \\ &\quad \left.47 - 2\cos(6\times 6\times 6^\circ) - 1\right)\right)^2 + 276 + \frac{2}{5} \text{ for } z = 1 \end{aligned}$$

$$\begin{aligned}
& \left(\frac{1}{27} \left(16 \left(16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \right. \right. \right. \\
& \quad \left. \left. \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + \right. \right. \right. \\
& \quad \left. \left. \left. 2 e^2 16^4 (-0.001437) 16 \right) \right) - 47 + \phi - 1 \right) \Big)^2 + 276 + \frac{2}{5} = \\
& \left(\frac{1}{27} \left(16 \left(16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 \exp(z) 16^2 \right. \right. \right. \right. \\
& \quad \left. \left. \left. (-0.001437)^2 + 2 \exp^2(z) 16^4 (-0.001437)^3 + \right. \right. \right. \\
& \quad \left. \left. \left. 2 \exp(z) 16^2 \times 16 + 2 \exp^2(z) 16^4 (-0.001437) 16 \right) \right) \right) - \\
& \quad 47 + \frac{1}{2} \csc\left(\frac{\pi}{10}\right) - 1 \Big)^2 + 276 + \frac{2}{5} \text{ for } z = 1
\end{aligned}$$

Series representations

$$\begin{aligned}
& \left(\frac{1}{27} \left(16 \left(16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \right. \right. \right. \\
& \quad \left. \left. \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + \right. \right. \right. \\
& \quad \left. \left. \left. 2 e^2 16^4 (-0.001437) 16 \right) \right) - 47 + \phi - 1 \right) \Big)^2 + \\
& \quad 276 + \frac{2}{5} = 0.00137174 \left(221\,709. - 284.35 \phi + \right. \\
& \quad \phi^2 - \\
& \quad 9.54122 \times 10^9 \sum_{k=0}^{\infty} \frac{1}{k!} + \\
& \quad 6.71089 \times 10^7 \phi \sum_{k=0}^{\infty} \frac{1}{k!} + \\
& \quad 1.1259 \times 10^{15} \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^2 - \\
& \quad 2.46875 \times 10^7 \phi \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^2 - \\
& \quad 8.28375 \times 10^{14} \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^3 + \\
& \quad \left. 1.52368 \times 10^{14} \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^4 \right)
\end{aligned}$$

$$\begin{aligned}
& \left(\frac{1}{27} \left(16 \left(16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \right. \right. \right. \\
& \quad \left. \left. \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + \right. \right. \right. \\
& \quad \left. \left. \left. 2 e^2 16^4 (-0.001437) 16 \right) \right) - 47 + \phi - 1 \right) \Big)^2 + \\
& 276 + \frac{2}{5} = 0.00137174 \left(221\,709. - 284.35 \phi + \right. \\
& \quad \phi^2 - \\
& \quad 4.77061 \times 10^9 \sum_{k=0}^{\infty} \frac{1+k}{k!} + \\
& \quad 3.35544 \times 10^7 \phi \sum_{k=0}^{\infty} \frac{1+k}{k!} + \\
& \quad 2.81476 \times 10^{14} \left(\sum_{k=0}^{\infty} \frac{1+k}{k!} \right)^2 - \\
& \quad 6.17187 \times 10^6 \phi \left(\sum_{k=0}^{\infty} \frac{1+k}{k!} \right)^2 - \\
& \quad 1.03547 \times 10^{14} \left(\sum_{k=0}^{\infty} \frac{1+k}{k!} \right)^3 + \\
& \quad \left. 9.52299 \times 10^{12} \left(\sum_{k=0}^{\infty} \frac{1+k}{k!} \right)^4 \right)
\end{aligned}$$

$$\begin{aligned}
& \left(\frac{1}{27} \left(16 \left(16^2 \left(\frac{4}{16} \times 16 (-0.001437) + 12 (-0.001437) + 6 e 16^2 (-0.001437)^2 + \right. \right. \right. \right. \\
& \quad \left. \left. \left. 2 e^2 16^4 (-0.001437)^3 + 2 e 16^2 \times 16 + \right. \right. \right. \\
& \quad \left. \left. \left. 2 e^2 16^4 (-0.001437) 16 \right) \right) - 47 + \phi - 1 \right) \Big)^2 + \\
& 276 + \frac{2}{5} = 0.00137174 \left(221\,709. - 284.35 \phi + \right. \\
& \quad \phi^2 - \\
& \quad 9.54122 \times 10^9 \sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} + \\
& \quad 6.71089 \times 10^7 \phi \sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} + \\
& \quad 1.1259 \times 10^{15} \left(\sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} \right)^2 - \\
& \quad 2.46875 \times 10^7 \phi \left(\sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} \right)^2 - \\
& \quad 8.28375 \times 10^{14} \left(\sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} \right)^3 + \\
& \quad \left. 1.52368 \times 10^{14} \left(\sum_{k=0}^{\infty} \frac{(-1+k)^2}{k!} \right)^4 \right)
\end{aligned}$$

From:

<https://twitter.com/royalsociety/status/997401631015751680/photo/1>

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vi. $\phi(x) + \phi(-x) = 2\phi(x^2)$
vii. $\phi(x) - \phi(-x) = 4x\psi(x^2)$
viii. $\phi(x)\phi(-x) = \phi^2(x^2)$
ix. $\phi(x)\psi(x^2) = \psi^2(x)$
x. $\phi^2(x) - \phi^2(-x) = 8x\psi^2(x^2)$
xi. $\phi^2(x) + \phi^2(-x) = 2\phi^2(x^2)$
xii. $\phi^4(x) - \phi^4(-x) = 16x\psi^4(x^2)$
xiii. $\psi^2(x) + \psi^2(-x) = 2\psi(x^2)\phi(x^2)$
xiv. If $\left(\frac{1-x}{1+x}\right)^2 = \left\{\frac{\phi(-x)}{\phi(x)}\right\}^4$ then $1-x^2 = \left\{\frac{\phi(-x^2)}{\phi(x^2)}\right\}^4$
Ex-1. $\frac{\psi(x)}{\psi(-x)} = \sqrt{\frac{\phi(x)}{\phi(-x)}}$
2. $\psi(x)\psi(-x) = \psi(x^2)\phi(-x^2)$
3. $\frac{\psi(x)\psi(-x)}{\psi(x^2)\psi(-x^2)} = \frac{\psi(x^2)}{\psi(-x^2)}$
18. If $F(x) = e^{-\pi \frac{(1/2)^2 x + (1/2)^2 x^2 + (1/2)^2 x^3 + \dots}{1 + (1/2)^2 x + (1/2)^2 x^2 + (1/2)^2 x^3 + \dots}}$
i. $F(x) = \frac{1}{16} e^{4 \frac{(1/2)^2 x + (1/2)^2 x^2 + (1/2)^2 x^3 + \dots}{1 + (1/2)^2 x + (1/2)^2 x^2 + (1/2)^2 x^3 + \dots}}$
ii. $F(1-x) + \theta = \frac{\log_e x}{10 + \sqrt{36 + (\log_e x)^2}}$ where θ is numerically
much less than $\frac{2}{135} F^5(1-x)$. $\theta = \frac{1}{2160} \left\{ \frac{\log_e x}{8 + \sqrt{36 + (\log_e x)^2}} \right\}$
iii. $\log_e F(x) \log_e F(1-x) = \pi^2$
iv. $F(1-x) + F(1-x^2) = 0$
v. $F\left\{\frac{4x}{(1+x)^2}\right\} = \sqrt{F(x^2)}$
N.B. If we know the expansion of $F\left(\frac{2x}{1+x}\right)$ to n terms, then
we can find its expansion to $2n$ terms as follows -
Suppose we know the expansion of $F\left(\frac{2x}{1+x}\right)$ to n terms.

From:

$$\psi(x) = \frac{\psi(x)}{1 + \left(\frac{1}{4}\right)^2(1-x) + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2(1-x)^2 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2(1-x)^3 + \dots}$$

$$e^{-\pi \cdot \frac{1 + \left(\frac{1}{4}\right)^2 x + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 x^2 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2 x^3 + \dots}{1 + \left(\frac{1}{4}\right)^2(1-x) + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2(1-x)^2 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2(1-x)^3 + \dots}}$$

for $x = 2$:

$$\text{Exp}(-\pi * (((1+1/4(1-2)+(3/8)^2(1-2)^2+(15/48)^2(1-2)^3)))/(((1+1/4*2+(3/8)^2*4+(15/48)^2*8))))$$

Input

$$\exp\left(-\pi \times \frac{1 + \frac{1}{4}(1-2) + \left(\frac{3}{8}\right)^2(1-2)^2 + \left(\frac{15}{48}\right)^2(1-2)^3}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8}\right)^2 \times 4 + \left(\frac{15}{48}\right)^2 \times 8}\right)$$

Exact result

$$e^{-(29\pi)/104}$$

Decimal approximation

0.4164366089307012351981966541617455222297165340221635985798712294

...

0.41643660893....

Property

$e^{-(29\pi)/104}$ is a transcendental number

Alternative representations

$$\exp\left(-\frac{\pi\left(1 + \frac{1-2}{4} + \left(\frac{3}{8}\right)^2(1-2)^2 + \left(\frac{15}{48}\right)^2(1-2)^3\right)}{1 + \frac{2}{4} + \left(\frac{3}{8}\right)^2 4 + \left(\frac{15}{48}\right)^2 8}\right) =$$

$$\exp\left(-\frac{180^\circ\left(1 - \frac{1}{4} + \left(\frac{3}{8}\right)^2 - \left(\frac{15}{48}\right)^2\right)}{1 + \frac{2}{4} + 4\left(\frac{3}{8}\right)^2 + 8\left(\frac{15}{48}\right)^2}\right)$$

$$\exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)=$$

$$\exp\left(\frac{i\log(-1)\left(1-\frac{1}{4}+\left(\frac{3}{8}\right)^2-\left(\frac{15}{48}\right)^2\right)}{1+\frac{2}{4}+4\left(\frac{3}{8}\right)^2+8\left(\frac{15}{48}\right)^2}\right)$$

$$\exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)=$$

$$\exp\left(-\frac{2i\log\left(\frac{1-i}{1+i}\right)\left(1-\frac{1}{4}+\left(\frac{3}{8}\right)^2-\left(\frac{15}{48}\right)^2\right)}{1+\frac{2}{4}+4\left(\frac{3}{8}\right)^2+8\left(\frac{15}{48}\right)^2}\right)$$

Series representations

$$\exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)=e^{-29/26\sum_{k=0}^{\infty}(-1)^k/(1+2k)}$$

$$\exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)=\left(\sum_{k=0}^{\infty}\frac{1}{k!}\right)^{-(29\pi)/104}$$

$$\exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)=\left(\frac{1}{\sum_{k=0}^{\infty}\frac{(-1)^k}{k!}}\right)^{-(29\pi)/104}$$

Integral representations

$$\exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)=e^{-29/26\int_0^1\sqrt{1-t^2}dt}$$

$$\exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)=e^{-29/52\times\int_0^1 1/\sqrt{1-t^2} dt}$$

$$\exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)=e^{-29/52\times\int_0^\infty 1/(1+t^2) dt}$$

From which:

$$(3+0.9568666373) * \text{Exp}(-\text{Pi} * (((1+1/4(1-2)+(3/8)^2(1-2)^2+(15/48)^2(1-2)^3)))/(((1+1/4*2+(3/8)^2*4+(15/48)^2*8))))$$

where 0.9568666373 is the value of the following Rogers-Ramanujan continued fraction:

$$\frac{e^{-\frac{\pi}{5}}}{\sqrt{(\varphi-1)\sqrt{5}}-\varphi+1}=1-\frac{e^{-\pi}}{1+\frac{e^{-2\pi}}{1+\frac{e^{-3\pi}}{1+\frac{e^{-4\pi}}{1+\dots}}}}\approx 0.9568666373$$

Input interpretation

$$(3+0.9568666373)\exp\left(-\pi\times\frac{1+\frac{1}{4}(1-2)+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3}{1+\frac{1}{4}\times 2+\left(\frac{3}{8}\right)^2\times 4+\left(\frac{15}{48}\right)^2\times 8}\right)$$

Result

1.6477841244...

$$1.6477841244\dots\approx\zeta(2)=\frac{\pi^2}{6}=1.644934\dots(\text{trace of the instanton shape})$$

$$(2 \times 0.9568666373 + 2 \times 0.9991104684) * \text{Exp}(-\text{Pi} * (((1 + 1/4(1-2) + (3/8)^2(1-2)^2 + (15/48)^2(1-2)^3)) / (((1 + 1/4 * 2 + (3/8)^2 * 4 + (15/48)^2 * 8))))))$$

where 0.9991104684 is the value of the following Rogers-Ramanujan continued fraction:

$$\frac{\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}}}{1 + \sqrt[5]{\sqrt{\varphi^5 \sqrt[4]{5^3} - 1}}} - \varphi + 1 = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

Input interpretation

$$(2 \times 0.9568666373 + 2 \times 0.9991104684) \exp\left(-\pi \times \frac{1 + \frac{1}{4}(1-2) + \left(\frac{3}{8}\right)^2(1-2)^2 + \left(\frac{15}{48}\right)^2(1-2)^3}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8}\right)^2 \times 4 + \left(\frac{15}{48}\right)^2 \times 8}\right)$$

Result

1.629080946...

1.629080946.... result very near to the mean between $\zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$ and the value of golden ratio 1.61803398..., i.e. 1.63148399

We have also:

$$5\text{Pi} * \text{Exp}(-\text{Pi} * (((1 + 1/4(1-2) + (3/8)^2(1-2)^2 + (15/48)^2(1-2)^3)) / (((1 + 1/4 * 2 + (3/8)^2 * 4 + (15/48)^2 * 8)))))) - 0.9568666373 + (((\sqrt{10 - 2\sqrt{5}} - 2)) / ((\sqrt{5} - 1))))$$

Input interpretation

$$5\pi \exp\left(-\pi \times \frac{1 + \frac{1}{4}(1-2) + \left(\frac{3}{8}\right)^2(1-2)^2 + \left(\frac{15}{48}\right)^2(1-2)^3}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8}\right)^2 \times 4 + \left(\frac{15}{48}\right)^2 \times 8}\right) - 0.9568666373 + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}$$

Result

5.8685833631...

5.8685833631....

Series representations

$$\begin{aligned} & 5 \pi \exp \left(- \frac{ \pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right) }{ 1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8 } \right) - \\ & 0.956867 + \frac{ \sqrt{10 - 2 \sqrt{5}} - 2 }{ \sqrt{5} - 1 } = \\ & \left(5 \left(-0.208627 - \pi \exp \left(- \frac{29 \pi}{104} \right) - 0.191373 \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} + \right. \right. \\ & \quad \left. \pi \exp \left(- \frac{29 \pi}{104} \right) \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} + \right. \\ & \quad \left. \left. 0.2 \sqrt{9 - 2 \sqrt{5}} \sum_{k=0}^{\infty} \binom{\frac{1}{2}}{k} (9 - 2 \sqrt{5})^{-k} \right) \right) / \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right) \end{aligned}$$

$$\begin{aligned} & 5 \pi \exp \left(- \frac{ \pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right) }{ 1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8 } \right) - \\ & 0.956867 + \frac{ \sqrt{10 - 2 \sqrt{5}} - 2 }{ \sqrt{5} - 1 } = \\ & \left(5 \left(-0.208627 - \pi \exp \left(- \frac{29 \pi}{104} \right) - 0.191373 \sqrt{4} \sum_{k=0}^{\infty} \frac{ \left(-\frac{1}{4} \right)^k \left(-\frac{1}{2} \right)_k }{ k! } + \right. \right. \\ & \quad \left. \pi \exp \left(- \frac{29 \pi}{104} \right) \sqrt{4} \sum_{k=0}^{\infty} \frac{ \left(-\frac{1}{4} \right)^k \left(-\frac{1}{2} \right)_k }{ k! } + 0.2 \sqrt{9 - 2 \sqrt{5}} \right. \\ & \quad \left. \left. \sum_{k=0}^{\infty} \frac{ (-1)^k \left(-\frac{1}{2} \right)_k (9 - 2 \sqrt{5})^{-k} }{ k! } \right) \right) / \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} \frac{ \left(-\frac{1}{4} \right)^k \left(-\frac{1}{2} \right)_k }{ k! } \right) \end{aligned}$$

$$\begin{aligned}
& 5 \pi \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \\
& 0.956867 + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} = \\
& \left(5 \left(-0.208627 - \pi \exp \left(- \frac{29\pi}{104} \right) - 0.191373 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k (5 - z_0)^k z_0^{-k}}{k!} + \right. \right. \\
& \quad \pi \exp \left(- \frac{29\pi}{104} \right) \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k (5 - z_0)^k z_0^{-k}}{k!} + \\
& \quad \left. \left. 0.2 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k (10 - 2\sqrt{5} - z_0)^k z_0^{-k}}{k!} \right) \right) / \\
& \left(-1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k (5 - z_0)^k z_0^{-k}}{k!} \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

And:

$$\ln(139503) * \exp(-\pi * (((1 + 1/4(1-2) + (3/8)^2(1-2)^2 + (15/48)^2(1-2)^3)) / (((1 + 1/4 * 2 + (3/8)^2 * 4 + (15/48)^2 * 8)))) - 1/\pi$$

where $\ln(139503) = 11.8458$ is a value of an entropy black hole

Input

$$\log(139503) \exp \left(-\pi \times \frac{1 + \frac{1}{4}(1-2) + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8} \right)^2 \times 4 + \left(\frac{15}{48} \right)^2 \times 8} \right) - \frac{1}{\pi}$$

$\log(x)$ is the natural logarithm

Exact result

$$e^{-(29\pi)/104} \log(139503) - \frac{1}{\pi}$$

Decimal approximation

4.6147321302784246050585399174982902615740429784097015956469484009

...

4.61473213....

Alternate forms

$$e^{-(29\pi)/104} (2 \log(7) + \log(2847)) - \frac{1}{\pi}$$

$$\frac{e^{-(29\pi)/104} (\pi \log(139503) - e^{(29\pi)/104})}{\pi}$$

$$\frac{e^{-(29\pi)/104} \pi (\log(3) + 2 \log(7) + \log(13) + \log(73)) - 1}{\pi}$$

$$e^{-(29\pi)/104} (\log(3) + 2 \log(7) + \log(13) + \log(73)) - \frac{1}{\pi}$$

Alternative representations

$$\log(139503) \exp\left(-\frac{\pi\left(1 + \frac{1-2}{4} + \left(\frac{3}{8}\right)^2 (1-2)^2 + \left(\frac{15}{48}\right)^2 (1-2)^3\right)}{1 + \frac{2}{4} + \left(\frac{3}{8}\right)^2 4 + \left(\frac{15}{48}\right)^2 8}\right) - \frac{1}{\pi} =$$
$$\exp\left(-\frac{\pi\left(1 - \frac{1}{4} + \left(\frac{3}{8}\right)^2 - \left(\frac{15}{48}\right)^2\right)}{1 + \frac{2}{4} + 4\left(\frac{3}{8}\right)^2 + 8\left(\frac{15}{48}\right)^2}\right) \log_e(139503) - \frac{1}{\pi}$$

$$\log(139503) \exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)-\frac{1}{\pi}=$$

$$\exp\left(-\frac{\pi\left(1-\frac{1}{4}+\left(\frac{3}{8}\right)^2-\left(\frac{15}{48}\right)^2\right)}{1+\frac{2}{4}+4\left(\frac{3}{8}\right)^2+8\left(\frac{15}{48}\right)^2}\right)\log(a)\log_a(139503)-\frac{1}{\pi}$$

Series representations

$$\log(139503) \exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)-\frac{1}{\pi}=$$

$$-\frac{1}{\pi}+e^{-(29\pi)/104}\log(139502)-e^{-(29\pi)/104}\sum_{k=1}^{\infty}\frac{\left(-\frac{1}{139502}\right)^k}{k}$$

$$\log(139503) \exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)-\frac{1}{\pi}=$$

$$-\frac{1}{\pi}+e^{-(29\pi)/104}\left(\log(z_0)+\left[\frac{\arg(139503-z_0)}{2\pi}\right]\left(\log\left(\frac{1}{z_0}\right)+\log(z_0)\right)-\sum_{k=1}^{\infty}\frac{(-1)^k(139503-z_0)^k z_0^{-k}}{k}\right)$$

$$\log(139503) \exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)-\frac{1}{\pi}=$$

$$-\frac{1}{\pi}+2ie^{-(29\pi)/104}\pi\left[\frac{\arg(139503-x)}{2\pi}\right]+e^{-(29\pi)/104}\log(x)-$$

$$e^{-(29\pi)/104}\sum_{k=1}^{\infty}\frac{(-1)^k(139503-x)^k x^{-k}}{k} \text{ for } x < 0$$

Integral representations

$$\log(139503) \exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)-\frac{1}{\pi}=-\frac{1}{\pi}+e^{-(29\pi)/104}\int_1^{139503}\frac{1}{t}dt$$

$$\log(139503) \exp\left(-\frac{\pi\left(1+\frac{1-2}{4}+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3\right)}{1+\frac{2}{4}+\left(\frac{3}{8}\right)^2 4+\left(\frac{15}{48}\right)^2 8}\right)-\frac{1}{\pi}=-\frac{1}{\pi}-\frac{ie^{-(29\pi)/104}}{2\pi}\int_{-i\infty+\gamma}^{i\infty+\gamma}\frac{139502^{-s}\Gamma(-s)^2\Gamma(1+s)}{\Gamma(1-s)}ds \text{ for } -1<\gamma<0$$

We have also:

$$((\ln(139503)*\text{Exp}(-\text{Pi}*(((1+1/4(1-2)+(3/8)^2(1-2)^2+(15/48)^2(1-2)^3)))/(((1+1/4*2+(3/8)^2*4+(15/48)^2*8)))-1/\text{Pi}))^{\text{Pi}+47+4}$$

Input

$$\left(\log(139503) \exp\left(-\pi\times\frac{1+\frac{1}{4}(1-2)+\left(\frac{3}{8}\right)^2(1-2)^2+\left(\frac{15}{48}\right)^2(1-2)^3}{1+\frac{1}{4}\times 2+\left(\frac{3}{8}\right)^2\times 4+\left(\frac{15}{48}\right)^2\times 8}\right)-\frac{1}{\pi}\right)^{\pi}+47+4$$

$\log(x)$ is the natural logarithm

Exact result

$$51+\left(e^{-(29\pi)/104}\log(139503)-\frac{1}{\pi}\right)^{\pi}$$

Decimal approximation

173.03313128691295455076392834722767247115187177998789369283412973
...

173.03313128...

Alternate forms

$$51 + \left(\frac{e^{-(29\pi)/104} \pi (2 \log(7) + \log(2847)) - 1}{\pi} \right)^\pi$$

$$51 + \left(\frac{e^{-(29\pi)/104} \pi (\log(3) + 2 \log(7) + \log(13) + \log(73)) - 1}{\pi} \right)^\pi$$

$$51 + \left(e^{-(29\pi)/104} (\log(3) + 2 \log(7) + \log(13) + \log(73)) - \frac{1}{\pi} \right)^\pi$$

Alternative representations

$$\left(\log(139503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + 47 + 4 =$$

$$51 + \left(\exp \left(- \frac{\pi \left(1 - \frac{1}{4} + \left(\frac{3}{8} \right)^2 - \left(\frac{15}{48} \right)^2 \right)}{1 + \frac{2}{4} + 4 \left(\frac{3}{8} \right)^2 + 8 \left(\frac{15}{48} \right)^2} \right) \log_e(139503) - \frac{1}{\pi} \right)^\pi$$

$$\left(\log(139503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + 47 + 4 =$$

$$51 + \left(\exp \left(- \frac{\pi \left(1 - \frac{1}{4} + \left(\frac{3}{8} \right)^2 - \left(\frac{15}{48} \right)^2 \right)}{1 + \frac{2}{4} + 4 \left(\frac{3}{8} \right)^2 + 8 \left(\frac{15}{48} \right)^2} \right) \log(a) \log_a(139503) - \frac{1}{\pi} \right)^\pi$$

Series representations

$$\left(\log(139\,503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + 47 + 4 =$$

$$51 + \left(-\frac{1}{\pi} + e^{-(29\pi)/104} \left(\log(139\,502) - \sum_{k=1}^{\infty} \frac{\left(-\frac{1}{139\,502} \right)^k}{k} \right) \right)^\pi$$

$$\left(\log(139\,503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + 47 + 4 =$$

$$51 + \left(-\frac{1}{\pi} + e^{-(29\pi)/104} \left(2i\pi \left[\frac{\arg(139\,503 - x)}{2\pi} \right] + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (139\,503 - x)^k x^{-k}}{k} \right) \right)^\pi \text{ for } x < 0$$

$$\left(\log(139\,503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + 47 + 4 =$$

$$51 + \left(-\frac{1}{\pi} + e^{-(29\pi)/104} \left(\log(z_0) + \left[\frac{\arg(139\,503 - z_0)}{2\pi} \right] \left(\log\left(\frac{1}{z_0}\right) + \log(z_0) \right) - \sum_{k=1}^{\infty} \frac{(-1)^k (139\,503 - z_0)^k z_0^{-k}}{k} \right) \right)^\pi$$

Integral representations

$$\left(\log(139\,503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + 47 + 4 =$$

$$51 + \left(-\frac{1}{\pi} + e^{-(29\pi)/104} \int_1^{139\,503} \frac{1}{t} dt \right)^\pi$$

$$\left(\log(139503) \exp \left(-\frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + 47 + 4 =$$

$$51 + \left(-\frac{1}{\pi} - \frac{i e^{-(29\pi)/104}}{2\pi} \int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{139502^{-s} \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds \right)^\pi \text{ for } -1 < \gamma < 0$$

From which:

$$((\ln(139503)*\text{Exp}(-\text{Pi}* (((1+1/4(1-2)+(3/8)^2(1-2)^2+(15/48)^2(1-2)^3))))/(((1+1/4*2+(3/8)^2*4+(15/48)^2*8))))-1/\text{Pi}))^\text{Pi}+\phi^2$$

Input

$$\left(\log(139503) \exp \left(-\pi \times \frac{1 + \frac{1}{4} (1-2) + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8} \right)^2 \times 4 + \left(\frac{15}{48} \right)^2 \times 8} \right) - \frac{1}{\pi} \right)^\pi + \phi^2$$

$\log(x)$ is the natural logarithm

ϕ is the golden ratio

Exact result

$$\phi^2 + \left(e^{-(29\pi)/104} \log(139503) - \frac{1}{\pi} \right)^\pi$$

Decimal approximation

124.65116527566284939896851518159331058887218095979365655496957835

...

124.65116527.... result very near to the value of Planck multipole spectrum data 124.55 and to the value of Higgs boson mass 125.10

Alternate forms

$$\phi^2 + \left(\frac{e^{-(29\pi)/104} \pi (2 \log(7) + \log(2847)) - 1}{\pi} \right)^\pi$$

$$\frac{3}{2} + \frac{\sqrt{5}}{2} + \left(e^{-(29\pi)/104} \log(139503) - \frac{1}{\pi} \right)^\pi$$

$$\frac{1}{2} \left(3 + \sqrt{5} + 2 \left(e^{-(29\pi)/104} \log(139503) - \frac{1}{\pi} \right)^\pi \right)$$

$$\phi^2 + \left(e^{-(29\pi)/104} (\log(3) + 2 \log(7) + \log(13) + \log(73)) - \frac{1}{\pi} \right)^\pi$$

Alternative representations

$$\left(\log(139503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + \phi^2 =$$

$$\phi^2 + \left(\exp \left(- \frac{\pi \left(1 - \frac{1}{4} + \left(\frac{3}{8} \right)^2 - \left(\frac{15}{48} \right)^2 \right)}{1 + \frac{2}{4} + 4 \left(\frac{3}{8} \right)^2 + 8 \left(\frac{15}{48} \right)^2} \right) \log_e(139503) - \frac{1}{\pi} \right)^\pi$$

$$\left(\log(139503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + \phi^2 =$$

$$\phi^2 + \left(\exp \left(- \frac{\pi \left(1 - \frac{1}{4} + \left(\frac{3}{8} \right)^2 - \left(\frac{15}{48} \right)^2 \right)}{1 + \frac{2}{4} + 4 \left(\frac{3}{8} \right)^2 + 8 \left(\frac{15}{48} \right)^2} \right) \log(a) \log_a(139503) - \frac{1}{\pi} \right)^\pi$$

Series representations

$$\left(\log(139503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + \phi^2 =$$

$$\phi^2 + \left(-\frac{1}{\pi} + e^{-(29\pi)/104} \left(\log(139502) - \sum_{k=1}^{\infty} \frac{\left(-\frac{1}{139502} \right)^k}{k} \right) \right)^\pi$$

$$\left(\log(139\,503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + \phi^2 =$$

$$\phi^2 + \left(-\frac{1}{\pi} + e^{-(29\pi)/104} \left(2i\pi \left[\frac{\arg(139\,503 - x)}{2\pi} \right] + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (139\,503 - x)^k x^{-k}}{k} \right) \right)^\pi \text{ for } x < 0$$

$$\left(\log(139\,503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + \phi^2 =$$

$$\phi^2 + \left(-\frac{1}{\pi} + e^{-(29\pi)/104} \left(\log(z_0) + \left[\frac{\arg(139\,503 - z_0)}{2\pi} \right] \left(\log\left(\frac{1}{z_0}\right) + \log(z_0) \right) - \sum_{k=1}^{\infty} \frac{(-1)^k (139\,503 - z_0)^k z_0^{-k}}{k} \right) \right)^\pi$$

Integral representations

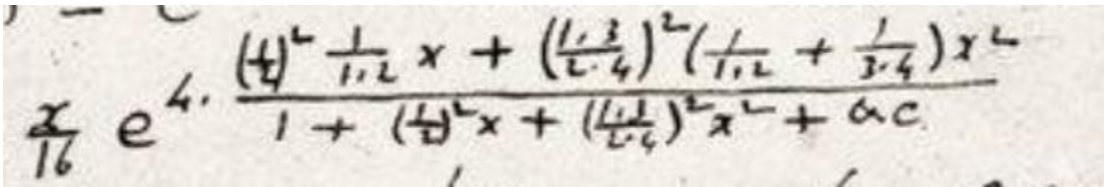
$$\left(\log(139\,503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + \phi^2 =$$

$$\phi^2 + \left(-\frac{1}{\pi} + e^{-(29\pi)/104} \int_1^{139\,503} \frac{1}{t} dt \right)^\pi$$

$$\left(\log(139\,503) \exp \left(- \frac{\pi \left(1 + \frac{1-2}{4} + \left(\frac{3}{8} \right)^2 (1-2)^2 + \left(\frac{15}{48} \right)^2 (1-2)^3 \right)}{1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 + \left(\frac{15}{48} \right)^2 8} \right) - \frac{1}{\pi} \right)^\pi + \phi^2 =$$

$$\phi^2 + \left(-\frac{1}{\pi} - \frac{i e^{-(29\pi)/104}}{2\pi} \int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{139\,502^{-s} \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds \right)^\pi \text{ for } -1 < \gamma < 0$$

From:



For $x = 2$:

$$\frac{2}{16} * e^4 * ((\frac{1}{4} * \frac{1}{2} * 2 + (\frac{3}{8})^2 (\frac{1}{2} + \frac{1}{12}) * 2^2)) / ((1 + \frac{1}{4} * 2 + (\frac{3}{8})^2 * 4))$$

Input

$$\frac{2}{16} e^4 \times \frac{\frac{1}{4} \times \frac{1}{2} \times 2 + \left(\frac{3}{8}\right)^2 \left(\frac{1}{2} + \frac{1}{12}\right) \times 2^2}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8}\right)^2 \times 4}$$

Result

$$\frac{37 e^4}{1056}$$

Decimal approximation

1.9130033629037280737595451368426633531280845363908338474011211898
...

1.9130033629....

Property

$\frac{37 e^4}{1056}$ is a transcendental number

Alternative representation

$$\frac{\left(e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8}\right)^2 \left(\frac{1}{2} + \frac{1}{12}\right) 2^2\right)\right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8}\right)^2 4\right) 16} = \frac{\left(\exp^4(z) \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8}\right)^2 \left(\frac{1}{2} + \frac{1}{12}\right) 2^2\right)\right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8}\right)^2 4\right) 16} \quad \text{for } z = 1$$

Series representations

$$\frac{\left(e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8}\right)^2 \left(\frac{1}{2} + \frac{1}{12}\right) 2^2\right)\right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8}\right)^2 4\right) 16} = \frac{37 \sum_{k=0}^{\infty} \frac{4^k}{k!}}{1056}$$

$$\frac{\left(e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8}\right)^2 \left(\frac{1}{2} + \frac{1}{12}\right) 2^2\right)\right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8}\right)^2 4\right) 16} = \frac{37 \left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^4}{1056}$$

$$\frac{\left(e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8}\right)^2 \left(\frac{1}{2} + \frac{1}{12}\right) 2^2\right)\right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8}\right)^2 4\right) 16} = \frac{37}{1056 \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\right)^4}$$

From which:

$$1 + \frac{1}{3} \left(\frac{2}{16} * e^4 * \left(\left(\frac{1}{4} * \frac{1}{2} * 2 + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) * 2^2 \right) \right) / \left(\left(1 + \frac{1}{4} * 2 + \left(\frac{3}{8} \right)^2 * 4 \right) \right) \right)$$

Input

$$1 + \frac{1}{3} \left(\frac{2}{16} e^4 \times \frac{\frac{1}{4} \times \frac{1}{2} \times 2 + \left(\frac{3}{8}\right)^2 \left(\frac{1}{2} + \frac{1}{12}\right) \times 2^2}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8}\right)^2 \times 4} \right)$$

Result

$$1 + \frac{37 e^4}{3168}$$

Decimal approximation

1.6376677876345760245865150456142211177093615121302779491337070632

...

1.63766778.... result very near to the mean between $\zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$, the value of golden ratio 1.61803398... and the 14th root of the Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578..., i.e. 1.63958266

Property

$1 + \frac{37 e^4}{3168}$ is a transcendental number

Alternate form

$$\frac{3168 + 37 e^4}{3168}$$

Alternative representation

$$1 + \frac{2 e^4 \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 3} = 1 + \frac{2 \exp^4(z) \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 3} \text{ for } z = 1$$

Series representations

$$1 + \frac{2 e^4 \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 3} = 1 + \frac{37 \sum_{k=0}^{\infty} \frac{4^k}{k!}}{3168}$$

$$1 + \frac{2 e^4 \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 3} = 1 + \frac{37 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^4}{3168}$$

$$1 + \frac{2e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 3} = 1 + \frac{37}{3168 \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \right)^4}$$

$$8 \left(\left(\left(\left(\left(\frac{2}{16} * e^4 * \left(\left(\frac{1}{4} * \frac{1}{2} * 2 + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) * 2^2 \right) \right) / \left(\left(1 + \frac{1}{4} * 2 + \left(\frac{3}{8} \right)^2 * 4 \right) \right) \right) \right) \right) \right)^4 - 7$$

Input

$$8 \left(\frac{2}{16} e^4 \times \frac{\frac{1}{4} \times \frac{1}{2} \times 2 + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) \times 2^2}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8} \right)^2 \times 4} \right)^4 - 7$$

Result

$$\frac{1874161 e^{16}}{155441037312} - 7$$

Decimal approximation

100.14031549981088104068316561960118741098726529869093276453648662
...

100.140315499....

Property

$-7 + \frac{1874161 e^{16}}{155441037312}$ is a transcendental number

Alternate form

$$\frac{1874161 e^{16} - 1088087261184}{155441037312}$$

Alternative representation

$$8 \left(\frac{\left(e^4 \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right) \right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) 16} \right)^4 - 7 =$$

$$8 \left(\frac{\left(\exp^4(z) \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right) \right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) 16} \right)^4 - 7 \text{ for } z = 1$$

Series representations

$$8 \left(\frac{\left(e^4 \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right) \right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) 16} \right)^4 - 7 = -7 + \frac{1874161 \sum_{k=0}^{\infty} \frac{16^k}{k!}}{155441037312}$$

$$8 \left(\frac{\left(e^4 \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right) \right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) 16} \right)^4 - 7 = -7 + \frac{1874161 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{16}}{155441037312}$$

$$8 \left(\frac{\left(e^4 \left(\frac{2}{4 \times 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right) \right) 2}{\left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) 16} \right)^4 - 7 = -7 + \frac{1874161}{155441037312 \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \right)^{16}}$$

And also:

$$1/78((2/16 * e^4 * ((1/4*1/2*2 + (3/8)^2 (1/2+1/12)*2^2))/((1+1/4*2+(3/8)^2*4))))$$

Input

$$\frac{1}{78} \left(\frac{2}{16} e^4 \times \frac{\frac{1}{4} \times \frac{1}{2} \times 2 + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) \times 2^2}{1 + \frac{1}{4} \times 2 + \left(\frac{3}{8} \right)^2 \times 4} \right)$$

Result

$$\frac{37 e^4}{82368}$$

Decimal approximation

0.0245256841397913855610198094467008122195908273896260749666810408
...

0.024525684....

Property

$\frac{37 e^4}{82368}$ is a transcendental number

Alternative representation

$$\frac{2 e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 78} = \frac{2 \exp^4(z) \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 78} \text{ for } z = 1$$

Series representations

$$\frac{2 e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 78} = \frac{37 \sum_{k=0}^{\infty} \frac{4^k}{k!}}{82368}$$

$$\frac{2 e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 78} = \frac{37 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^4}{82368}$$

$$\frac{2e^4 \left(\frac{2}{4 \cdot 2} + \left(\frac{3}{8} \right)^2 \left(\frac{1}{2} + \frac{1}{12} \right) 2^2 \right)}{\left(16 \left(1 + \frac{2}{4} + \left(\frac{3}{8} \right)^2 4 \right) \right) 78} = \frac{37}{82368 \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \right)^4}$$

From

(<https://www.buymeacoffee.com/jorgenveisdal/ramanujan-early-work-continued-fractions>)

Theorem (5)

$$\frac{1}{1} + \frac{e^{-2\pi}}{1} + \frac{e^{-4\pi}}{1} + \frac{e^{-6\pi}}{1} + \dots = \left(\sqrt{\frac{5p + \sqrt{5}}{2}} - \frac{\sqrt{5} + 1}{2} \right) \sqrt[5]{e^{\pi}}.$$

Theorem (6)

$$\frac{1}{1} - \frac{e^{-\pi}}{1} + \frac{e^{-2\pi}}{1} - \frac{e^{-3\pi}}{1} + \dots = \left(\sqrt{\frac{5p - \sqrt{5}}{2}} - \frac{\sqrt{5} - 1}{2} \right) \sqrt[5]{e^{\pi}}.$$

Theorem (7)

$$\frac{1}{1} + \frac{e^{-\pi\sqrt{n}}}{1} + \frac{e^{-2\pi\sqrt{n}}}{1} + \frac{e^{-3\pi\sqrt{n}}}{1} + \dots$$

can be exactly found if n be any positive rational quantity.

We consider the first two expressions as follows:

From Theorem 5:

$$1/(e^{(-2\pi)} + 1/(e^{(-4\pi)} + 1/e^{(-6\pi)}))$$

Input

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}}$$

Exact result

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}}$$

Decimal approximation

535.48978808854541798179453368147781365538863590427837683767021937

...

$$535.489788\dots \approx e^{2\pi}$$

Property

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \text{ is a transcendental number}$$

Alternate forms

$$\frac{2e^{2\pi}}{2 + e^\pi \operatorname{sech}(5\pi)}$$

$$\frac{e^{2\pi} (1 + e^{10\pi})}{1 + e^{6\pi} + e^{10\pi}}$$

$$e^{2\pi} - \frac{e^{8\pi}}{1 + e^{6\pi} + e^{10\pi}}$$

$\operatorname{sech}(x)$ is the hyperbolic secant function

Alternative representations

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}}}$$

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \frac{1}{e^{2i \log(-1)} + \frac{1}{e^{4i \log(-1)} + \frac{1}{e^{6i \log(-1)}}}}$$

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \frac{1}{\exp^{-2\pi}(z) + \frac{1}{\exp^{-4\pi}(z) + \frac{1}{\exp^{-6\pi}(z)}}} \quad \text{for } z = 1$$

Series representations

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \frac{\left(e^{8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \left(1 + e^{8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \right) \left(1 - e^{8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{32 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \right) \right)}{\left(1 + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{40 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \right)}$$

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \frac{\left(\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{2\pi} \left(1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{2\pi} \right) \left(1 - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{2\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{4\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{8\pi} \right) \right)}{\left(1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{10\pi} \right)}$$

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \left(\left(1 + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{2\pi} \right) \right. \\ \left. \left(1 - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{2\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{4\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{8\pi} \right) \right. \\ \left. \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{2\pi} \right) / \left(1 + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{10\pi} \right)$$

Integral representations

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \frac{1}{e^{-6 \int_0^{\infty} \sin^4(t)/t^4 dt} + \frac{1}{e^{-12 \int_0^{\infty} \sin^4(t)/t^4 dt} + e^{18 \int_0^{\infty} \sin^4(t)/t^4 dt}}$$

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \frac{1}{e^{-16/3 \int_0^{\infty} \sin^3(t)/t^3 dt} + \frac{1}{e^{-32/3 \int_0^{\infty} \sin^3(t)/t^3 dt} + e^{16 \int_0^{\infty} \sin^3(t)/t^3 dt}}$$

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} = \frac{1}{e^{-2 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{t-t^2} dt \right)} + \frac{1}{e^{-4 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{t-t^2} dt \right)} + e^{6 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{t-t^2} dt \right)}}$$

$$(((\sqrt{((5p+\sqrt{5})/2)-((\sqrt{5}+1)/2)})) (e^{\pi})^{1/5} = ((1/(e^{(-2\pi)}+1/(e^{(-4\pi)}+1/e^{(-6\pi)}))))$$

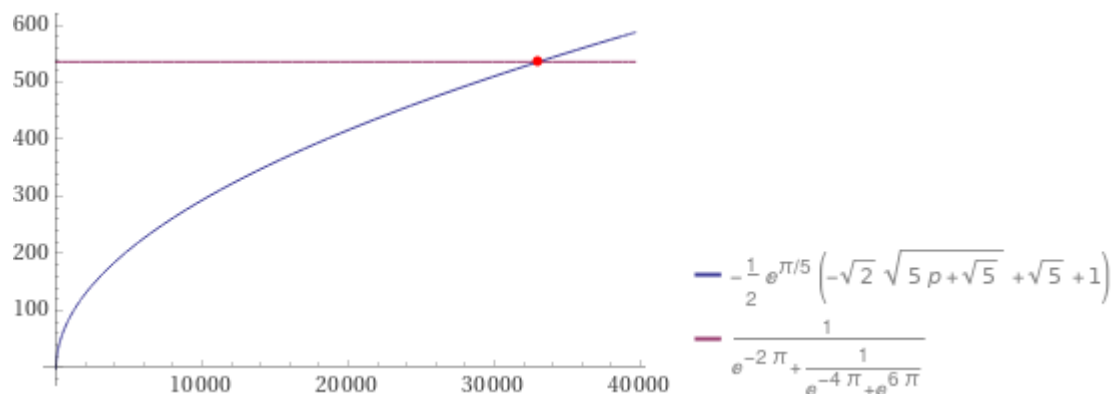
Input

$$\left(\sqrt{\frac{1}{2}(5p+\sqrt{5})}-\frac{1}{2}(\sqrt{5}+1)\right)^5\sqrt[5]{e^{\pi}}=\frac{1}{e^{-2\pi}+\frac{1}{e^{-4\pi}+\frac{1}{e^{-6\pi}}}}$$

Exact result

$$e^{\pi/5}\left(\frac{\sqrt{5p+\sqrt{5}}}{\sqrt{2}}+\frac{1}{2}(-1-\sqrt{5})\right)=\frac{1}{e^{-2\pi}+\frac{1}{e^{-4\pi}+e^{6\pi}}}$$

Plot



$$((\sqrt{((5(33015.025)+\sqrt{5})/2)-((\sqrt{5}+1)/2)})) (e^{\pi})^{1/5}$$

Input interpretation

$$\left(\sqrt{\frac{1}{2}(5\times 33\,015.025+\sqrt{5})}-\frac{1}{2}(\sqrt{5}+1)\right)^5\sqrt[5]{e^{\pi}}$$

Result

535.48979...

535.48979.... as above

Series representations

$$\left(\sqrt{\frac{1}{2} (5 \times 33015. + \sqrt{5})} - \frac{1}{2} (\sqrt{5} + 1) \right) \sqrt[5]{e^\pi} = -\frac{\sqrt[5]{e^\pi}}{2} + \sum_{k=0}^{\infty} \frac{1}{2} \sqrt[5]{e^\pi} \left(\frac{1}{2} \right)_k \left(-4^{-k} \sqrt{4} + 2 (82536.6 + 0.5 \sqrt{5})^{-k} \sqrt{82536.6 + 0.5 \sqrt{5}} \right)$$

$$\left(\sqrt{\frac{1}{2} (5 \times 33015. + \sqrt{5})} - \frac{1}{2} (\sqrt{5} + 1) \right) \sqrt[5]{e^\pi} = -\frac{\sqrt[5]{e^\pi}}{2} + \sum_{k=0}^{\infty} \frac{1}{k!} 2^{-1-2k} \sqrt[5]{e^\pi} \left(-\frac{1}{2} \right)_k (165073. + \sqrt{5})^{-k} \left((-1)^{1+k} \sqrt{4} (165073. + \sqrt{5})^k + (-2)^k 2^{1+2k} \sqrt{82536.6 + 0.5 \sqrt{5}} \right)$$

$$\left(\sqrt{\frac{1}{2} (5 \times 33015. + \sqrt{5})} - \frac{1}{2} (\sqrt{5} + 1) \right) \sqrt[5]{e^\pi} = -\frac{\sqrt[5]{e^\pi}}{2} + \sum_{k=0}^{\infty} -\frac{(-1)^k \sqrt[5]{e^\pi} \left(-\frac{1}{2} \right)_k \sqrt{z_0} \left((5 - z_0)^k - 2 (82537.6 + 0.5 \sqrt{5} - z_0)^k \right) z_0^{-k}}{2 k!}$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

From Theorem 6:

$$1/(e^{(-\pi)} + 1/(e^{(-2\pi)} - 1/e^{(-3\pi)}))$$

Input

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}}$$

Exact result

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}}$$

Decimal approximation

23.183987408081097016167838156507419785948064807099113251388567983
...

23.1839874....

Property

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} \text{ is a transcendental number}$$

Alternate forms

$$\frac{e^{\pi} \left(e^{5\pi} - 1 \right)}{2 e^{4\pi} \sinh(\pi) - 1}$$

$$e^{\pi} + \frac{e^{4\pi}}{-1 - e^{3\pi} + e^{5\pi}}$$

$$\frac{e^{\pi} \left(e^{5\pi} - 1 \right)}{-1 - e^{3\pi} + e^{5\pi}}$$

$\sinh(x)$ is the hyperbolic sine function

Alternative representations

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} = \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}}}$$

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} = \frac{1}{e^{i \log(-1)} + \frac{1}{e^{2i \log(-1)} - \frac{1}{e^{3i \log(-1)}}}}$$

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} = \frac{1}{\exp^{-\pi}(z) + \frac{1}{\exp^{-2\pi}(z) - \frac{1}{\exp^{-3\pi}(z)}}} \quad \text{for } z = 1$$

Series representations

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} = \frac{1}{e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{-12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}}$$

$$\begin{aligned} \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} &= \left(\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{\pi} \left(-1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{\pi} \right) \right. \\ &\quad \left(1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{2\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{3\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{4\pi} \right) \Big/ \\ &\quad \left(-1 - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{3\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{5\pi} \right) \end{aligned}$$

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} = \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-4} \sum_{k=0}^{\infty} \frac{(-1)^k}{(1+2k)} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-8} \sum_{k=0}^{\infty} \frac{(-1)^k}{(1+2k)} - \left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{12} \sum_{k=0}^{\infty} \frac{(-1)^k}{(1+2k)}}$$

Integral representations

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} = \frac{1}{e^{-2 \times \int_0^{\infty} \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-4 \times \int_0^{\infty} \frac{1}{(1+t^2)} dt} - e^{-6 \times \int_0^{\infty} \frac{1}{(1+t^2)} dt}}}$$

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} = \frac{1}{e^{-2 \int_0^{\infty} \frac{\sin(t)}{t} dt} + \frac{1}{e^{-4 \int_0^{\infty} \frac{\sin(t)}{t} dt} - e^{-6 \int_0^{\infty} \frac{\sin(t)}{t} dt}}}$$

$$\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} = \frac{1}{e^{-4 \int_0^1 \sqrt{1-t^2} dt} + \frac{1}{e^{-8 \int_0^1 \sqrt{1-t^2} dt} - e^{-12 \int_0^1 \sqrt{1-t^2} dt}}}$$

$$(((\sqrt{(5p-\sqrt{5})/2}) - ((\sqrt{5}-1)/2)))) (e^{\pi})^{1/5} = 1/(e^{(-\pi)} + 1/(e^{(-2\pi)} - 1/e^{(-3\pi)}))$$

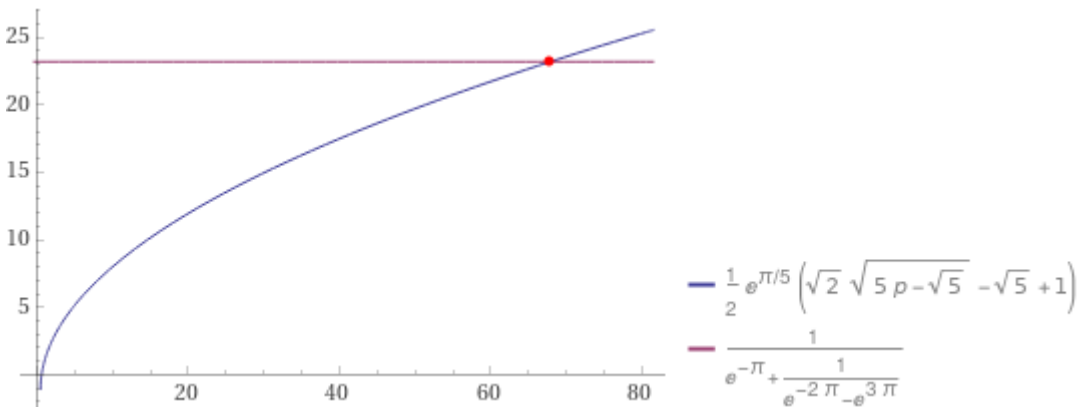
Input

$$\left(\sqrt{\frac{1}{2}(5p-\sqrt{5})} - \frac{1}{2}(\sqrt{5}-1)\right)^5 \sqrt[5]{e^{\pi}} = \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}}$$

Exact result

$$e^{\pi/5} \left(\frac{\sqrt{5p - \sqrt{5}}}{\sqrt{2}} + \frac{1}{2} (1 - \sqrt{5}) \right) = \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}}$$

Plot



Solution

$p \approx 67.9060052341479$

67.9060052341479

$((\sqrt{5(67.906005234) - \sqrt{5}}) - ((\sqrt{5} - 1)/2)) (e^{\pi})^{1/5}$

Input interpretation

$$\left(\sqrt{\frac{1}{2} (5 \times 67.906005234 - \sqrt{5})} - \frac{1}{2} (\sqrt{5} - 1) \right) \sqrt[5]{e^{\pi}}$$

Result

23.183987408...

23.183987408....

Series representations

$$\left(\sqrt{\frac{1}{2} (5 \times 67.9060052340000 - \sqrt{5})} - \frac{1}{2} (\sqrt{5} - 1) \right) \sqrt[5]{e^\pi} =$$

$$\frac{\sqrt[5]{e^\pi}}{2} + \sum_{k=0}^{\infty} \frac{1}{2} \sqrt[5]{e^\pi} \left(\frac{1}{2} \right) \left(-4^{-k} \sqrt{4} + 2 \left(168.765013085000 - \frac{\sqrt{5}}{2} \right)^{-k} \right.$$

$$\left. \sqrt{168.765013085000 - 0.5000000000000000 \sqrt{5}} \right)$$

$$\left(\sqrt{\frac{1}{2} (5 \times 67.9060052340000 - \sqrt{5})} - \frac{1}{2} (\sqrt{5} - 1) \right) \sqrt[5]{e^\pi} =$$

$$\frac{\sqrt[5]{e^\pi}}{2} + \sum_{k=0}^{\infty} \frac{1}{k!} (-1)^k 2^{-1-2k} \sqrt[5]{e^\pi} \left(-\frac{1}{2} \right)_k$$

$$\left(168.765013085000 - \frac{\sqrt{5}}{2} \right)^{-k} \left(-\sqrt{4} \left(168.765013085000 - \frac{\sqrt{5}}{2} \right)^k + \right.$$

$$\left. 2^{1+2k} \sqrt{168.765013085000 - 0.5000000000000000 \sqrt{5}} \right)$$

$$\left(\sqrt{\frac{1}{2} (5 \times 67.9060052340000 - \sqrt{5})} - \frac{1}{2} (\sqrt{5} - 1) \right) \sqrt[5]{e^\pi} =$$

$$\frac{\sqrt[5]{e^\pi}}{2} + \sum_{k=0}^{\infty} -\frac{1}{2 k!} (-1)^k \sqrt[5]{e^\pi} \left(-\frac{1}{2} \right)_k \sqrt{z_0} \left((5 - z_0)^k - 2 \left(169.765013085000 - \right. \right.$$

$$\left. \left. 0.5000000000000000 \sqrt{5} - 1.0000000000000000 z_0 \right)^k \right)$$

$$z_0^{-k} \text{ for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))$$

From which:

$$\left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}}\right) - \left[\left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}}\right) * \frac{1}{\left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}}\right)}\right]^2$$

Input

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} - \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} \times \frac{1}{\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}}} \right)^2$$

Exact result

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} - \frac{\left(e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}\right)^2}{\left(e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}\right)^2}$$

Decimal approximation

1.9999863515420925465156589797374673589825737937379181716115588014
...

$$1.99998635154..... \approx 2$$

Property

$$-\frac{\left(e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}\right)^2}{\left(e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}\right)^2} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \text{ is a transcendental number}$$

Alternate forms

$$\frac{\cosh(5\pi) \left(1 + \coth\left(\frac{5\pi}{2}\right)\right) \left(4 \cosh(5\pi) - e^{\pi/2} \operatorname{csch}\left(\frac{5\pi}{2}\right)\right)}{(\sinh(\pi) + \cosh(\pi) + 2 \cosh(5\pi))^2}$$

$$\frac{2 e^{5 \pi} \left(1+e^{10 \pi}\right) \left(-1+e^{5 \pi}-e^{8 \pi}-e^{10 \pi}+e^{15 \pi}\right)}{\left(e^{5 \pi}-1\right)^2 \left(1+e^{6 \pi}+e^{10 \pi}\right)^2}$$

$$\frac{2 e^{5 \pi} \left(1+e^{2 \pi}\right) \left(1-e^{2 \pi}+e^{4 \pi}-e^{6 \pi}+e^{8 \pi}\right) \left(-1+e^{5 \pi}-e^{8 \pi}-e^{10 \pi}+e^{15 \pi}\right)}{\left(e^{\pi}-1\right)^2 \left(1+e^{\pi}+e^{2 \pi}+e^{3 \pi}+e^{4 \pi}\right)^2 \left(1+e^{6 \pi}+e^{10 \pi}\right)^2}$$

Expanded form

$$-\frac{e^{-2 \pi}}{\left(e^{-2 \pi}+\frac{1}{e^{-4 \pi}+e^{6 \pi}}\right)^2}-\frac{1}{\left(e^{-2 \pi}-e^{3 \pi}\right)^2\left(e^{-2 \pi}+\frac{1}{e^{-4 \pi}+e^{6 \pi}}\right)^2}-\frac{2 e^{-\pi}}{\left(e^{-2 \pi}-e^{3 \pi}\right)\left(e^{-2 \pi}+\frac{1}{e^{-4 \pi}+e^{6 \pi}}\right)^2}+\frac{1}{e^{-2 \pi}+\frac{1}{e^{-4 \pi}+e^{6 \pi}}}$$

Alternative representations

$$\frac{1}{e^{-2 \pi}+\frac{1}{e^{-4 \pi}+\frac{1}{e^{-6 \pi}}}}-\left(\frac{\frac{1}{e^{-2 \pi}+\frac{1}{e^{-4 \pi}+\frac{1}{e^{-6 \pi}}}}}{\frac{1}{e^{-\pi}+\frac{1}{e^{-2 \pi}-\frac{1}{e^{-3 \pi}}}}}\right)^2=\frac{1}{e^{-360^{\circ}}+\frac{1}{e^{-720^{\circ}}+\frac{1}{e^{-1080^{\circ}}}}}-\left(\frac{\frac{1}{e^{-360^{\circ}}+\frac{1}{e^{-720^{\circ}}+\frac{1}{e^{-1080^{\circ}}}}}}{\frac{1}{e^{-180^{\circ}}+\frac{1}{e^{-360^{\circ}}-\frac{1}{e^{-540^{\circ}}}}}}}\right)^2$$

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} - \left(\frac{\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}}}{\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}}} \right)^2 =$$

$$\frac{1}{e^{2i\log(-1)} + \frac{1}{e^{4i\log(-1)} + \frac{1}{e^{6i\log(-1)}}}} - \left(\frac{\frac{1}{e^{2i\log(-1)} + \frac{1}{e^{4i\log(-1)} + \frac{1}{e^{6i\log(-1)}}}}}{\frac{1}{e^{i\log(-1)} + \frac{1}{e^{2i\log(-1)} - \frac{1}{e^{3i\log(-1)}}}}} \right)^2$$

$$\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} - \left(\frac{\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}}}{\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}}} \right)^2 =$$

$$\frac{1}{\exp^{-2\pi}(z) + \frac{1}{\exp^{-4\pi}(z) + \frac{1}{\exp^{-6\pi}(z)}}}} - \left(\frac{\frac{1}{\exp^{-2\pi}(z) + \frac{1}{\exp^{-4\pi}(z) + \frac{1}{\exp^{-6\pi}(z)}}}}}{\frac{1}{\exp^{-\pi}(z) + \frac{1}{\exp^{-2\pi}(z) - \frac{1}{\exp^{-3\pi}(z)}}}}} \right)^2 \quad \text{for } z = 1$$

Series representations

$$\begin{aligned}
 & \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} - \left(\frac{1}{\frac{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}}}{\frac{e^{-\pi} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-3\pi}}}}}} \right)^2 = \left(2 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{5\pi} \left(1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{2\pi} \right) \right. \\
 & \left(1 - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{2\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{4\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{8\pi} \right) \\
 & \left. \left(-1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{5\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{8\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{10\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{15\pi} \right) \right) / \\
 & \left(\left(-1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{\pi} \right)^2 \left(1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{2\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{3\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{4\pi} \right)^2 \right. \\
 & \left. \left(1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{10\pi} \right)^2 \right)
 \end{aligned}$$

$$\begin{aligned}
& \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} - \left(\frac{1}{\frac{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}}}{\frac{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}}}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}} \right)^2 = \\
& - \frac{\left(e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}} \right)^2}{1 / \left(\left(e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \right)^2 \right.} \\
& \quad \left. \left(e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}} \right)^2 \right) - \\
& \quad \left(2 e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \right) / \left(\left(e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \right) \right. \\
& \quad \left. \left(e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}} \right)^2 \right) + \\
& \quad \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}}
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} - \left(\frac{1}{\frac{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}}} \right)^2 = \\
& \left(2 \left(1 + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{2\pi} \right) \left(1 - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{2\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{4\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi} + \right. \right. \\
& \quad \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{8\pi} \left(-1 + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{5\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{8\pi} - \right. \\
& \quad \left. \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{10\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{15\pi} \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{5\pi} \right) / \\
& \quad \left(\left(-1 + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{\pi} \right)^2 \left(1 + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{2\pi} + \right. \right. \\
& \quad \left. \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{3\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{4\pi} \right)^2 \\
& \quad \left. \left. \left(1 + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{10\pi} \right)^2 \right) \right)
\end{aligned}$$

$$2[(((1/(e^(-2Pi))+1/(e^(-4Pi))+1/e^(-6Pi)))))+((1/(e^(-Pi))+1/(e^(-2Pi))-1/e^(-3Pi))))]$$

Input

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)$$

Exact result

$$2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)$$

Decimal approximation

1117.3475509932530299959247436759704668826734014227549801781175747

...

1117.3475509.... result very near to the Lambda baryon mass 1115.683

Property

$$2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right) \text{ is a transcendental number}$$

Alternate forms

$$2e^{\pi} + 2e^{2\pi} + \frac{2e^{4\pi}}{-1 - e^{3\pi} + e^{5\pi}} - \frac{2e^{8\pi}}{1 + e^{6\pi} + e^{10\pi}}$$

$$\frac{2e^{\pi}(-1 - e^{\pi} - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi})}{(-1 - e^{3\pi} + e^{5\pi})(1 + e^{6\pi} + e^{10\pi})}$$

Expanded form

$$\frac{2}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{2}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}}$$

Alternative representations

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) =$$

$$2 \left(\frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}}} + \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}}} \right)$$

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) =$$

$$2 \left(\frac{1}{e^{i \log(-1)} + \frac{1}{e^{2i \log(-1)} - \frac{1}{e^{3i \log(-1)}}}} + \frac{1}{e^{2i \log(-1)} + \frac{1}{e^{4i \log(-1)} + \frac{1}{e^{6i \log(-1)}}}} \right)$$

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) =$$

$$2 \left(\frac{1}{\exp^{-2\pi}(z) + \frac{1}{\exp^{-4\pi}(z) + \frac{1}{\exp^{-6\pi}(z)}}}} + \frac{1}{\exp^{-\pi}(z) + \frac{1}{\exp^{-2\pi}(z) - \frac{1}{\exp^{-3\pi}(z)}}}} \right) \text{ for } z = 1$$

Series representations

$$\begin{aligned}
& 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) = \\
& \frac{e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}}{2} + \\
& \frac{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}}{2}
\end{aligned}$$

$$\begin{aligned}
& 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) = \\
& \left(2 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{\pi} \left(-1 - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{4\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{5\pi} - \right. \right. \\
& \quad \left. \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{10\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{14\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{15\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{16\pi} \right) / \\
& \left(\left(-1 - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{3\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{5\pi} \right) \left(1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{10\pi} \right) \right)
\end{aligned}$$

$$\begin{aligned}
& 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) = \\
& \left(2 \left(-1 - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{4\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{5\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{10\pi} - \right. \right. \\
& \quad \left. \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{14\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{15\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{16\pi} \right) \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{\pi} \right) / \\
& \left(\left(-1 - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{3\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{5\pi} \right) \right. \\
& \quad \left. \left(1 + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{10\pi} \right) \right)
\end{aligned}$$

Integral representations

$$\begin{aligned}
& 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) = \\
& \frac{2}{e^{-4} \int_0^1 \sqrt{1-t^2} dt + \frac{1}{e^{-8} \int_0^1 \sqrt{1-t^2} dt - e^{-12} \int_0^1 \sqrt{1-t^2} dt}} + \\
& \frac{2}{e^{-8} \int_0^1 \sqrt{1-t^2} dt + \frac{1}{e^{-16} \int_0^1 \sqrt{1-t^2} dt + e^{-24} \int_0^1 \sqrt{1-t^2} dt}}
\end{aligned}$$

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) =$$

$$\frac{2}{e^{-8/3 \int_0^\infty \sin^3(t)/t^3 dt} + \frac{1}{e^{-16/3 \int_0^\infty \sin^3(t)/t^3 dt} - e^{8 \int_0^\infty \sin^3(t)/t^3 dt}}} +$$

$$\frac{2}{e^{-16/3 \int_0^\infty \sin^3(t)/t^3 dt} + \frac{1}{e^{-32/3 \int_0^\infty \sin^3(t)/t^3 dt} + e^{16 \int_0^\infty \sin^3(t)/t^3 dt}}}$$

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) =$$

$$\left(2 e^{2 \int_0^\infty \sin(t)/t dt} \left(-1 - e^{2 \int_0^\infty \sin(t)/t dt} - e^{8 \int_0^\infty \sin(t)/t dt} + e^{10 \int_0^\infty \sin(t)/t dt} - \right. \right.$$

$$\left. e^{20 \int_0^\infty \sin(t)/t dt} - e^{28 \int_0^\infty \sin(t)/t dt} + e^{30 \int_0^\infty \sin(t)/t dt} + e^{32 \int_0^\infty \sin(t)/t dt} \right) \Bigg/$$

$$\left(\left(-1 - e^{6 \int_0^\infty \sin(t)/t dt} + e^{10 \int_0^\infty \sin(t)/t dt} \right) \left(1 + e^{12 \int_0^\infty \sin(t)/t dt} + e^{20 \int_0^\infty \sin(t)/t dt} \right) \right)$$

From which:

$$(2[(((1/(e^{(-2\pi)})+1/(e^{(-4\pi)})+1/e^{(-6\pi)}))))) + ((1/(e^{(-\pi)})+1/(e^{(-2\pi)})-1/e^{(-3\pi)})))]^{1/14}$$

Input

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)}$$

Exact result

$$\sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}$$

Decimal approximation

1.6509264835923620214714995019084454593438192078816210027075340368

...

1.6509264835..... result very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Property

$$\sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}$$
 is a transcendental number

Alternate form

$$e^{\pi/14} \sqrt[14]{\frac{2(-1 - e^{\pi} - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi})}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}}}$$

All 14th roots of $2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right) + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} + e^{6\pi}$

$$\sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)} e^0 \approx 1.6509 \text{ (real, principal root)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)} e^{(i\pi)/7} \approx 1.4874 + 0.7163 i$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)} e^{(2i\pi)/7} \approx 1.0293 + 1.2907 i$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)} e^{(3i\pi)/7} \approx 0.3674 + 1.6095 i$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)} e^{(4i\pi)/7} \approx -0.3674 + 1.6095 i$$

Alternative representations

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} =$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}} + \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}} \right)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} =$$

$$\sqrt[14]{2 \left(\frac{1}{e^{i \log(-1)} + \frac{1}{e^{2i \log(-1)} - \frac{1}{e^{3i \log(-1)}}}} + \frac{1}{e^{2i \log(-1)} + \frac{1}{e^{4i \log(-1)} + \frac{1}{e^{6i \log(-1)}}}} \right)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} =$$

$$\sqrt[14]{2 \left(\frac{1}{\exp^{-2\pi}(z) + \frac{1}{\exp^{-4\pi}(z) + \frac{1}{\exp^{-6\pi}(z)}}}} + \frac{1}{\exp^{-\pi}(z) + \frac{1}{\exp^{-2\pi}(z) - \frac{1}{\exp^{-3\pi}(z)}}}} \right)} \text{ for } z = 1$$

Series representations

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} =$$

$$\sqrt[14]{2 \left(\frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-2\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{3\pi}}} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-2\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-4\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi}}} \right)}^{\wedge (1/14)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} =$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}} + \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}} \right)}^{\wedge (1/14)}$$

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} = \\
& \sqrt[14]{2 \left(\frac{1}{\frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-4\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi}} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi} + \right.} \\
& \quad \left. \frac{1}{\frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{3\pi}} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-\pi}} \right)^{\wedge (1/14)}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} = \\
& \sqrt[14]{2 \left(\frac{1}{e^{-2 \int_0^{\infty} \sin(t)/t \, dt} + \frac{1}{e^{-4 \int_0^{\infty} \sin(t)/t \, dt} - e^{-6 \int_0^{\infty} \sin(t)/t \, dt}} + \right.} \\
& \quad \left. \frac{1}{e^{-4 \int_0^{\infty} \sin(t)/t \, dt} + \frac{1}{e^{-8 \int_0^{\infty} \sin(t)/t \, dt} + e^{-12 \int_0^{\infty} \sin(t)/t \, dt}} \right)^{\wedge (1/14)}
\end{aligned}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} =$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-4 \times \int_0^\infty \frac{1}{(1+t^2)} dt} - e^{-6 \times \int_0^\infty \frac{1}{(1+t^2)} dt}} + \right.}$$

$$\left. \frac{1}{e^{-4 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-8 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + e^{12 \times \int_0^\infty \frac{1}{(1+t^2)} dt}} \right) \wedge (1/14)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} =$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2 \int_0^\infty \frac{\sin^2(t)}{t^2} dt} + \frac{1}{e^{-4 \int_0^\infty \frac{\sin^2(t)}{t^2} dt} - e^{-6 \int_0^\infty \frac{\sin^2(t)}{t^2} dt}} + \right.}$$

$$\left. \frac{1}{e^{-4 \int_0^\infty \frac{\sin^2(t)}{t^2} dt} + \frac{1}{e^{-8 \int_0^\infty \frac{\sin^2(t)}{t^2} dt} + e^{12 \int_0^\infty \frac{\sin^2(t)}{t^2} dt}} \right) \wedge (1/14)}$$

$$(((2[(((1/(e^{(-2\text{Pi})}+1/(e^{(-4\text{Pi})}+1/e^{(-6\text{Pi}})))))))+(1/(e^{(-\text{Pi})}+1/(e^{(-2\text{Pi})}-1/e^{(-3\text{Pi}}))))))\wedge(1/14))\wedge(15-89-21-3-\phi^2)$$

Input

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)}^{15} - 89 - 21 - 3 - \phi^2$$

ϕ is the golden ratio

Exact result

$$-\phi^2 - 113 + 2^{14}\sqrt[14]{2} \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)^{15/14}$$

Decimal approximation

1729.0406293230787406846536905399805465311382512931752888109035138

...

1729.040629....

This result is very near to the mass of candidate glueball **f₀(1710) scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. (1728 = 8² * 3³) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

Property

$$-113 + 2^{14}\sqrt[14]{2} \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)^{15/14} - \phi^2$$

is a transcendental number

Alternate forms

$$-\phi^2 - 113 + \frac{2^{14}\sqrt[14]{2} e^{(15\pi)/14}}{\left(\frac{(-1 - e^{3\pi} + e^{5\pi})(1 + e^{6\pi} + e^{10\pi})}{-1 + e^{5\pi} - e^{6\pi} - e^{10\pi} + e^{11\pi} + e^{15\pi} + e^{\pi}(-1 - e^{3\pi} + e^{5\pi})(1 + e^{10\pi})} \right)^{15/14}}$$

$$\frac{1}{2} \left(-229 - \sqrt{5} + 4^{14}\sqrt[14]{2} e^{(15\pi)/14} \left(\frac{-1 - e^{\pi} - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi}}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}} \right)^{15/14} \right)$$

$$\begin{aligned}
& \frac{1}{2}(-229 - \sqrt{5}) + \\
& \frac{2e^{(15\pi)/14} \sqrt[14]{\frac{2(-1 - e^\pi - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi})}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}}}}{2e^{(29\pi)/14} \sqrt[14]{\frac{2(-1 - e^\pi - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi})}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}}}} + \\
& \frac{2e^{(57\pi)/14} \sqrt[14]{\frac{2(-1 - e^\pi - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi})}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}}}}{-1 - e^{3\pi} + e^{5\pi}} - \\
& \frac{2e^{(113\pi)/14} \sqrt[14]{\frac{2(-1 - e^\pi - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi})}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}}}}{1 + e^{6\pi} + e^{10\pi}}
\end{aligned}$$

Expanded forms

$$\begin{aligned}
& -\frac{229}{2} - \frac{\sqrt{5}}{2} + \frac{2^{14} \sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \\
& \frac{2^{14} \sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}}
\end{aligned}$$

$$-\frac{229}{2} - \frac{\sqrt{5}}{2} + 2^{14} \sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}^{15/14}$$

Alternative representations

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 =}$$

$$-113 - \left(2 \cos\left(\frac{\pi}{5}\right) \right)^2 + \sqrt[14]{2 \left(\frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}}} + \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}}} \right)^{15}}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 =}$$

$$-113 + \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} -}$$

root of $-1 - x + x^2$ near $x = 1.61803$

²

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 =}$$

$$-113 + \sqrt[14]{2 \left(\frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}}} + \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}}} \right)^{15} -}$$

root of $-1 - x + x^2$ near $x = 1.61803$

²

Series representations

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)}^{15} - 89 - 21 - 3 - \phi^2 = -\frac{229}{2} - \frac{\sqrt{5}}{2} + \\
& \frac{2^{\frac{14}{\sqrt{2}}} \sqrt[14]{\frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-2\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{3\pi}}} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-2\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-4\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{6\pi}}}}}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-2\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{3\pi}}} + \\
& \frac{2^{\frac{14}{\sqrt{2}}} \sqrt[14]{\frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-2\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{3\pi}}} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-2\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-4\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{6\pi}}}}}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-2\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{-4\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!}\right)^{6\pi}}}
\end{aligned}$$

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)}^{15} - 89 - 21 - 3 - \phi^2 = -\frac{229}{2} - \frac{\sqrt{5}}{2} + \\
& \left(2 \sqrt[14]{2} \left(\frac{1}{e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{-12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}} + \right. \right. \\
& \quad \left. \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}} \right)^{\wedge} \\
& \quad (1/14) \left/ \left(e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \right. \right. \\
& \quad \left. \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{-12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}} \right)^+ + \\
& \left(2 \sqrt[14]{2} \left(\frac{1}{e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{-12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}} + \right. \right. \\
& \quad \left. \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}} \right)^{\wedge} \\
& \quad (1/14) \left/ \left(e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \right. \right. \\
& \quad \left. \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}} \right)
\end{aligned}$$

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 =} \\
& -\frac{229}{2} - \frac{\sqrt{5}}{2} + \left(2^{\frac{14}{\sqrt{2}}} \left[\frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-4\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi}} + \right. \right. \\
& \left. \left. \frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{3\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-\pi}} \right]^{\wedge (1/14)} / \right. \\
& \left. \left(\frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-4\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi}} + \right. \right. \\
& \left. \left. 2^{\frac{14}{\sqrt{2}}} \left[\frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-4\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi}} + \right. \right. \right. \\
& \left. \left. \frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{3\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-\pi}} \right]^{\wedge (1/14)} / \right. \\
& \left. \left(\frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{3\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-\pi}} \right) \right)
\end{aligned}$$

Integral representations

$$\begin{aligned}
 & \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)}^{15} - 89 - 21 - 3 - \phi^2 = \\
 & \frac{1}{2} \left(-229 - \sqrt{5} + 4 \sqrt[14]{2} \left(1 / \left(\exp \left(-\frac{3\sqrt{3}}{4} - 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} dt \right) + \right. \right. \right. \\
 & \quad \left. \left. 1 / \left(\exp \left(-2 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} dt \right) \right) - \exp \left(\right. \right. \right. \right. \\
 & \quad \left. \left. \left. 3 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} dt \right) \right) \right) \right) + \right. \\
 & \quad \left. 1 / \left(\exp \left(-2 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} dt \right) \right) + \right. \right. \\
 & \quad \left. \left. 1 / \left(\exp \left(-4 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} dt \right) \right) + \exp \left(\right. \right. \right. \right. \\
 & \quad \left. \left. \left. 6 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} dt \right) \right) \right) \right) \right)^{15/14} \Bigg)
 \end{aligned}$$

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 =} \\
& -\frac{229}{2} - \frac{\sqrt{5}}{2} + \left(2 \sqrt[14]{2 \left(\frac{1}{e^{-2 \int_0^\infty \sin(t)/t \, dt} + \frac{1}{e^{-4 \int_0^\infty \sin(t)/t \, dt} - e^{6 \int_0^\infty \sin(t)/t \, dt}} \right.} \right. \\
& \quad \left. \left. \frac{1}{e^{-4 \int_0^\infty \sin(t)/t \, dt} + \frac{1}{e^{-8 \int_0^\infty \sin(t)/t \, dt} + e^{12 \int_0^\infty \sin(t)/t \, dt}}} \right)^{\wedge (1/14)} \right) / \\
& \left(e^{-2 \int_0^\infty \sin(t)/t \, dt} + \frac{1}{e^{-4 \int_0^\infty \sin(t)/t \, dt} - e^{6 \int_0^\infty \sin(t)/t \, dt}} \right) + \\
& \left(2 \sqrt[14]{2 \left(\frac{1}{e^{-2 \int_0^\infty \sin(t)/t \, dt} + \frac{1}{e^{-4 \int_0^\infty \sin(t)/t \, dt} - e^{6 \int_0^\infty \sin(t)/t \, dt}}} \right.} \right. \\
& \quad \left. \left. \frac{1}{e^{-4 \int_0^\infty \sin(t)/t \, dt} + \frac{1}{e^{-8 \int_0^\infty \sin(t)/t \, dt} + e^{12 \int_0^\infty \sin(t)/t \, dt}}} \right)^{\wedge (1/14)} \right) / \\
& \left(e^{-4 \int_0^\infty \sin(t)/t \, dt} + \frac{1}{e^{-8 \int_0^\infty \sin(t)/t \, dt} + e^{12 \int_0^\infty \sin(t)/t \, dt}} \right)
\end{aligned}$$

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 =} \\
& -\frac{229}{2} - \frac{\sqrt{5}}{2} + \left(2^{\frac{14}{\sqrt{2}}} \left(\frac{1}{e^{-2 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-4 \times \int_0^\infty \frac{1}{(1+t^2)} dt} - e^{6 \times \int_0^\infty \frac{1}{(1+t^2)} dt}} \right. \right. \\
& \left. \left. \frac{1}{e^{-4 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-8 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + e^{12 \times \int_0^\infty \frac{1}{(1+t^2)} dt}} \right)^{\wedge (1/14)} \right) / \\
& \left(e^{-2 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-4 \times \int_0^\infty \frac{1}{(1+t^2)} dt} - e^{6 \times \int_0^\infty \frac{1}{(1+t^2)} dt}} \right) + \\
& \left(2^{\frac{14}{\sqrt{2}}} \left(\frac{1}{e^{-2 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-4 \times \int_0^\infty \frac{1}{(1+t^2)} dt} - e^{6 \times \int_0^\infty \frac{1}{(1+t^2)} dt}} \right. \right. \\
& \left. \left. \frac{1}{e^{-4 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-8 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + e^{12 \times \int_0^\infty \frac{1}{(1+t^2)} dt}} \right)^{\wedge (1/14)} \right) / \\
& \left(e^{-4 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + \frac{1}{e^{-8 \times \int_0^\infty \frac{1}{(1+t^2)} dt} + e^{12 \times \int_0^\infty \frac{1}{(1+t^2)} dt}} \right)
\end{aligned}$$

$$(((1/27((((2[(((1/(e^(-2Pi))+1/(e^(-4Pi))+1/e^(-6Pi))])))+((1/(e^(-Pi))+1/(e^(-2Pi))-1/e^(-3Pi)))]))^\wedge(1/14))^\wedge(15-89-21-3-\phi^2-1))))^\wedge(2+276)$$

Input

$$\left(\frac{1}{27} \left(\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 - 1} \right)^2 + 276 \right)$$

ϕ is the golden ratio

Exact result

$$\frac{1}{729} \left(-\phi^2 - 114 + 2 \sqrt[14]{2} \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)^{15/14} \right)^2 + 276$$

Decimal approximation

4372.1926153515802766019939379672707412505187501602686703270310344
...

$$4372.192615\dots \approx 4372$$

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

Property

$$276 + \frac{1}{729} \left(-114 + 2 \sqrt[14]{2} \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)^{15/14} - \phi^2 \right)^2$$

is a transcendental number

Alternate forms

$$276 + \frac{1}{729} \left(\frac{1}{2} (-231 - \sqrt{5}) + 2 \sqrt[14]{2} \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)^{15/14} \right)^2$$

$$\frac{1}{729} \left(228 \phi^2 + \phi^4 - \frac{4 \sqrt[14]{2} e^{(15\pi)/14} (\phi^2 + 114)}{\left(\frac{(-1-e^{3\pi}+e^{5\pi})(1+e^{6\pi}+e^{10\pi})}{-1+e^{5\pi}-e^{6\pi}-e^{10\pi}+e^{11\pi}+e^{15\pi}+e^{\pi}(-1-e^{3\pi}+e^{5\pi})(1+e^{10\pi})} \right)^{15/14}} + \right. \\ \left. 214200 + \frac{4 \sqrt[7]{2} e^{(15\pi)/7}}{\left(\frac{(-1-e^{3\pi}+e^{5\pi})(1+e^{6\pi}+e^{10\pi})}{-1+e^{5\pi}-e^{6\pi}-e^{10\pi}+e^{11\pi}+e^{15\pi}+e^{\pi}(-1-e^{3\pi}+e^{5\pi})(1+e^{10\pi})} \right)^{15/7}} \right)$$

$$\frac{1}{1458} \left(429091 + 231 \sqrt{5} - 924 \sqrt[14]{2} e^{(15\pi)/14} \right. \\ \left(\frac{-1 - e^{\pi} - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi}}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}} \right)^{15/14} - 4 \sqrt[14]{2} \sqrt{5} \\ e^{(15\pi)/14} \left(\frac{-1 - e^{\pi} - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi}}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}} \right)^{15/14} + \\ 8 \sqrt[7]{2} e^{(15\pi)/7} \left(\frac{-1 - e^{\pi} - e^{4\pi} + e^{5\pi} - e^{10\pi} - e^{14\pi} + e^{15\pi} + e^{16\pi}}{-1 - e^{3\pi} + e^{5\pi} - e^{6\pi} - e^{9\pi} - e^{10\pi} + e^{11\pi} - e^{13\pi} + e^{15\pi}} \right)^{15/7} \Bigg)$$

Expanded forms

$$\begin{aligned}
& \frac{429091}{1458} + \frac{77\sqrt{5}}{486} - \frac{154 \sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{243 \left(e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}} \right)} - \\
& \frac{2\sqrt{5} \sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{729 \left(e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}} \right)} - \\
& \frac{154 \sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{243 \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}} \right)} - \\
& \frac{2\sqrt{5} \sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{729 \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}} \right)} + \\
& \frac{4 \sqrt[7]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{729 \left(e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}} \right)^2} + \\
& \frac{4 \sqrt[7]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{729 \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}} \right)^2} + \frac{8 \sqrt[7]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}}{729 \left(e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}} \right) \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}} \right)}
\end{aligned}$$

$$\begin{aligned}
& \frac{429091}{1458} + \frac{77\sqrt{5}}{486} - \frac{154}{243} \sqrt[14]{2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}^{15/14} - \\
& \frac{2}{729} \sqrt[14]{2} \sqrt{5} \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)^{15/14} + \\
& \frac{4}{729} \sqrt[7]{2} \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)^{15/7}
\end{aligned}$$

Alternative representations

$$\left(\frac{1}{27} \left(\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 - 1} \right) \right)^2 +$$

$$276 = 276 + \left(\frac{1}{27} \left(-114 - \left(2 \cos\left(\frac{\pi}{5}\right) \right)^2 + \right.$$

$$\left. \sqrt[14]{2 \left(\frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}}} + \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}}} \right)^{15}} \right) \right)^2$$

$$\left(\frac{1}{27} \left(\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15} - 89 - 21 - 3 - \phi^2 - 1} \right) \right)^2 +$$

$$276 = 276 + \left(\frac{1}{27} \left(-114 + \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^{15}} - \right.$$

$$\left. \left. \left[\text{root of } -1 - x + x^2 \text{ near } x = 1.61803 \right]^2 \right) \right)^2$$

$$\left(\frac{1}{27} \left(\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)} - 89 - 21 - 3 - \phi^2 - 1 \right) \right)^2 +$$

$$276 =$$

$$276 + \left(\frac{1}{27} \left(-114 + \sqrt[14]{2 \left(\frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}}} + \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}}} \right)} - \right.$$

$$\left. \left. \left(\text{root of } -1 - x + x^2 \text{ near } x = 1.61803 \right)^2 \right) \right)^2$$

But, we have also:

$$2[(((1/(e^{(-2\pi)}+1/(e^{(-4\pi)}+1/e^{(-6\pi)})))))+((1/(e^{(-\pi)}+1/(e^{(-2\pi)}-1/e^{(-3\pi)})))))]+47$$

Input

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47$$

Exact result

$$47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)$$

Decimal approximation

1164.3475509932530299959247436759704668826734014227549801781175747
...

1164.3475509..... result very near to the following Ramanujan's class invariant

$$Q = (G_{505}/G_{101/5})^3 = 1164.2696$$

Property

$$47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right) \text{ is a transcendental number}$$

Alternate forms

$$47 + 2e^\pi + 2e^{2\pi} + \frac{2e^{4\pi}}{-1 - e^{3\pi} + e^{5\pi}} - \frac{2e^{8\pi}}{1 + e^{6\pi} + e^{10\pi}}$$

$$\frac{(-47 - 2e^\pi - 2e^{2\pi} - 47e^{3\pi} + 45e^{5\pi} - 45e^{6\pi} - 47e^{9\pi} - 47e^{10\pi} + 45e^{11\pi} - 47e^{13\pi} + 45e^{15\pi} + 2e^{16\pi} + 2e^{17\pi})}{((-1 - e^{3\pi} + e^{5\pi})(1 + e^{6\pi} + e^{10\pi}))}$$

Alternative representations

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 =$$

$$47 + 2 \left(\frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}}} + \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}}} \right)$$

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 =$$

$$47 + 2 \left(\frac{1}{e^{i \log(-1)} + \frac{1}{e^{2i \log(-1)} - \frac{1}{e^{3i \log(-1)}}}} + \frac{1}{e^{2i \log(-1)} + \frac{1}{e^{4i \log(-1)} + \frac{1}{e^{6i \log(-1)}}}} \right)$$

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 =$$

$$2 \left(\frac{1}{\exp^{-2\pi}(z) + \frac{1}{\exp^{-4\pi}(z) + \frac{1}{\exp^{-6\pi}(z)}}}} + \frac{1}{\exp^{-\pi}(z) + \frac{1}{\exp^{-2\pi}(z) - \frac{1}{\exp^{-3\pi}(z)}}}} \right) + 47 \text{ for } z = 1$$

Series representations

$$2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 =$$

$$47 + \frac{e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}{2} + \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}} +$$

$$\frac{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}{2} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}$$

$$\begin{aligned}
& 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 = \\
& 47 + 2 / \left(\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-4} \sum_{k=0}^{\infty} (-1)^k / (1+2k) \right. \\
& \left. \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-8} \sum_{k=0}^{\infty} (-1)^k / (1+2k) - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{12} \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \right) + \\
& 2 / \left(\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-8} \sum_{k=0}^{\infty} (-1)^k / (1+2k) \right. \\
& \left. \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-16} \sum_{k=0}^{\infty} (-1)^k / (1+2k) + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{24} \sum_{k=0}^{\infty} (-1)^k / (1+2k)} \right)
\end{aligned}$$

$$\begin{aligned}
& 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 = \\
& \left(-47 - 2 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{\pi} - 2 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{2\pi} - 47 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{3\pi} + 45 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{5\pi} - \right. \\
& 45 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi} - 47 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{9\pi} - 47 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{10\pi} + 45 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{11\pi} - \\
& 47 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{13\pi} + 45 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{15\pi} + 2 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{16\pi} + 2 \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{17\pi} \Big) / \\
& \left(\left(-1 - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{3\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{5\pi} \right) \left(1 + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{10\pi} \right) \right)
\end{aligned}$$

Integral representations

$$\begin{aligned}
 & 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 = \\
 & 47 + \frac{2}{e^{-4} \int_0^1 \sqrt{1-t^2} dt + \frac{1}{e^{-8} \int_0^1 \sqrt{1-t^2} dt - e^{12} \int_0^1 \sqrt{1-t^2} dt}} + \\
 & \frac{e^{-8} \int_0^1 \sqrt{1-t^2} dt + \frac{1}{e^{-16} \int_0^1 \sqrt{1-t^2} dt + e^{24} \int_0^1 \sqrt{1-t^2} dt}}{2}
 \end{aligned}$$

$$\begin{aligned}
 & 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 = \\
 & 47 + \frac{2}{e^{-8/3} \int_0^\infty \sin^3(t)/t^3 dt + \frac{1}{e^{-16/3} \int_0^\infty \sin^3(t)/t^3 dt - e^8 \int_0^\infty \sin^3(t)/t^3 dt}} + \\
 & \frac{e^{-16/3} \int_0^\infty \sin^3(t)/t^3 dt + \frac{1}{e^{-32/3} \int_0^\infty \sin^3(t)/t^3 dt + e^{16} \int_0^\infty \sin^3(t)/t^3 dt}}{2}
 \end{aligned}$$

$$\begin{aligned}
& 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 = \\
& 47 + 2 / \left(\exp \left(-\frac{3\sqrt{3}}{4} - 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} \, dt \right) + \right. \\
& \quad \left. 1 / \left(\exp \left(-2 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} \, dt \right) \right) - \right. \right. \\
& \quad \quad \left. \left. \exp \left(3 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} \, dt \right) \right) \right) \right) + \\
& \quad 2 / \left(\exp \left(-2 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} \, dt \right) \right) + \right. \\
& \quad \quad \left. 1 / \left(\exp \left(-4 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} \, dt \right) \right) + \right. \right. \\
& \quad \quad \quad \left. \left. \exp \left(6 \left(\frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{-(-1+t)t} \, dt \right) \right) \right) \right) \right)
\end{aligned}$$

$$(2[(((1/(e^{(-2\pi)}+1/(e^{(-4\pi)}+1/e^{(-6\pi)}))))) + ((1/(e^{(-\pi)}+1/(e^{(-2\pi)}-1/e^{(-3\pi)}))))) + 47)^{1/14}$$

Input

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47}$$

Exact result

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)}$$

Decimal approximation

1.6557924669474465120220720626513880685440922328109264620053251735

...

1.655792466..... result that is very near to the 14th root of the following
Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Indeed, from:

$$G_{505} = P^{-1/4} Q^{1/6} = (\sqrt{5} + 2)^{1/2} \left(\frac{\sqrt{5} + 1}{2} \right)^{1/4} (\sqrt{101} + 10)^{1/4} \\ \times \left((130\sqrt{5} + 29\sqrt{101}) + \sqrt{169440 + 7540\sqrt{505}} \right)^{1/6}.$$

Thus, it remains to show that

$$(130\sqrt{5} + 29\sqrt{101}) + \sqrt{169440 + 7540\sqrt{505}} = \left(\sqrt{\frac{113 + 5\sqrt{505}}{8}} + \sqrt{\frac{105 + 5\sqrt{505}}{8}} \right)^3,$$

which is straightforward. $\square$

$$\sqrt[14]{\left(\sqrt{\frac{113 + 5\sqrt{505}}{8}} + \sqrt{\frac{105 + 5\sqrt{505}}{8}} \right)^3} = 1.65578 \dots$$

Property

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - e^{3\pi}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + e^{6\pi}}} \right)} \text{ is a transcendental number}$$

Alternate form

$$1 / \left(\left((-1 - e^{3\pi} + e^{5\pi})(1 + e^{6\pi} + e^{10\pi}) \right) / \left(-47 - 47e^{3\pi} + 47e^{5\pi} - 47e^{6\pi} - 47e^{9\pi} - \right. \right. \\ \left. \left. 47e^{10\pi} - 47e^{13\pi} + 47e^{15\pi} + 2e^{\pi}(e^{5\pi} - 1) + 2e^{7\pi}(e^{5\pi} - 1) + \right. \right. \\ \left. \left. e^{11\pi}(45 + 2e^{5\pi}) + 2e^{2\pi}(-1 - e^{3\pi} + e^{5\pi})(1 + e^{10\pi}) \right) \right)^{(1/14)}$$

All 14th roots of $47 + 2 \left(\frac{1}{e^{-\pi}} + \frac{1}{e^{(-2\pi - e^{(3\pi)})}} + \frac{1}{e^{(-2\pi)} + \frac{1}{e^{(-4\pi + e^{(6\pi)})}} \right)$

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi - e^{3\pi}}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi + e^{6\pi}}}} \right)} e^0 \approx 1.6558 \text{ (real, principal root)}$$

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi - e^{3\pi}}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi + e^{6\pi}}}} \right)} e^{(i\pi)/7} \approx 1.4918 + 0.7184 i$$

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi - e^{3\pi}}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi + e^{6\pi}}}} \right)} e^{(2i\pi)/7} \approx 1.0324 + 1.2946 i$$

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi - e^{3\pi}}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi + e^{6\pi}}}} \right)} e^{(3i\pi)/7} \approx 0.3684 + 1.6143 i$$

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi - e^{3\pi}}}} + \frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi + e^{6\pi}}}} \right)} e^{(4i\pi)/7} \approx -0.3684 + 1.6143 i$$

Alternative representations

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47} =$$

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}}} + \frac{1}{e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}}} \right)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47} =$$

$$\sqrt[14]{47 + 2 \left(\frac{1}{e^{i \log(-1)} + \frac{1}{e^{2i \log(-1)} - \frac{1}{e^{3i \log(-1)}}}} + \frac{1}{e^{2i \log(-1)} + \frac{1}{e^{4i \log(-1)} + \frac{1}{e^{6i \log(-1)}}}} \right)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47} =$$

$$\sqrt[14]{2 \left(\frac{1}{\exp^{-2\pi}(z) + \frac{1}{\exp^{-4\pi}(z) + \frac{1}{\exp^{-6\pi}(z)}}}} + \frac{1}{\exp^{-\pi}(z) + \frac{1}{\exp^{-2\pi}(z) - \frac{1}{\exp^{-3\pi}(z)}}}} \right) + 47} \text{ for}$$

$z = 1$

Series representations

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47} =$$

$$\left(47 + 2 \left(\frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-2\pi} - \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{3\pi}}} + \right.$$

$$\left. \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-2\pi} + \frac{1}{\left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{-4\pi} + \left(\sum_{k=0}^{\infty} \frac{1}{k!} \right)^{6\pi}}} \right) \right)^{(1/14)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47} =$$

$$\left(47 + 2 \left(\frac{1}{e^{-4 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} - e^{12 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}} + \right.$$

$$\left. \frac{1}{e^{-8 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + \frac{1}{e^{-16 \sum_{k=0}^{\infty} (-1)^k / (1+2k)} + e^{24 \sum_{k=0}^{\infty} (-1)^k / (1+2k)}}} \right) \right)^{(1/14)}$$

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47} = \\
& \left(47 + 2 \left(\frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-4\pi} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{6\pi}} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi}} \right. \right. \\
& \quad \left. \left. \frac{1}{\left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-2\pi} - \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{3\pi}} + \left(\frac{1}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}} \right)^{-\pi}} \right) \right)^{\wedge (1/14)}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& \sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47} = \\
& \left(47 + 2 \left(\frac{1}{e^{-2 \int_0^{\infty} \sin(t)/t \, dt} + \frac{1}{e^{-4 \int_0^{\infty} \sin(t)/t \, dt} - e^{-6 \int_0^{\infty} \sin(t)/t \, dt}}} \right. \right. \\
& \quad \left. \left. \frac{1}{e^{-4 \int_0^{\infty} \sin(t)/t \, dt} + \frac{1}{e^{-8 \int_0^{\infty} \sin(t)/t \, dt} + e^{12 \int_0^{\infty} \sin(t)/t \, dt}}} \right) \right)^{\wedge (1/14)}
\end{aligned}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 = \left(47 + 2 \left(\frac{1}{e^{-2 \times \int_0^\infty 1/(1+t^2) dt} + \frac{1}{e^{-4 \times \int_0^\infty 1/(1+t^2) dt} - e^{-6 \times \int_0^\infty 1/(1+t^2) dt}} + \frac{1}{e^{-4 \times \int_0^\infty 1/(1+t^2) dt} + \frac{1}{e^{-8 \times \int_0^\infty 1/(1+t^2) dt} + e^{-12 \times \int_0^\infty 1/(1+t^2) dt}} \right) \right)^{(1/14)}$$

$$\sqrt[14]{2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} + \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right) + 47 = \left(47 + 2 \left(\frac{1}{e^{-2 \int_0^\infty \sin^2(t)/t^2 dt} + \frac{1}{e^{-4 \int_0^\infty \sin^2(t)/t^2 dt} - e^{-6 \int_0^\infty \sin^2(t)/t^2 dt}} + \frac{1}{e^{-4 \int_0^\infty \sin^2(t)/t^2 dt} + \frac{1}{e^{-8 \int_0^\infty \sin^2(t)/t^2 dt} + e^{-12 \int_0^\infty \sin^2(t)/t^2 dt}} \right) \right)^{(1/14)}$$

And again:

$$0.9725502648 * 2[(((((((1/(e^{(-2\pi)}+1/(e^{(-4\pi)}+1/e^{(-6\pi)})))))))*((1/(e^{(-\pi)}+1/(e^{(-2\pi)}-1/e^{(-3\pi)})))))))]^2$$

where 0.9725502648 is in the range of the following Meson Regge slope:

$$\omega/\omega_3 \mid 5 + 3 \mid m_{u/d} = 240 - 345 \mid 0.937 - 1.000$$

Input interpretation

$$0.9725502648 \times 2 \left(\frac{1}{e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}}} \times \frac{1}{e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}}} \right)^2$$

Result

$$2.997924580... \times 10^8$$

$$2.997924580... * 10^8 \approx c = \text{speed of light} = 299792458$$

Alternative representations

$$0.97255 \times 2 \left(\frac{1}{\left(e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}} \right) \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}} \right)} \right)^2 =$$

$$1.9451 \left(\frac{1}{\left(e^{-360^\circ} + \frac{1}{e^{-720^\circ} + \frac{1}{e^{-1080^\circ}}} \right) \left(e^{-180^\circ} + \frac{1}{e^{-360^\circ} - \frac{1}{e^{-540^\circ}}} \right)} \right)^2$$

$$0.97255 \times 2 \left(\frac{1}{\left(e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}} \right) \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}} \right)} \right)^2 =$$

$$1.9451 \left(\frac{1}{\left(e^{i \log(-1)} + \frac{1}{e^{2i \log(-1)} - \frac{1}{e^{3i \log(-1)}}} \right) \left(e^{2i \log(-1)} + \frac{1}{e^{4i \log(-1)} + \frac{1}{e^{6i \log(-1)}}} \right)} \right)^2$$

$$0.97255 \times 2 \left(\frac{1}{\left(e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}} \right) \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}} \right)} \right)^2 = 0.97255 \times 2$$

$$\left(\frac{1}{\left(\exp^{-\pi}(z) + \frac{1}{\exp^{-2\pi}(z) - \frac{1}{\exp^{-3\pi}(z)}} \right) \left(\exp^{-2\pi}(z) + \frac{1}{\exp^{-4\pi}(z) + \frac{1}{\exp^{-6\pi}(z)}} \right)} \right)^2 \text{ for } z = 1$$

Integral representations

$$0.97255 \times 2 \left(\frac{1}{\left(e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}} \right) \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}} \right)} \right)^2 =$$

$$1.9451 / \left(\left(e^{-2 \int_0^\infty \sin(t)/t dt} + \frac{1}{e^{-4 \int_0^\infty \sin(t)/t dt} - e^{6 \int_0^\infty \sin(t)/t dt}} \right)^2 \right.$$

$$\left. \left(e^{-4 \int_0^\infty \sin(t)/t dt} + \frac{1}{e^{-8 \int_0^\infty \sin(t)/t dt} + e^{12 \int_0^\infty \sin(t)/t dt}} \right)^2 \right)$$

$$\begin{aligned}
& 0.97255 \times 2 \left(\frac{1}{\left(e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}} \right) \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}} \right)} \right)^2 = \\
& 1.9451 / \left(\left(e^{-2 \times \int_0^\infty 1/(1+t^2) dt} + \frac{1}{e^{-4 \times \int_0^\infty 1/(1+t^2) dt} - e^{6 \times \int_0^\infty 1/(1+t^2) dt}} \right)^2 \right. \\
& \left. \left(e^{-4 \times \int_0^\infty 1/(1+t^2) dt} + \frac{1}{e^{-8 \times \int_0^\infty 1/(1+t^2) dt} + e^{12 \times \int_0^\infty 1/(1+t^2) dt}} \right)^2 \right)
\end{aligned}$$

$$\begin{aligned}
& 0.97255 \times 2 \left(\frac{1}{\left(e^{-\pi} + \frac{1}{e^{-2\pi} - \frac{1}{e^{-3\pi}}} \right) \left(e^{-2\pi} + \frac{1}{e^{-4\pi} + \frac{1}{e^{-6\pi}}} \right)} \right)^2 = \\
& 1.9451 / \left(\left(e^{-4 \int_0^1 \sqrt{1-t^2} dt} + \frac{1}{e^{-8 \int_0^1 \sqrt{1-t^2} dt} - e^{12 \int_0^1 \sqrt{1-t^2} dt}} \right)^2 \right. \\
& \left. \left(e^{-8 \int_0^1 \sqrt{1-t^2} dt} + \frac{1}{e^{-16 \int_0^1 \sqrt{1-t^2} dt} + e^{24 \int_0^1 \sqrt{1-t^2} dt}} \right)^2 \right)
\end{aligned}$$

Mathematical connections with some sectors of String Theory

From:

Modular equations and approximations to π - Srinivasa Ramanujan
Quarterly Journal of Mathematics, XLV, 1914, 350 – 372

We have that:

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

Again

$$G_{37} = (6 + \sqrt{37})^{\frac{1}{4}},$$

$$\begin{aligned} 64G_{37}^{24} &= e^{\pi\sqrt{37}} + 24 + 276e^{-\pi\sqrt{37}} + \dots, \\ 64G_{37}^{-24} &= 4096e^{-\pi\sqrt{37}} - \dots, \end{aligned}$$

so that

$$64(G_{37}^{24} + G_{37}^{-24}) = e^{\pi\sqrt{37}} + 24 + 4372e^{-\pi\sqrt{37}} - \dots = 64\{(6 + \sqrt{37})^6 + (6 - \sqrt{37})^6\}.$$

Hence

$$e^{\pi\sqrt{37}} = 199148647.999978\dots$$

Similarly, from

$$g_{58} = \sqrt{\left(\frac{5 + \sqrt{29}}{2}\right)},$$

we obtain

$$64(g_{58}^{24} + g_{58}^{-24}) = e^{\pi\sqrt{58}} - 24 + 4372e^{-\pi\sqrt{58}} + \dots = 64\left\{\left(\frac{5 + \sqrt{29}}{2}\right)^{12} + \left(\frac{5 - \sqrt{29}}{2}\right)^{12}\right\}.$$

Hence

$$e^{\pi\sqrt{58}} = 24591257751.99999982\dots$$

From:

An Update on Brane Supersymmetry Breaking

J. Mourad and A. Sagnotti - arXiv:1711.11494v1 [hep-th] 30 Nov 2017

From the following vacuum equations:

$$T e^{\gamma_E \phi} = - \frac{\beta_E^{(p)} h^2}{\gamma_E} e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

$$16 k' e^{-2C} = \frac{h^2 \left(p + 1 - \frac{2\beta_E^{(p)}}{\gamma_E} \right) e^{-2(8-p)C + 2\beta_E^{(p)} \phi}}{(7-p)}$$

$$(A')^2 = k e^{-2A} + \frac{h^2}{16(p+1)} \left(7 - p + \frac{2\beta_E^{(p)}}{\gamma_E} \right) e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

we have obtained, from the results almost equals of the equations, putting

$4096 e^{-\pi\sqrt{18}}$ instead of

$$e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

a new possible mathematical connection between the two exponentials. Thence, also the values concerning p , C , β_E and ϕ correspond to the exponents of e (i.e. of exp). Thence we obtain for $p = 5$ and $\beta_E = 1/2$:

$$e^{-6C+\phi} = 4096 e^{-\pi\sqrt{18}}$$

Therefore, with respect to the exponentials of the vacuum equations, the Ramanujan's exponential has a coefficient of 4096 which is equal to 64^2 , while $-6C+\phi$ is equal to $-\pi\sqrt{18}$. From this it follows that it is possible to establish mathematically, the dilaton value.

For

$\exp((- \pi \sqrt{18}))$ we obtain:

Input:

$$\exp\left(-\pi \sqrt{18}\right)$$

Exact result:

$$e^{-3 \sqrt{2} \pi}$$

Decimal approximation:

$$1.6272016226072509292942156739117979541838581136954016... \times 10^{-6}$$

$$1.6272016... * 10^{-6}$$

Property:

$$e^{-3 \sqrt{2} \pi} \text{ is a transcendental number}$$

Series representations:

$$e^{-\pi \sqrt{18}} = e^{-\pi \sqrt{17} \sum_{k=0}^{\infty} 17^{-k} \binom{1/2}{k}}$$

$$e^{-\pi \sqrt{18}} = \exp\left(-\pi \sqrt{17} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{17}\right)^k \left(-\frac{1}{2}\right)_k}{k!}\right)$$

$$e^{-\pi \sqrt{18}} = \exp\left(-\frac{\pi \sum_{j=0}^{\infty} \operatorname{Res}_{s=-\frac{1}{2}+j} 17^{-s} \Gamma\left(-\frac{1}{2}-s\right) \Gamma(s)}{2 \sqrt{\pi}}\right)$$

Now, we have the following calculations:

$$e^{-6C+\phi} = 4096e^{-\pi\sqrt{18}}$$

$$e^{-\pi\sqrt{18}} = 1.6272016... * 10^{-6}$$

from which:

$$\frac{1}{4096} e^{-6C+\phi} = 1.6272016... * 10^{-6}$$

$$0.000244140625 e^{-6C+\phi} = e^{-\pi\sqrt{18}} = 1.6272016... * 10^{-6}$$

Now:

$$\ln(e^{-\pi\sqrt{18}}) = -13.328648814475 = -\pi\sqrt{18}$$

And:

$$(1.6272016 * 10^{-6}) * 1 / (0.000244140625)$$

Input interpretation:

$$\frac{1.6272016}{10^6} \times \frac{1}{0.000244140625}$$

Result:

0.0066650177536

0.006665017...

Thence:

$$0.000244140625 e^{-6C+\phi} = e^{-\pi\sqrt{18}}$$

Dividing both sides by 0.000244140625, we obtain:

$$\frac{0.000244140625}{0.000244140625} e^{-6C+\phi} = \frac{1}{0.000244140625} e^{-\pi\sqrt{18}}$$

$$e^{-6C+\phi} = 0.0066650177536$$

$$((((\exp((-Pi*\sqrt{18})))))))*1/0.000244140625$$

Input interpretation:

$$\exp\left(-\pi\sqrt{18}\right)\times\frac{1}{0.000244140625}$$

Result:

0.00666501785...

0.00666501785...

Series representations:

$$\frac{\exp(-\pi\sqrt{18})}{0.000244141} = 4096 \exp\left(-\pi\sqrt{17} \sum_{k=0}^{\infty} 17^{-k} \left(\frac{1}{2}\right)_k\right)$$

$$\frac{\exp(-\pi\sqrt{18})}{0.000244141} = 4096 \exp\left(-\pi\sqrt{17} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{17}\right)^k \left(-\frac{1}{2}\right)_k}{k!}\right)$$

$$\frac{\exp(-\pi \sqrt{18})}{0.000244141} = 4096 \exp\left(-\frac{\pi \sum_{j=0}^{\infty} \operatorname{Res}_{s=-\frac{1}{2}+j} 17^{-s} \Gamma\left(-\frac{1}{2}-s\right) \Gamma(s)}{2 \sqrt{\pi}}\right)$$

Now:

$$e^{-6C+\phi} = 0.0066650177536$$

$$\exp(-\pi \sqrt{18}) \times \frac{1}{0.000244140625} =$$

$$e^{-\pi \sqrt{18}} \times \frac{1}{0.000244140625}$$

$$= 0.00666501785\dots$$

From:

$$\ln(0.00666501784619)$$

Input interpretation:

$$\log(0.00666501784619)$$

Result:

$$-5.010882647757\dots$$

$$-5.010882647757\dots$$

Alternative representations:

$$\log(0.006665017846190000) = \log_e(0.006665017846190000)$$

$$\log(0.006665017846190000) = \log(a) \log_a(0.006665017846190000)$$

$$\log(0.006665017846190000) = -\text{Li}_1(0.993334982153810000)$$

Series representations:

$$\log(0.006665017846190000) = -\sum_{k=1}^{\infty} \frac{(-1)^k (-0.993334982153810000)^k}{k}$$

$$\log(0.006665017846190000) = 2i\pi \left\lfloor \frac{\arg(0.006665017846190000 - x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (0.006665017846190000 - x)^k x^{-k}}{k} \quad \text{for } x < 0$$

$$\log(0.006665017846190000) = \left\lfloor \frac{\arg(0.006665017846190000 - z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(0.006665017846190000 - z_0)}{2\pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (0.006665017846190000 - z_0)^k z_0^{-k}}{k}$$

Integral representation:

$$\log(0.006665017846190000) = \int_1^{0.006665017846190000} \frac{1}{t} dt$$

In conclusion:

$$-6C + \phi = -5.010882647757 \dots$$

and for $C = 1$, we obtain:

$$\phi = -5.010882647757 + 6 = \mathbf{0.989117352243} = \phi$$

Note that the values of n_s (spectral index) 0.965, of the average of the Omega mesons Regge slope 0.987428571 and of the dilaton 0.989117352243, are also connected to the following two Rogers-Ramanujan continued fractions:

$$\frac{e^{-\frac{\pi}{5}}}{\sqrt{(\varphi-1)\sqrt{5}} - \varphi + 1} = 1 - \frac{e^{-\pi}}{1 + \frac{e^{-2\pi}}{1 + \frac{e^{-3\pi}}{1 + \frac{e^{-4\pi}}{1 + \dots}}}} \approx 0.9568666373$$

$$\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

(<http://www.bitman.name/math/article/102/109/>)

The mean between the two results of the above Rogers-Ramanujan continued fractions is 0.97798855285, value very near to the ψ Regge slope 0.979:

Ψ	3	$m_c = 1500$	0.979	-0.09
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Also performing the 512th root of the inverse value of the Pion meson rest mass 139.57, we obtain:

$$((1/(139.57)))^{1/512}$$

Input interpretation:

$$\sqrt[512]{\frac{1}{139.57}}$$

Result:

0.990400732708644027550973755713301415460732796178555551684...

0.99040073.... result very near to the dilaton value **0.989117352243 = ϕ** and to the value of the following Rogers-Ramanujan continued fraction:

$$\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

From

AdS Vacua from Dilaton Tadpoles and Form Fluxes - *J. Mourad and A. Sagnotti*
- arXiv:1612.08566v2 [hep-th] 22 Feb 2017 - March 27, 2018

We have:

$$e^{2C} = \frac{2\xi e^{\frac{\phi}{2}}}{1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}}$$

$$\frac{h^2}{32} = \frac{\xi^7 e^{4\phi}}{\left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right)^7} \left[\frac{42}{\xi} \left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right) + 5 T e^{2\phi} \right]. \quad (2.7)$$

For

$$T = \frac{16}{\pi^2}$$

$$\xi = 1$$

we obtain:

$$(2 \cdot e^{(0.989117352243/2)}) / (1 + \sqrt{((1 - 1/3 \cdot 16/(\pi)^2 \cdot e^{(2 \cdot 0.989117352243)}))})$$

Input interpretation:

$$\frac{2 e^{0.989117352243/2}}{1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}}$$

Result:

$$0.83941881822... - \\ 1.4311851867... i$$

Polar coordinates:

$$r = 1.65919106525 \text{ (radius), } \theta = -59.607521917^\circ \text{ (angle)}$$

1.65919106525..... result very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Series representations:

$$\frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \frac{2 e^{0.4945586761215000}}{1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \binom{\frac{1}{2}}{k}}$$

$$\begin{aligned}
& \frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \\
& \frac{2 e^{0.4945586761215000}}{1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!}} \\
& \frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \\
& \frac{2 e^{0.4945586761215000}}{1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!}} \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From

$$\frac{h^2}{32} = \frac{\xi^7 e^{4\phi}}{\left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right)^7} \left[\frac{42}{\xi} \left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right) + 5 T e^{2\phi} \right]$$

we obtain:

$$\begin{aligned}
& e^{(4 \times 0.989117352243) / (((1 + \sqrt{1 - 1/3 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}}))^{1/7} \\
& [42(1 + \sqrt{1 - 1/3 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}}) + 5 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}]
\end{aligned}$$

Input interpretation:

$$\begin{aligned}
& \frac{e^{4 \times 0.989117352243}}{\left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}\right)^7} \\
& \left(42 \left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}\right) + 5 \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}\right)
\end{aligned}$$

Result:

50.84107889... –
20.34506335... i

Polar coordinates:

$r = 54.76072411$ (radius), $\theta = -21.80979492^\circ$ (angle)

54.76072411.....

Series representations:

$$\left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 =$$

$$\left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \right.$$

$$\left. \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)^k \right) \right) /$$

$$\left(\pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)^k \right) \right)^7$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 = \\
& \left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \right. \\
& \quad \left. \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right) / \\
& \left(\pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right)^7 \\
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 = \\
& \left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \right. \right. \\
& \quad \left. \left. \pi^2 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \\
& \left(\pi^2 \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) \right)^7 \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From which:

$$\begin{aligned}
& e^{(4 \times 0.989117352243)} / (((1 + \sqrt{1 - 1/3 \times 16 / (\pi^2 \times e^{(2 \times 0.989117352243)})}))^7 \\
& [42(1 + \sqrt{1 - 1/3 \times 16 / (\pi^2 \times e^{(2 \times 0.989117352243)})}) + 5 \times 16 / (\pi^2 \times e^{(2 \times 0.989117352243)})] \times 1/34
\end{aligned}$$

Input interpretation:

$$\frac{e^{4 \times 0.989117352243}}{\left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}\right)^7} \left(42 \left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}\right) + 5 \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}\right) \times \frac{1}{34}$$

Result:

1.495325850... -
0.5983842161... *i*

Polar coordinates:

$r = 1.610609533$ (radius), $\theta = -21.80979492^\circ$ (angle)

1.610609533.... result that is a good approximation to the value of the golden ratio
1.618033988749...

Series representations:

$$\left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 =$$

$$\left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right.$$

$$\left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)^{\binom{1}{k}} \right) /$$

$$\left(17 \pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)^{\binom{1}{k}} \right) \right)^7$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 = \\
& \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \\
& \quad \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) / \\
& \quad \left(17 \pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right)^7 \\
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 = \\
& \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \right. \\
& \quad \left. \pi^2 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) / \\
& \quad \left(17 \pi^2 \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) \right)^7 \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

Now, we have:

$$e^{2C} = \frac{2\xi e^{-\frac{\phi}{2}}}{1 + \sqrt{1 + \frac{\xi\Lambda}{3} e^{2\phi}}}, \quad (2.9)$$

$$\frac{h^2}{32} = \frac{e^{-4\phi}}{\left[1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right]^7} \left[42 \left(1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right) - 13\Lambda e^{2\phi}\right]. \quad (2.10)$$

For:

$$\xi = 1$$

$$\Lambda \simeq \frac{4\pi^2}{25}$$

$$\phi = 0.989117352243$$

From

$$e^{2C} = \frac{2\xi e^{-\frac{\phi}{2}}}{1 + \sqrt{1 + \frac{\xi\Lambda}{3} e^{2\phi}}},$$

we obtain:

$$\frac{(2e^{(-0.989117352243/2)})}{((((1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)}))))))}$$

Input interpretation:

$$\frac{2e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4\pi^2) \right) e^{2*0.989117352243}}}$$

Result:

0.382082347529...

0.382082347529....

Series representations:

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = 2 / \left(e^{0.4945586761215000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)_k \right) \right)$$

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = 2 / \left(e^{0.4945586761215000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)$$

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = \frac{2}{e^{0.4945586761215000} \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} z_0 \right)^k z_0^{-k}}{k!} \right)}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

From which:

$$1 + 1 / (((4 * (2 * e^{(-0.989117352243/2)})) / (((1 + \sqrt{((1 + 1/3 * (4 \pi^2)/25 * e^{(2 * 0.989117352243))}))))))))))$$

Input interpretation:

$$1 + \frac{1}{4 \times \frac{2 e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}}}}$$

Result:

1.65430921270...

1.6543092..... We note that, the result 1.6543092... is very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Indeed:

$$G_{505} = P^{-1/4} Q^{1/6} = (\sqrt{5} + 2)^{1/2} \left(\frac{\sqrt{5} + 1}{2} \right)^{1/4} (\sqrt{101} + 10)^{1/4} \\ \times \left((130\sqrt{5} + 29\sqrt{101}) + \sqrt{169440 + 7540\sqrt{505}} \right)^{1/6}.$$

Thus, it remains to show that

$$(130\sqrt{5} + 29\sqrt{101}) + \sqrt{169440 + 7540\sqrt{505}} = \left(\sqrt{\frac{113 + 5\sqrt{505}}{8}} + \sqrt{\frac{105 + 5\sqrt{505}}{8}} \right)^3,$$

which is straightforward. $\square$

$$\sqrt[14]{\left(\sqrt{\frac{113 + 5\sqrt{505}}{8}} + \sqrt{\frac{105 + 5\sqrt{505}}{8}} \right)^3} = 1.65578 \dots$$

Series representations:

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = \frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{1 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \binom{\frac{1}{2}}{k}}$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = \frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{1 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!}}$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = 1 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0 \right)^k}{k!} z_0^{-k}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

And from

$$\frac{h^2}{32} = \frac{e^{-4\phi}}{\left[1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right]^7} \left[42 \left(1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right) - 13 \Lambda e^{2\phi}\right].$$

we obtain:

$$e^{(-4 \times 0.989117352243) / [1 + \sqrt{((1 + 1/3 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243))})}]^7 \times [42(1 + \sqrt{((1 + 1/3 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243))})} - 13 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243)})]$$

Input interpretation:

$$\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4\pi^2)\right) e^{2 \times 0.989117352243}}\right)^7 \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4\pi^2)\right) e^{2 \times 0.989117352243}}\right) - 13 \left(\frac{1}{25} (4\pi^2)\right) e^{2 \times 0.989117352243}\right)}$$

Result:

-0.034547055658...

-0.034547055658...

Series representations:

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right. \right. \\
& \quad \left. \left. e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \right. \\
& \quad - \left(\left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\
& \quad \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right)_k \right) \right) / \left(25 e^{5.934704113458000} \right. \right. \\
& \quad \left. \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right)_k \right) \right)^7 \right) \right) \\
& \left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right. \\
& \quad \left. e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& \quad - \left(\left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\
& \quad \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right)^7 \right) \right)
\end{aligned}$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right. \\
& \quad \left. e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& - \left(\left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \right. \\
& \quad \left. \left. \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From which:

$$\begin{aligned}
& 47 * 1 / (((-1 / (((e^{(-4 * 0.989117352243)} / \\
& [1 + \sqrt{((1 + 1/3 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}))])^7 * \\
& [42(1 + \sqrt{((1 + 1/3 * (4\pi^2)/25 * e^{(2 * 0.989117352243)})) - \\
& 13 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}])])])])
\end{aligned}$$

Input interpretation:

$$\begin{aligned}
& 47 \left(- \left(1 / 1 / \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4\pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \right. \right. \\
& \quad \left. \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4\pi^2) \right) e^{2 \times 0.989117352243}} - \right. \right. \right. \\
& \quad \left. \left. \left. 13 \left(\frac{1}{25} (4\pi^2) \right) e^{2 \times 0.989117352243} \right) \right) \right) \right)
\end{aligned}$$

Result:

1.6237116159...

1.6237116159.... result that is an approximation to the value of the golden ratio
1.618033988749...

Series representations:

$$\begin{aligned}
 & - \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
 & \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
 & \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
 & \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \\
 & \quad \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
 & \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right) \right) \right) / \left(25 e^{5.934704113458000} \right. \\
 & \quad \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right) \right) \right)^7
 \end{aligned}$$

$$\begin{aligned}
& - \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \\
& \quad 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \\
& \quad \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k \left(e^{1.978234704486000} \pi^2\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \Bigg) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k \left(e^{1.978234704486000} \pi^2\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^7 \right) \\
& - \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \\
& \quad \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \Bigg) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

And again:

$$32((((e^{(-4*0.989117352243)} / [1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)})})}]^7 * [42(1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)})})})-13*(4\pi^2)/25*e^{(2*0.989117352243)})])])])$$

Input interpretation:

$$32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right) \right)$$

Result:

-1.1055057810...

-1.1055057810....

We note that the result -1.1055057810.... is very near to the value of Cosmological Constant, less 10^{-52} , thence 1.1056, with minus sign

Series representations:

$$\begin{aligned}
& \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& - \left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \\
& \quad \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \binom{\frac{1}{2}}{k} \right) \right) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \binom{\frac{1}{2}}{k} \right)^7 \right) \right)
\end{aligned}$$

$$\begin{aligned}
& \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& - \left(\left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\
& \quad \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right)^7 \right) \right)
\end{aligned}$$

$$\begin{aligned}
& \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& - \left(\left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \right. \\
& \quad \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \left(25 \right. \\
& \quad \left. e^{5.934704113458000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

And:

$$- [32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \\ \left. [42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}] \right)^5$$

Input interpretation:

$$- \left[32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \right. \\ \left. \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - \right. \right. \\ \left. \left. \left. 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right) \right) \right)^5$$

Result:

1.651220569...

1.651220569.... result very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Series representations:

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg)^5 = \\
& \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)^5 \right) \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)^{35} \right) \right)
\end{aligned}$$

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg)^5 = \\
& \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^5 \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^{35} \right)
\end{aligned}$$

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg)^5 = \\
& \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - \right. \right. \\
& \quad \left. \left. 25 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^5 \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \right. \\
& \quad \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^{35} \right)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

We obtain also:

$$-\left[32\left(\frac{e^{-4 \times 0.989117352243}}{\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)^7}\right)}\right)^7 \times \left[42\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)}\right)-13\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}\right)\right]\right)^{1 / 2}$$

Input interpretation:

$$-\sqrt{\left(32\left(\frac{e^{-4 \times 0.989117352243}}{\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)^7}\right)}\right)^7 \times \left[42\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)}\right)-13\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}\right)\right]\right)^{1 / 2}}$$

Result:

$$-0.10514303501 \dots i$$

Polar coordinates:

$$r=1.05143035007 \text { (radius), } \theta=-90^{\circ} \text { (angle)}$$

$$1.05143035007$$

Series representations:

$$\begin{aligned}
 & - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
 & \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \right. \\
 & \quad \left. \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \right) = -\frac{8}{5} \sqrt{21} \\
 & \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
 & \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right) / \left(e^{3.956469408972000} \right. \\
 & \quad \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right)^7 \right) \right)
 \end{aligned}$$

$$\begin{aligned}
& - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = -\frac{8}{5} \sqrt{21} \\
& \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) / \right. \\
& \quad \left. \left(e^{3.956469408972000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \right. \\
& \quad \left. \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^7 \right) \right) \Bigg) \\
& - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& -\frac{8}{5} \sqrt{21} \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + \right. \right. \\
& \quad \left. \left. 25 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) / \right. \\
& \quad \left(e^{3.956469408972000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right) \Bigg)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

$$1 / -[32((((e^{(-4*0.989117352243)} / [1+\sqrt{(1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)}})]^7 * [42(1+\sqrt{(1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)}})]-13*(4\pi^2)/25*e^{(2*0.989117352243)})))]^1/2$$

Input interpretation:

$$- \left[1 / \left(\sqrt[32]{ \frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right) \right)} \right]$$

Result:

0.95108534763... *i*

Polar coordinates:

$r = 0.95108534763$ (radius), $\theta = 90^\circ$ (angle)

0.95108534763

We know that the primordial fluctuations are consistent with Gaussian purely adiabatic scalar perturbations characterized by a power spectrum with a spectral index $n_s = 0.965 \pm 0.004$, consistent with the predictions of slow-roll, single-field, inflation.

Thence 0.95108534763 is a result very near to the spectral index n_s , to the mesonic Regge slope, to the inflaton value at the end of the inflation 0.9402 and to the value of the following Rogers-Ramanujan continued fraction:

$$\frac{e^{-\frac{\pi}{5}}}{\sqrt{(\varphi-1)\sqrt{5}}-\varphi+1} = 1 - \frac{e^{-\pi}}{1 + \frac{e^{-2\pi}}{1 + \frac{e^{-3\pi}}{1 + \frac{e^{-4\pi}}{1 + \dots}}}} \approx 0.9568666373$$

Series representations:

$$\begin{aligned} & - \left[1 / \left(\sqrt{\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right. \right. \right. \right. \right. \right. \\ & \qquad \qquad \qquad \left. \left. \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) \right) \right] / \\ & \qquad \qquad \qquad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) \Bigg) = \\ & - \left[5 / \left(8 \sqrt{21} \sqrt{\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \right. \\ & \qquad \qquad \qquad \left. \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right) \right] / \\ & \qquad \qquad \qquad \left(e^{3.956469408972000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right) \right. \\ & \qquad \qquad \qquad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right)^7 \Bigg) \Bigg) \Bigg) \Bigg) \end{aligned}$$

$$\left(1 + \sqrt{1 + \frac{(4\pi^2)e^{2 \times 0.9891173522430000}}{3 \times 25}}\right)^7 \Bigg) =$$

$$- \left(5 / \left(8 \sqrt{21} \sqrt{\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}}\right.}\right.\right.$$

$$\left.\left.\sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k \left(e^{1.978234704486000} \pi^2\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!}\right) / \right.$$

$$\left.\left(e^{3.956469408972000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}}\right.\right.\right.$$

$$\left.\left.\sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k \left(e^{1.978234704486000} \pi^2\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!}\right)^7 \Bigg) \Bigg) \Bigg)$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

From the previous expression

$$\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}}\right)^7 \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}}\right) - 13 \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}\right)}$$

$$= -0.034547055658...$$

we have also:

$$1+1/(((4((2*e^{(-0.989117352243/2)})) / (((1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243))})))))) + (-0.034547055658)$$

Input interpretation:

$$1 + \frac{1}{4 \times \frac{2 e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}}}} - 0.034547055658$$

Result:

1.61976215705...

1.61976215705..... result that is a very good approximation to the value of the golden ratio 1.618033988749...

Series representations:

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 =$$

$$\frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000}}$$

$$\sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)_k}$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 =$$

$$\frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000}}$$

$$\sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!}}$$

$$\begin{aligned}
& 1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 = \\
& \frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} +} \\
& \frac{1}{8} e^{0.4945586761215000} \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From

Properties of Nilpotent Supergravity

E. Dudas, S. Ferrara, A. Kehagias and A. Sagnotti - arXiv:1507.07842v2 [hep-th] 14 Sep 2015

We have that:

Cosmological inflation with a tiny tensor-to-scalar ratio r , consistently with PLANCK data, may also be described within the present framework, for instance choosing

$$\alpha(\Phi) = i M \left(\Phi + b \Phi e^{ik\Phi} \right). \quad (4.35)$$

This potential bears some similarities with the Kähler moduli inflation of [32] and with the poly-instanton inflation of [33]. One can verify that $\chi = 0$ solves the field equations, and that the potential along the $\chi = 0$ trajectory is now

$$V = \frac{M^2}{3} \left(1 - a \phi e^{-\gamma \phi} \right)^2. \quad (4.36)$$

We analyzing the following equation:

$$V = \frac{M^2}{3} \left(1 - a \phi e^{-\gamma \phi} \right)^2 .$$

$$\phi = \varphi - \frac{\sqrt{6}}{k} ,$$

$$a = \frac{b\gamma}{e} < 0 , \quad \gamma = \frac{k}{\sqrt{6}} < 0 .$$

We have:

$$(M^2)/3*[1-(b/\text{euler number} * k/\text{sqrt}6) * (\varphi - \text{sqrt}6/k) * \exp(-(k/\text{sqrt}6)(\varphi - \text{sqrt}6/k))]^2$$

i.e.

$$V = (M^2)/3*[1-(b/\text{euler number} * k/\text{sqrt}6) * (\varphi - \text{sqrt}6/k) * \exp(-(k/\text{sqrt}6)(\varphi - \text{sqrt}6/k))]^2$$

For $k = 2$ and $\varphi = 0.9991104684$, that is the value of the scalar field that is equal to the value of the following Rogers-Ramanujan continued fraction:

$$\frac{\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}}}{1 + \sqrt[5]{\sqrt{\varphi^5 \sqrt[4]{5^3} - 1}} - \varphi + 1} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

we obtain:

$$V = (M^2)/3*[1-(b/\text{euler number} * 2/\text{sqrt}6) * (0.9991104684 - \text{sqrt}6/2) * \exp(-(2/\text{sqrt}6)(0.9991104684 - \text{sqrt}6/2))]^2$$

Input interpretation:

$$V = \frac{M^2}{3} \left(1 - \left(\frac{b}{e} \times \frac{2}{\sqrt{6}} \right) \left(0.9991104684 - \frac{\sqrt{6}}{2} \right) \exp \left(- \frac{2}{\sqrt{6}} \left(0.9991104684 - \frac{\sqrt{6}}{2} \right) \right) \right)^2$$

Result:

$$V = \frac{1}{3} (0.0814845 b + 1)^2 M^2$$

Solutions:

$$b = \frac{225.913 \left(-0.054323 M^2 \pm 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} \quad (M \neq 0)$$

Alternate forms:

$$V = 0.00221324 (b + 12.2723)^2 M^2$$

$$V = 0.00221324 (b^2 M^2 + 24.5445 b M^2 + 150.609 M^2)$$

$$-0.00221324 b^2 M^2 - 0.054323 b M^2 - \frac{M^2}{3} + V = 0$$

Expanded form:

$$V = 0.00221324 b^2 M^2 + 0.054323 b M^2 + \frac{M^2}{3}$$

Alternate form assuming b, M, and V are positive:

$$V = 0.00221324 (b + 12.2723)^2 M^2$$

Alternate form assuming b, M, and V are real:

$$V = 0.00221324 b^2 M^2 + 0.054323 b M^2 + 0.333333 M^2 + 0$$

Derivative:

$$\frac{\partial}{\partial b} \left(\frac{1}{3} (0.0814845 b + 1)^2 M^2 \right) = 0.054323 (0.0814845 b + 1) M^2$$

Implicit derivatives:

$$\frac{\partial b(M, V)}{\partial V} = \frac{154317775011120075}{36961748 (226802245 + 18480874 b) M^2}$$

$$\frac{\partial b(M, V)}{\partial M} = - \frac{\frac{226802245}{18480874} + b}{M}$$

$$\frac{\partial M(b, V)}{\partial V} = \frac{154317775011120075}{2 (226802245 + 18480874 b)^2 M}$$

$$\frac{\partial M(b, V)}{\partial b} = - \frac{18480874 M}{226802245 + 18480874 b}$$

$$\frac{\partial V(b, M)}{\partial M} = \frac{2 (226802245 + 18480874 b)^2 M}{154317775011120075}$$

$$\frac{\partial V(b, M)}{\partial b} = \frac{36961748 (226802245 + 18480874 b) M^2}{154317775011120075}$$

Global minimum:

$$\min\left\{\frac{1}{3} (0.0814845 b + 1)^2 M^2\right\} = 0 \text{ at } (b, M) = (-16, 0)$$

Global minima:

$$\min\left\{\frac{1}{3} M^2 \left(1 - \frac{(b+2) \left(0.9991104684 - \frac{\sqrt{6}}{2}\right) \exp\left(-\frac{2 \left(0.9991104684 - \frac{\sqrt{6}}{2}\right)}{\sqrt{6}}\right)}{e \sqrt{6}}\right)^2\right\} = 0$$

for $b = -\frac{226802245}{18480874}$

$$\min\left\{\frac{1}{3} M^2 \left(1 - \frac{(b+2) \left(0.9991104684 - \frac{\sqrt{6}}{2}\right) \exp\left(-\frac{2 \left(0.9991104684 - \frac{\sqrt{6}}{2}\right)}{\sqrt{6}}\right)}{e \sqrt{6}}\right)^2\right\} = 0$$

for $M = 0$

From:

$$b = \frac{225.913 \left(-0.054323 M^2 \pm 6.58545 \times 10^{-10} \sqrt{M^4}\right)}{M^2} \quad (M \neq 0)$$

we obtain

$$(225.913 (-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4}))/M^2$$

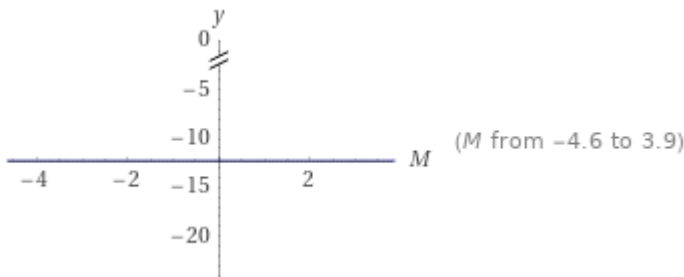
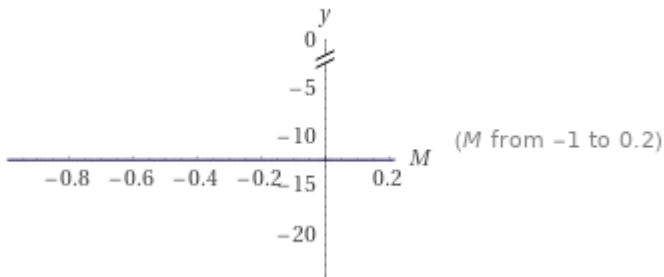
Input interpretation:

$$\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4}\right)}{M^2}$$

Result:

$$\frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2}$$

Plots:



Alternate form assuming M is real:

$$-12.2723$$

-12.2723 result very near to the black hole entropy value $12.1904 = \ln(196884)$

Alternate forms:

$$-\frac{12.2723 \left(M^2 - 1.21228 \times 10^{-8} \sqrt{M^4} \right)}{M^2}$$

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4} - 12.2723 M^2}{M^2}$$

Expanded form:

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M^2} - 12.2723$$

Property as a function:

Parity

even

Series expansion at $M = 0$:

$$\left(\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M^2} - 12.2723 \right) + O(M^6)$$

(generalized Puiseux series)

Series expansion at $M = \infty$:

$$-12.2723$$

Derivative:

$$\frac{d}{dM} \left(\frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right) = \frac{3.55271 \times 10^{-15}}{M}$$

Indefinite integral:

$$\int \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} dM =$$

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M} - 12.2723 M + \text{constant}$$

Global maximum:

$$\max \left\{ \frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right\} =$$

$$-\frac{140\,119\,826\,723\,990\,341\,497\,649}{11\,417\,594\,849\,251\,000\,000\,000} \text{ at } M = -1$$

Global minimum:

$$\min \left\{ \frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right\} =$$

$$-\frac{140\,119\,826\,723\,990\,341\,497\,649}{11\,417\,594\,849\,251\,000\,000\,000} \text{ at } M = -1$$

Limit:

$$\lim_{M \rightarrow \pm\infty} \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} = -12.2723$$

Definite integral after subtraction of diverging parts:

$$\int_0^\infty \left(\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} - -12.2723 \right) dM = 0$$

From b that is equal to

$$\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2}$$

From:

$$V = \frac{1}{3} (0.0814845 b + 1)^2 M^2$$

we obtain:

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4}))/M^2) + 1)^2 M^2$$

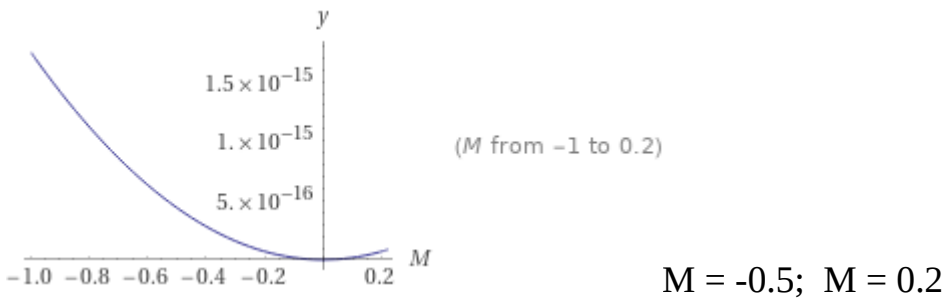
Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

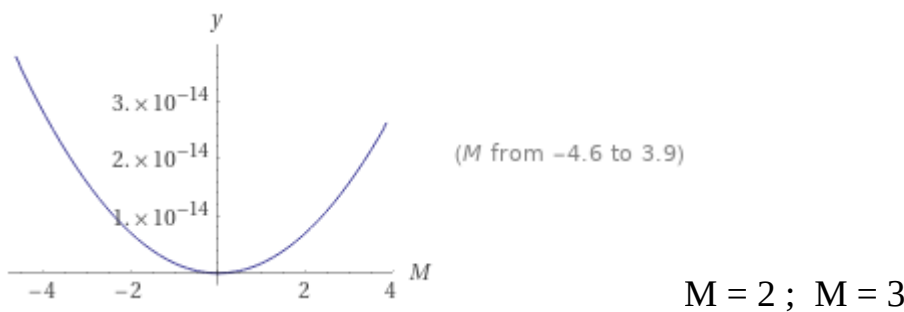
Result:

0

Plots: (possible mathematical connection with an open string)



(possible mathematical connection with an open string)



Root:

$M = 0$

Property as a function:

Parity

even

Series expansion at $M = 0$:

$$O(M^{62194})$$

(Taylor series)

Series expansion at $M = \infty$:

$$1.75541 \times 10^{-15} M^2 + O\left(\left(\frac{1}{M}\right)^{62194}\right)$$

(Taylor series)

Definite integral after subtraction of diverging parts:

$$\int_0^\infty \left(\frac{1}{3} M^2 \left(1 + \frac{18.4084 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} \right)^2 - 1.75541 \times 10^{-15} M^2 \right) dM = 0$$

For $M = -0.5$, we obtain:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right) M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 (-0.5)^2 + 6.58545 \times 10^{-10} \sqrt{(-0.5)^4}))/(-0.5)^2) + 1)^2 * (-0.5^2)$$

Input interpretation:

$$\frac{1}{3} \left(\frac{0.0814845 \times \frac{225.913 \left(-0.054323 (-0.5)^2 + 6.58545 \times 10^{-10} \sqrt{(-0.5)^4} \right)}{(-0.5)^2} + 1}{(-0.5^2)} \right)^2$$

Result:

$$-4.38851344947464545348970783378088020833333333333333333... \times 10^{-16}$$

$$-4.38851344947 \times 10^{-16}$$

For $M = 0.2$:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 \cdot 0.2^2 + 6.58545 \times 10^{-10} \sqrt{0.2^4}))) / 0.2^2) + 1)^2 \cdot 0.2^2$$

Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 \times 0.2^2 + 6.58545 \times 10^{-10} \sqrt{0.2^4} \right)}{0.2^2} + 1 \right)^2 \times 0.2^2$$

Result:

7.0216215191594327255835325340494083333333333333333333333... $\times 10^{-17}$

$7.021621519159 \times 10^{-17}$

For $M = 3$:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 3^2 + 6.58545 \times 10^{-10} \sqrt{3^4}))) / 3^2 + 1)^2 3^2$

Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 \times 3^2 + 6.58545 \times 10^{-10} \sqrt{3^4} \right)}{3^2} + 1 \right)^2 \times 3^2$$

Result:

$$1.579864841810872363256294820161116875 \times 10^{-14}$$

$$1.57986484181 \times 10^{-14}$$

For $M = 2$:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 \cdot 2^2 + 6.58545 \times 10^{-10} \sqrt{2^4}))/2^2) + 1)^2 \cdot 2^2$$

Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 \times 2^2 + 6.58545 \times 10^{-10} \sqrt{2^4} \right)}{2^2} + 1 \right)^2 \times 2^2$$

Result:

$$7.0216215191594327255835325340494083333333333333333333333 \times 10^{-15}$$

$$7.021621519 \times 10^{-15}$$

From the four results

7.021621519*10⁻¹⁵; 1.57986484181*10⁻¹⁴; 7.021621519159*10⁻¹⁷;

$-4.38851344947 \times 10^{-16}$

we obtain, after some calculations:

$$\text{sqrt}[1/(2\pi)(7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16})]$$

Input interpretation:

$$\sqrt{\left(\frac{1}{2\pi} (7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16})\right)}$$

Result:

$$5.9776991059... \times 10^{-8}$$

5.9776991059*10⁻⁸ result very near to the Planck's electric flow 5.975498 × 10⁻⁸ that is equal to the following formula:

$$\phi_P^E = \mathbf{E}_P l_P^2 = \phi_P l_P = \sqrt{\frac{\hbar c}{\epsilon_0}}$$

We note that:

$$1/55 * (((((1/[(7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16})]))))^{1/7} - ((\log^{5/8}(2))/(2 \cdot 2^{1/8} \cdot 3^{1/4} \cdot e \log^{3/2}(3))))))$$

Input interpretation:

$$\frac{1}{55} \left(\left(1 / (7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16}) \right)^{(1/7)} - \frac{\log^{5/8}(2)}{2 \sqrt[8]{2} \sqrt[4]{3} e \log^{3/2}(3)} \right)$$

$\log(x)$ is the natural logarithm

Result:

1.6181818182...

1.6181818182... result that is a very good approximation to the value of the golden ratio 1.618033988749...

From the Planck units:

Planck Length

$$l_P = \sqrt{\frac{4\pi\hbar G}{c^3}}$$

$5.729475 * 10^{-35}$ Lorentz-Heaviside value

Planck's Electric field strength

$$\mathbf{E}_P = \frac{F_P}{q_P} = \sqrt{\frac{c^7}{16\pi^2 \epsilon_0 \hbar G^2}}$$

$1.820306 * 10^{61}$ V*m Lorentz-Heaviside value

Planck's Electric flux

$$\phi_P^E = \mathbf{E}_P l_P^2 = \phi_P l_P = \sqrt{\frac{\hbar c}{\epsilon_0}}$$

$5.975498 * 10^{-8}$ V*m Lorentz-Heaviside value

Planck's Electric potential

$$\phi_P = V_P = \frac{E_P}{q_P} = \sqrt{\frac{c^4}{4\pi\epsilon_0 G}}$$

1.042940×10^{27} V Lorentz-Heaviside value

Relationship between Planck's Electric Flux and Planck's Electric Potential

$$E_P * I_P = (1.820306 * 10^{61}) * 5.729475 * 10^{-35}$$

Input interpretation:

$$\frac{(1.820306 \times 10^{61}) \times 5.729475}{10^{35}}$$

Result:

1 042 939 771 935 000 000 000 000 000

Scientific notation:

$$1.042939771935 \times 10^{27}$$

$$1.042939771935 \times 10^{27} \approx 1.042940 \times 10^{27}$$

Or:

$$E_P * I_P^2 / I_P = (5.975498 \times 10^{-8}) * 1 / (5.729475 * 10^{-35})$$

Input interpretation:

$$5.975498 \times 10^{-8} \times \frac{1}{\frac{5.729475}{10^{35}}}$$

Result:

1.04293988541707573556041347592929544155441816222254220500133... $\times 10^{27}$

$$1.042939885417 \times 10^{27} \approx 1.042940 \times 10^{27}$$

Observations

We note that, from the number 8, we obtain as follows:

$$8^2$$

$$64$$

$$8^2 \times 2 \times 8$$

$$1024$$

$$8^4 = 8^2 \times 2^6$$

True

$$8^4 = 4096$$

$$8^2 \times 2^6 = 4096$$

$$2^{13} = 2 \times 8^4$$

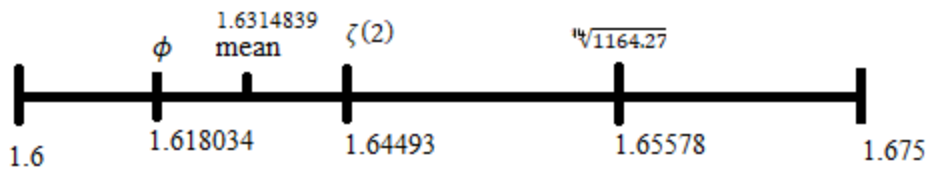
True

$$2^{13} = 8192$$

$$2 \times 8^4 = 8192$$

We notice how from the numbers 8 and 2 we get 64, 1024, 4096 and 8192, and that 8 is the fundamental number. In fact $8^2 = 64$, $8^3 = 512$, $8^4 = 4096$. We define it "fundamental number", since 8 is a Fibonacci number, which by rule, divided by the previous one, which is 5, gives 1.6, a value that tends to the golden ratio, as for all numbers in the Fibonacci sequence

“Golden” Range



Finally we note how $8^2 = 64$, multiplied by 27, to which we add 1, is equal to 1729, the so-called "Hardy-Ramanujan number". Then taking the 15th root of 1729, we obtain a value close to $\zeta(2)$ that 1.6438 ..., which, in turn, is included in the range of what we call "golden numbers"

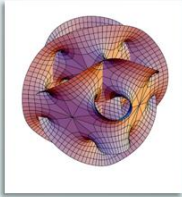
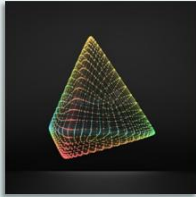
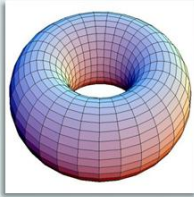
Furthermore for all the results very near to 1728 or 1729, adding $64 = 8^2$, one obtain values about equal to 1792 or 1793. These are values almost equal to the Planck multipole spectrum frequency 1792.35 and to the hypothetical Gluino mass

Appendix

Outlook

Remarkably rich (apparently **UNIQUE**) framework

BUT :



Why a given **“shape” of the extra dimensions** ?
[**CRUCIAL**, it determines the predictions for α , ...]

✎ ✂ ✖ ➡A. Sagnotti – AstronomiAmo, 23.4.202021

From: A. Sagnotti – AstronomiAmo, 23.04.2020

In the above figure, it is said that: “why a given shape of the extra dimensions? Crucial, it determines the predictions for α ”.

We propose that whatever shape the compactified dimensions are, their geometry must be based on the values of the golden ratio and $\zeta(2)$, (the latter connected to 1728 or 1729, whose fifteenth root provides an excellent approximation to the above mentioned value) which are recurrent as solutions of the equations that we are going to develop. It is important to specify that the initial conditions are **always** values belonging to a fundamental chapter of the work of S. Ramanujan "Modular equations and Approximations to Pi" (see references). These values are some multiples of 8 (64 and 4096), 276, which added to 4096, is equal to 4372, and finally $e^{\pi\sqrt{22}}$

We have, in certain cases, the following connections:

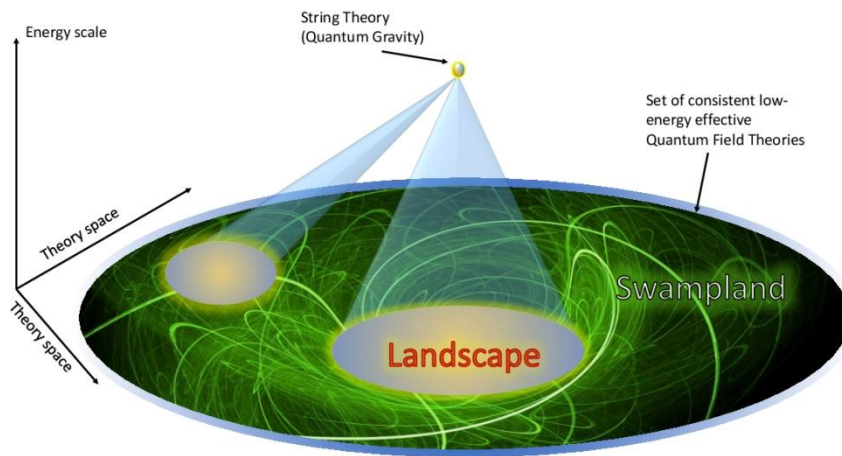


Fig. 1

The String Theory “Landscape”

- Graph axes show only 2 out of hundreds of parameters (“moduli”) that determine the exact Calabi-Yau manifolds and how strings wrap around them

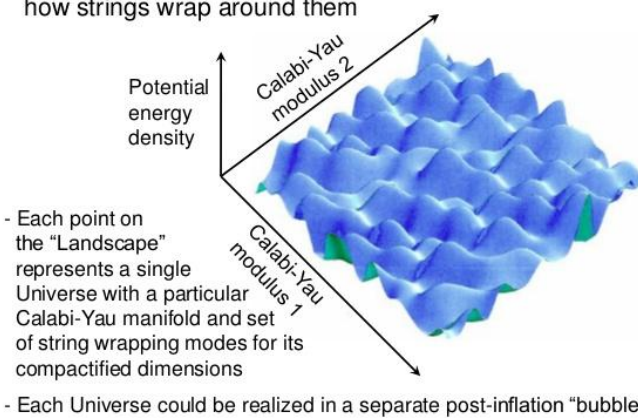


Fig. 2

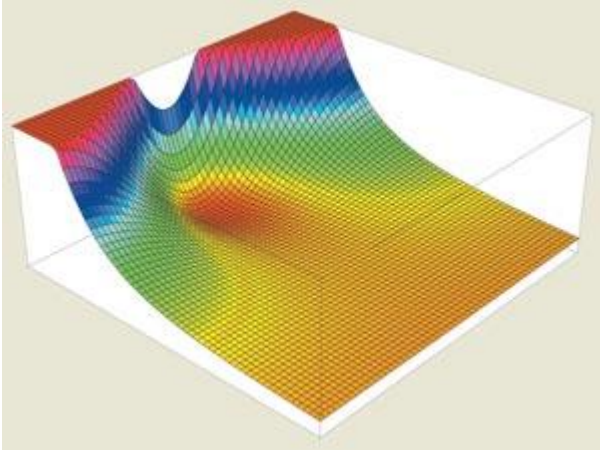


Fig. 3

Stringscape - a small part of the string-theory landscape showing the new de Sitter solution as a local minimum of the energy (vertical axis). The global minimum occurs at the infinite size of the extra dimensions on the extreme right of the figure.

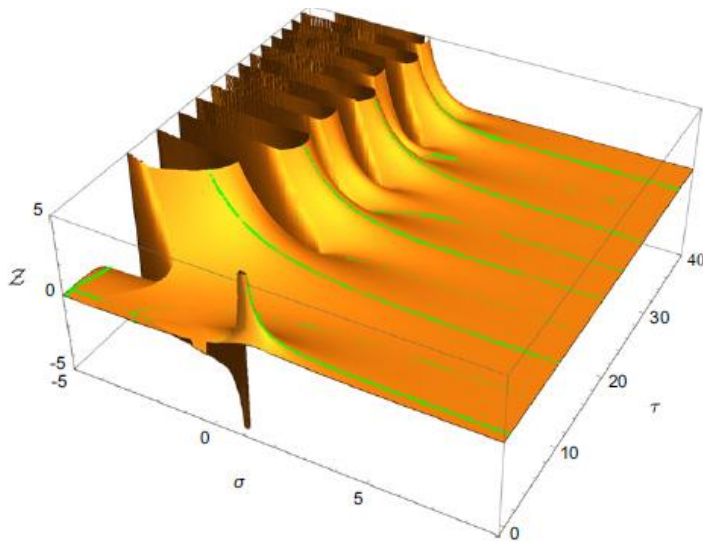


Figure 2. Lines in the complex plane where the Riemann zeta function ζ is real (green) depicted on a relief representing the positive absolute value of ζ for arguments $s \equiv \sigma + i\tau$ where the real part of ζ is positive, and the negative absolute value of ζ where the real part of ζ is negative. This representation brings out most clearly that the lines of constant phase corresponding to phases of integer multiples of 2π run down the hills on the left-hand side, turn around on the right and terminate in the non-trivial zeros. This pattern repeats itself infinitely many times. The points of arrival and departure on the right-hand side of the picture are equally spaced and given by equation (11).

Fig. 4

With regard the Fig. 4 the points of arrival and departure on the right-hand side of the picture are equally spaced and given by the following equation:

$$\tau'_k \equiv k \frac{\pi}{\ln 2},$$

with $k = \dots, -2, -1, 0, 1, 2, \dots$

we obtain:

$$2\pi/(\ln(2))$$

Input:

$$2 \times \frac{\pi}{\log(2)}$$

Exact result:

$$\frac{2\pi}{\log(2)}$$

Decimal approximation:

9.0647202836543876192553658914333336203437229354475911683720330958
...

9.06472028365....

Alternative representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log_e(2)}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log(a) \log_a(2)}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2 \coth^{-1}(3)}$$

Series representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2i\pi \left\lfloor \frac{\arg(2-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k}} \quad \text{for } x < 0$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log(z_0) + \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \left(\log\left(\frac{1}{z_0}\right) + \log(z_0) \right) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k}}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2i\pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k}}$$

Integral representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\int_1^2 \frac{1}{t} dt}$$

$$\frac{2\pi}{\log(2)} = \frac{4i\pi^2}{\int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{\Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds} \quad \text{for } -1 < \gamma < 0$$

From which:

$$(2\pi/(\ln(2))) \cdot (1/12 \pi \log(2))$$

Input:

$$\left(2 \times \frac{\pi}{\log(2)}\right) \left(\frac{1}{12} \pi \log(2)\right)$$

$\log(x)$ is the natural logarithm

Exact result:

$$\frac{\pi^2}{6}$$

Decimal approximation:

1.6449340668482264364724151666460251892189499012067984377355582293
...

$$1.6449340668\dots = \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$$

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